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United States Patent Application Publication

20250277152

Kind Code

A1

Publication Date

September 04, 2025

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LIQUID-CRYSTAL MEDIUM

Abstract

A liquid-crystal (LC) medium (as a subcategory of liquid crystal material), based on a mixture of polar compounds, its use for optical, electro-optical and electronic purposes, in particular in LC displays, especially in LC displays of the vertically aligned mode, an LC display of the vertically aligned mode comprising the LC medium, especially an energy-saving LC display, and a process of manufacturing the LC display.

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Appl. No.: 19/060246

Filed: February 21, 2025

Foreign Application Priority Data

EP

24160433

Feb. 29, 2024

Publication Classification

Int. Cl.: C09K19/46 (20060101); C09K19/04 (20060101); G02F1/1343 (20060101); G02F1/137 (20060101)

U.S. Cl.:

CPC C09K19/46 (20130101); G02F1/13439 (20130101); G02F1/13712 (20210101); C09K2019/0448 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a U.S. national application filed under 35 U.S.C. § 111 (a), claiming priority benefit under 35 U.S.C. § 119 (a) of and to EP patent application Ser. No. 24/160,433.9, filed Feb. 29, 2024, the entire contents of which are incorporated herein by reference in their entirety and for all purposes.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to a liquid-crystal (LC) medium (as a subcategory of liquid crystal material), based on a mixture of polar compounds, to its use for optical, electro-optical and electronic purposes, in particular in LC displays, especially in LC displays of the vertically aligned mode, to an LC display of the vertically aligned mode comprising the LC medium, especially an energy-saving LC display and to a process of manufacturing the LC display.

[0003] The popularity of 8K and gaming monitors leads to an increased need for LC display (LCD) panels having higher refresh rates and thus for LC media having faster response times. Many of these LCD panels are using display modes wherein the LC molecules are aligned substantially perpendicular or slightly tilted relative to the electrode surface in the switched-off state.

[0004] Thus, so-called VA (“vertically aligned”) displays are known which have a broad viewing angle and fast response times. The LC cell of a VA display contains a layer of an LC medium between two transparent electrodes, where the LC medium usually has a negative value of the dielectric anisotropy ($\Delta\epsilon$). In the switched-off state, the molecules of the LC layer are aligned perpendicular to the electrode surfaces (homeotropically) or have a tilted homeotropic alignment. On application of a voltage to the two electrodes, a realignment of the LC molecules parallel to the electrode surfaces takes place.

[0005] Furthermore, so-called FFS (“fringe-field switching”) displays have been reported (see, inter alia, S. H. Jung et al., Jpn. J. Appl. Phys., Volume 43, No. 3, 2004, 1028), which contain two electrodes on the same substrate, one of which is structured in a comb-shaped manner and the other is unstructured. A strong, so-called “fringe field” is thereby generated, i.e. a strong electric field close to the edge of the electrodes, and, throughout the cell, an electric field which has both a strong vertical component and also a strong horizontal component. FFS displays have a low viewing-angle dependence of the contrast. FFS displays usually contain an LC medium with positive dielectric anisotropy, and an alignment layer, usually of polyimide, which provides planar alignment to the molecules of the LC medium.

[0006] FFS displays can be operated as active-matrix or passive-matrix displays. In the case of active-matrix displays, individual pixels are usually addressed by integrated, non-linear active elements, such as, for example, transistors (for example thin-film transistors (“TFTs”)), while in the case of passive-matrix displays, individual pixels are usually addressed by the multiplex method, as known from the prior art.

[0007] Also known are so-called IPS (“in-plane switching”) displays, which contain an LC layer between two substrates with planar orientation, where the two electrodes are arranged on only one of the two substrates and preferably have interdigitated, comb-shaped structures. On application of a voltage to the electrodes an electric field with a significant component parallel to the LC layer is generated between them. This causes realignment of the LC molecules in the layer plane.

[0008] Furthermore, FFS displays have been disclosed (see S. H. Lee et al., Appl. Phys. Lett. 73 (20), 1998, 2882-2883 and S. H. Lee et al., Liquid Crystals 39 (9), 2012, 1141-1148), which have similar electrode design and layer thickness as FFS displays, but comprise a layer of an LC medium with negative dielectric anisotropy instead of an LC medium with positive dielectric anisotropy. The LC medium with negative dielectric anisotropy shows a more favourable director orientation that has less tilt and more twist orientation compared to the LC medium with positive

dielectric anisotropy, as a result of which these displays have a higher transmission.

[0009] Furthermore, VA displays have been disclosed which use an alignment layer that is prepared by photoalignment, also known as UV.sup.2A mode (see, e.g., Q. Tang et al., SID Symposium Digest of Technical Papers 2018, 414-417). These displays utilize an alignment layer prepared from crosslinkable and photo orientable monomers or prepolymers, e.g., cinnamate chromophores which are irradiated obliquely with linearly polarized UV light. As a result, a crosslinked alignment layer is formed which induces uniaxial alignment with a pretilt angle in the LC molecules close to its surface. By changing the irradiation direction, a multidomain configuration with different pretilt directions can be obtained.

[0010] However, the use of LC media with negative dielectric anisotropy in VA or FFS displays has also several drawbacks. For example, they have a significantly lower reliability compared to LC media with positive dielectric anisotropy.

[0011] The term “reliability” as used hereinafter means the quality of the performance of the display during time and with different stress loads, such as light load, temperature, humidity, or voltage which cause display defects such as image sticking (area and line image sticking), mura, yogore etc. and which are known to the skilled person in the field of LC displays. As a standard parameter for categorising the reliability usually the voltage holding ration (VHR) value is used, which is a measure for maintaining a constant electrical voltage in a test display. The higher the VHR value, the better the reliability of the LC medium.

[0012] The reduced reliability of an LC medium with negative dielectric anisotropy in a VA or FFS display can be explained by an interaction of the LC molecules with the polyimide of the alignment layer, as a result of which ions are extracted from the polyimide alignment layer, and wherein LC molecules with negative dielectric anisotropy do more effectively extract such ions.

[0013] This results in new requirements for LC media to be used in VA or FFS displays. In particular, the LC medium has to show a high reliability and a high VHR value after UV exposure. Further requirements are a high specific resistance, a large working-temperature range, short response times even at low temperatures, a low threshold voltage, a multiplicity of grey levels, high contrast and a broad viewing angle, and reduced image sticking.

[0014] Thus, in displays known from prior art often the undesired effect of so-called “image sticking” or “image burn” is observed, wherein the image produced in the LC display by temporary addressing of individual pixels still remains visible even after the electric field in these pixels has been switched off, or after other pixels have been addressed.

[0015] This “image sticking” can occur on the one hand if LC media having a low VHR are used. The UV component of daylight or the backlight can cause undesired decomposition reactions of the LC molecules therein and thus initiate the production of ionic or free-radical impurities. These may accumulate, in particular, at the electrodes or the alignment layers, where they may reduce the effective applied voltage.

[0016] Another problem observed in prior art is that LC media for use in displays, including but not limited to VA and FFS displays, do often exhibit high viscosities and, as a consequence, high switching times. In order to reduce the viscosity and switching time of the LC medium, it has been suggested in prior art to add LC compounds with an alkenyl group. However, it was observed that LC media containing alkenyl compounds often show a decrease of the reliability and stability, and a decrease of the VHR especially after exposure to UV radiation but also to visible light from the backlight of a display, that usually does not emit UV light.

[0017] Especially for some FFS display modes, like the UB-FFS (“Ultra Bright FFS”) and UBplus mode, apart from fast switching times, there is a strong desire in the market for a high contrast ratio in order to achieve a superior black state. A possible approach for the generation of a high contrast ratio could be to provide LC materials with a high value of the average of the elastic constants K_{11} (splay), K_{22} (twist) and K_{33} (bend). However, it is difficult to find LC single compounds which have the potential to increase the average of the elastic constants without

negatively affecting the other physical parameters. An alternative approach could be to increase the average of the elastic constants by increasing the clearing point of the LC mixture. However, this would also result in an increase of the rotational viscosity of the LC mixture, and thus in slower switching times, which is not desirable.

[0018] Further displays for achieving faster response times are using the polymer stabilized (PS) or polymer sustained alignment modes (PSA) modes, like the PS-VA (vertically aligned), PS-IPS (in-plane switching) or PS-FFS (fringe-field switching) mode or modes derived therefrom, or self-aligned (SA) modes like SA-VA which are polymer stabilized.

[0019] In the PS or PSA mode a small amount, typically from 0.1 to 1% of one or more polymerizable mesogenic compounds, also known as RMs (reactive mesogens), is added to the LC medium. After filling the LC medium into the display, the RMs are then polymerized in situ by UV photopolymerization, while a voltage is applied to the electrodes of the display. Thereby a small tilt angle is generated in the LC molecules of the LC medium, which is stabilized by the polymerized RMs. The UV polymerization process, also referred to as “PSA process”, is usually carried out in two steps, a first UV exposure step (“UV1 step”), with application of a voltage, to generate the tilt angle, and a second UV exposure step (“UV2 step”), without application of a voltage, to complete polymerization of the RMs.

[0020] In the SA-VA mode the alignment layers are omitted in the display. Instead, a small amount, typically 0.1 to 2.5%, of a self-alignment (SA) additive is added to the LC medium, which induces the desired alignment, for example homeotropic or planar alignment, in situ by a self-assembling mechanism. The SA additive usually contains an organic, mesogenic core group and attached thereto one or more polar anchor groups, for example hydroxy, carboxy, amino or thiol groups, which can interact with the substrate surface, causing the additives on the substrate surface to align and induce the desired alignment also in the LC molecules. The SA additive may also contain one or more polymerizable groups that can be polymerized under similar conditions as the RMs used in the PSA process. The LC medium may in addition to the SA additive also contain one or more RMs.

[0021] One method to reduce the response times in LC media for the PSA mode is for example by using compounds with an alkenyl group as components of the LC host mixture. However, this may lead to a decrease of the reliability of the mixture when being exposed to the UV light need to polymerize the RMs additives, which is believed to be caused by a reaction of the alkenyl compound with the polyimide of the alignment layer, which is especially problematic when using shorter UV wavelengths of less than 320 nm. Therefore, there is a tendency to use longer UV wavelengths for the PSA process.

[0022] UV-LED lamps have also been proposed for use in the PSA process, as they show less energy consumption, longer lifetime and more effective optical energy transfer to the LC medium due to the narrower emission peak, which allows to reduce the UV intensity and/or UV irradiation time. This enables a reduced tact time and savings in energy and production costs. The UV lamps currently available have higher wavelength emission, for example at 365 nm.

[0023] Therefore, there is also need for polymerizable LC media which enables the RMs to be effectively polymerized at longer UV wavelengths.

[0024] A further problem in the production of PSA displays is the presence or removal of residual amounts of unpolymerized RMs, in particular after the polymerization step for production of the tilt angle in the display. For example, unreacted RMs of this type may adversely affect the properties of the display by, for example, polymerizing in an uncontrolled manner during operation after finishing of the display.

[0025] Thus, in PSA displays of prior art often additional image sticking can be observed which is caused by the presence of unpolymerized RMs. Uncontrolled polymerization of the residual RMs is initiated here by UV light from the environment or by the backlighting. In the switched display areas, this changes the tilt angle after a number of addressing cycles. As a result, a change in

transmission in the switched areas may occur, while it remains unchanged in the unswitched areas.

[0026] It is therefore desirable for the polymerization of the RMs to proceed as completely as possible during production of the PSA display and for the presence of unpolymerized RMs in the display to be excluded as far as possible or reduced to a minimum. Thus, RMs and LC mixtures are required which enable or support highly effective and complete polymerization of the RMs. In addition, controlled reaction of the residual RM amounts would be desirable. This would be simpler if the RM polymerized more rapidly and effectively than the compounds known to date.

[0027] A further problem that has been observed in the operation of PSA displays is the stability of the tilt angle. Thus, it was observed that the tilt angle, which was generated during display manufacture by polymerizing the RM as described above, does not remain constant but can deteriorate after the display was subjected to voltage stress during its operation. This can negatively affect the display performance, e.g., by increasing the black state transmission, hence lowering the contrast.

SUMMARY OF THE INVENTION

[0028] It is therefore an object of the present invention to provide improved LC media for use in VA-, IPS- or FFS displays, and VA-, IPS- and FFS displays comprising them, which do not exhibit the disadvantages described above or only do so to a small extent and have improved properties, including but not limited to a high reliability and high VHR especially after backlight exposure, a high specific resistance, a large working-temperature range, fast response times even at low temperatures, a low threshold voltage, a multiplicity of grey levels, high contrast ratio, superior black state, broad viewing angle, good transmission and reduced image sticking.

[0029] Another object of the invention it to provide improved LC media comprising RMs for use in PSA or SA displays, and PSA or SA displays comprising them, which show very high specific resistance values, high VHR values, high reliability, low threshold voltages, short response times, high birefringence, good UV absorption especially at longer UV wavelengths, preferably in the range from 340 to 380 nm, quick and complete polymerization of the RMs, controlled generation of a low tilt angle, high tilt angle stability after UV exposure, reduced image sticking and ODF mura, and wherein the RMs show a high solubility in the LC host mixture.

[0030] It was found that one or more of these objects could be achieved by providing LC media, which optionally comprise one or more polymerizable compounds, as disclosed and claimed hereinafter.

[0031] The invention relates to an LC medium having negative dielectric anisotropy and comprising one or more compounds of formula L1:

##STR00001## [0032] wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings: [0033] R^{sup}.L1 straight chain, branched or cyclic alkyl having 1 to 15 C atoms, wherein one or more non-adjacent CH₂-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, CR^{sup}.0—CR^{sup}.00—, —C≡C—,

##STR00002## [0034] in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F or Cl, preferably alkyl or alkoxy having 1 to 6 C atoms, [0035] X^{sup}.L CH₃, F, Cl, CF₃ or CHF₂, preferably F, CF₃ or CH₃, very preferably F, [0036] Y^{sup}.L H, F, Cl or CH₃, preferably H or F, very preferably F, [0037] Y^{sup}.0 H, F, Cl, CF₃, CHF₂ or CH₃, preferably H or CH₃, very preferably H,

[0038] The invention further relates to an LC medium as described above and below, which additionally comprises one or more additives selected from the group consisting of stabilisers, chiral dopants, polymerization initiators and self-alignment additives.

[0039] The invention further relates to the use of the LC medium as described above and below in LC displays of the VA, IPS, FFS, UB-FFS, PS-VA, PS-IPS, PS-FFS, PS-UB-FFS or SA-VA mode.

[0040] The invention furthermore relates to a process for preparing an LC medium as described

above and below, comprising the steps of mixing one or more compounds of formula L1, and optionally one, two or more polymerizable compounds of formula M, with further LC compounds and/or additives.

[0041] The invention furthermore relates to an LC display comprising an LC medium according to the invention as described above and below, which is preferably a VA, IPS, FFS, UB-FFS, PS-VA, PS-IPS, PS-FFS, PS-UB-FFS or SA-VA display.

[0042] The invention furthermore relates to an LC display comprising an LC medium as described above and below wherein the polymerizable compounds are present in polymerized form, which is preferably a PSA or SA display, very preferably a PS-VA, PS-IPS, PS-FFS or SA-VA display.

[0043] The invention furthermore relates to an LC display of the PSA type comprising two substrates, at least one which is transparent to light, an electrode provided on each substrate or two electrodes provided on only one of the substrates, and located between the substrates a layer of an LC medium as described above and below, wherein the polymerizable compounds are polymerized between the substrates of the display by UV photopolymerization.

[0044] The invention furthermore relates to a process for manufacturing an LC display as described above and below, comprising the steps of filling or otherwise providing an LC medium as described above and below between the substrates of the display, and polymerizing the polymerizable compounds, preferably by irradiation with UV light, preferably having a wavelength >340 nm, preferably >360 nm, preferably in the range from 340 to 400 nm, more preferably in the range from 350 to 390 nm, very preferably in the range from 360 to 380 nm, most preferably in the range from 360 to 368 nm, and preferably while a voltage is applied to the electrodes of the display.

[0045] The invention furthermore relates to a process for manufacturing a PSA display as described above and below, wherein irradiation of the polymerizable compounds is carried out using a UV-LED lamp.

[0046] The LC media according to the present invention show the following advantageous properties when used in LC displays: [0047] the VHR and reliability can be improved, while maintaining other properties like suitable dielectric anisotropy and birefringence, low viscosity and fast response times, [0048] they show low image sticking, [0049] they show good stability against heat, [0050] they show good solubility in organic solvents typically used in display manufacture.

[0051] In addition, the LC media according to the present invention show one or more of the following advantageous properties when used in PSA displays: [0052] they generate a tilt angle to a desired degree after exposure to UV-light, [0053] they enable fast polymerization of the RMs leading to minimal residues of RM after the UV-process, [0054] they show a high VHR and high reliability, both before and after the UV-process, [0055] they provide a high tilt stability, [0056] they show low image sticking, [0057] they show good UV absorption especially at longer UV wavelengths, preferably in the range from 340 to 400 nm, more preferably in the range from 350 to 390 nm, very preferably in the range from 360 to 380 nm, most preferably in the range from 360 to 368 nm, and enable quick and complete polymerization of the RMs at these wavelengths, [0058] they are suitable for use in PSA displays prepared by a polymerization process using UV-LED lamps, [0059] they enable to control the time range of the first UV-step in which the tilt angle is generated during UV-processing, [0060] they keep the time range of the second UV-step as short as possible to minimize production cost, [0061] after the first and second UV-exposure step, they reduce or avoid any negative effects of the residual RM on the LC mixture performance parameters, such as VHR, tilt stability, etc.

[0062] An alkenyl group in the compounds of formula II, III, IV, V or their subformulae, or in other components of the LC medium as disclosed above and below, is not considered to be within the meaning of the term “polymerizable group” as used herein. The conditions for the polymerization of the polymerizable compounds of the LC medium are preferably selected such that alkenyl substituents do not participate in the polymerization reaction. Preferably the LC media disclosed and claimed in the present application do not contain an additive that initiates or enhances the

participation of the alkenyl group in a polymerization reaction.

[0063] Unless stated otherwise, the polymerizable compounds and the compounds of formula II are preferably selected from achiral compounds.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0064] As used herein, the expression “UV light having a wavelength of” followed by a given range of wavelengths (in nm), or by a given lower or upper wavelength limit (in nm), means that the UV emission spectrum of the respective radiation source has an emission peak, which is preferably the highest peak in the respective spectrum, in the given wavelength range or above the given lower wavelength limit or below the given upper wavelength limit and/or that the UV absorption spectrum of the respective chemical compound has a long or short wavelength tail that extends into the given wavelength range or above the given lower wavelength limit or below the given upper wavelength limit.

[0065] As used herein, the term “full width half maximum” or “FWHM” means the width of a spectrum curve measured between those points on the y-axis which are half the maximum amplitude.

[0066] As used herein, the term “substantially transmissive” means that the filter transmits a substantial part, preferably at least 50% of the intensity, of incident light of the desired wavelength(s). As used herein, the term “substantially blocking” means that the filter does not transmit a substantial part, preferably at least 50% of the intensity, of incident light of the undesired wavelengths. As used herein, the term “desired (undesired) wavelength” e.g. in case of a band pass filter means the wavelengths inside (outside) the given range of λ , and in case of a cut-off filter means the wavelengths above (below) the given value of λ .

[0067] As used herein, the terms “active layer” and “switchable layer” mean a layer in an electrooptical display, for example an LC display, that comprises one or more molecules having structural and optical anisotropy, like for example LC molecules, which change their orientation upon an external stimulus like an electric or magnetic field, resulting in a change of the transmission of the layer for polarized or unpolarized light.

[0068] As used herein, the terms “tilt” and “tilt angle” will be understood to mean a tilted alignment of the LC molecules of an LC medium relative to the surfaces of the cell in an LC display (here preferably a PSA display), and will be understood to be inclusive of “pretilt” and “pretilt angle”. The tilt angle here denotes the average angle ($<90^\circ$) between the longitudinal molecular axes of the LC molecules (LC director) and the surface of the plane-parallel outer plates which form the LC cell. A low absolute value for the tilt angle (i.e. a large deviation from the 90° angle) corresponds to a large tilt here. A suitable method for measurement of the tilt angle is given in the examples. Unless indicated otherwise, tilt angle values disclosed above and below relate to this measurement method.

[0069] As used herein, the terms “reactive mesogen” and “RM” will be understood to mean a compound containing a mesogenic or liquid crystalline skeleton, and one or more functional groups attached thereto which are suitable for polymerization and are also referred to as “polymerizable group” or “P”.

[0070] Unless stated otherwise, the term “polymerizable compound” as used herein will be understood to mean a polymerizable monomeric compound.

[0071] An SA-VA display according to the present invention will be of the polymer stabilised mode as it contains, or is manufactured by use of, an LC medium containing RMs of formula I and II. Consequently, as used herein, the term “SA-VA display” when referring to a display according to the present invention will be understood to refer to a polymer stabilised SA-VA display even if not

explicitly mentioned.

[0072] As used herein, the term “low-molecular-weight compound” will be understood to mean to a compound that is monomeric and/or is not prepared by a polymerization reaction, as opposed to a “polymeric compound” or a “polymer”.

[0073] As used herein, the term “unpolymerizable compound” will be understood to mean a compound that does not contain a functional group that is suitable for polymerization under the conditions usually applied for the polymerization of the RMs.

[0074] The term “mesogenic group” as used herein is known to the person skilled in the art and described in the literature, and means a group which, due to the anisotropy of its attracting and repelling interactions, essentially contributes to causing a liquid-crystal (LC) phase in low-molecular-weight or polymeric substances. Compounds containing mesogenic groups (mesogenic compounds) do not necessarily have to have an LC phase themselves. It is also possible for mesogenic compounds to exhibit LC phase behaviour only after mixing with other compounds and/or after polymerization. Typical mesogenic groups are, for example, rigid rod- or disc-shaped units. An overview of the terms and definitions used in connection with mesogenic or LC compounds is given in *Pure Appl. Chem.* 2001, 73 (5), 888 and C. Tschierske, G. Pelzl, S. Diele, *Angew. Chem.* 2004, 116, 6340-6368.

[0075] The term “spacer group”, hereinafter also referred to as “Sp”, as used herein is known to the person skilled in the art and is described in the literature, see, for example, *Pure Appl. Chem.* 2001, 73 (5), 888 and C. Tschierske, G. Pelzl, S. Diele, *Angew. Chem.* 2004, 116, 6340-6368. As used herein, the terms “spacer group” or “spacer” mean a flexible group, for example an alkylene group, which connects the mesogenic group and the polymerizable group(s) in a polymerizable mesogenic compound.

[0076] Above and below,

##STR00003##

denotes a trans-1,4-cyclohexylene ring, and

##STR00004##

denotes a 1,4-phenylene ring.

[0077] In a group

##STR00005##

the single bond shown between the two ring atoms can be attached to any free position of the benzene ring.

[0078] If in the formulae shown above and below a terminal group like R.sup.1A,2A, R.sup.1, R.sup.2, R.sup.11,12,13, R.sup.21,22, R.sup.31,32, R.sup.41,42, R.sup.51,52, R.sup.61,62, R.sup.71,72, R.sup.81,82,83, R.sup.Q, R.sup.0, R, R.sup.M, R.sup.S, R.sup.S1,S2,S3,S4 or L denotes an alkyl radical and/or an alkoxy radical, this may be straight-chain or branched. It is preferably straight-chain, has 2, 3, 4, 5, 6 or 7 C atoms and accordingly preferably denotes ethyl, propyl, butyl, pentyl, hexyl, heptyl, ethoxy, propoxy, butoxy, pentoxy, hexyloxy or heptyloxy, furthermore methyl, octyl, nonyl, decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, methoxy, octyloxy, nonyloxy, decyloxy, undecyloxy, dodecyloxy, tridecyloxy or tetradecyloxy.

[0079] If one of the aforementioned terminal groups denotes an alkyl radical wherein one or more CH.sub.2 groups are replaced by S, this may be straight-chain or branched. It is preferably straight-chain, has 1, 2, 3, 4, 5, 6 or 7 C atoms and accordingly preferably denotes thiomethyl, thioethyl, thiopropyl, thiobutyl, thiopentyl, thiohexyl or thioheptyl.

[0080] Oxaalkyl preferably denotes straight-chain 2-oxapropyl (=methoxymethyl), 2- (=ethoxymethyl) or 3-oxabutyl (=2-methoxyethyl), 2-, 3- or 4-oxapentyl, 2-, 3-, 4- or 5-oxahexyl, 2-, 3-, 4-, 5- or 6-oxaheptyl, 2-, 3-, 4-, 5-, 6- or 7-oxaoctyl, 2-, 3-, 4-, 5-, 6-, 7- or 8-oxanonyl, 2-, 3-, 4-, 5-, 6-, 7-, 8- or 9-oxadecyl.

[0081] If one of the aforementioned terminal groups denotes an alkoxy or oxaalkyl group it may also contain one or more additional oxygen atoms, provided that oxygen atoms are not linked

directly to one another.

[0082] If one of the aforementioned terminal groups denotes an alkyl radical in which one CH.sub.2 group has been replaced by —CH=CH—, this may be straight-chain or branched. It is preferably straight-chain and has 2 to 10 C atoms. Accordingly, it denotes, in particular, vinyl, prop-1- or -2-enyl, but-1-, -2- or -3-enyl, pent-1-, -2-, -3- or -4-enyl, hex-1-, -2-, -3-, -4- or -5-enyl, hept-1-, -2-, -3-, -4-, -5- or -6-enyl, oct-1-, -2-, -3-, -4-, -5-, -6- or -7-enyl, non-1-, -2-, -3-, -4-, -5-, -6-, -7- or -8-enyl, dec-1-, -2-, -3-, -4-, -5-, -6-, -7-, -8- or -9-enyl.

[0083] If one of the aforementioned terminal groups denotes an alkyl or alkenyl radical which is at least monosubstituted by halogen, this radical is preferably straight-chain, and halogen is preferably F or Cl. In the case of polysubstitution, halogen is preferably F. The resultant radicals also include perfluorinated radicals. In the case of mono-substitution, the fluorine or chlorine substituent may be in any desired position, but is preferably in the ω -position.

[0084] In another preferred embodiment, one or more of the aforementioned terminal groups, like R.sup.1A,2A, R.sup.1, R.sup.2, R.sup.11,12,13, R.sup.31,32, R.sup.41,42, R.sup.51,52, R.sup.61,62, R.sup.71,72, R.sup.81,82,83, R.sup.Q, R.sup.0, R, R.sup.M, R.sup.S, R.sup.S1,S2,S3,S4 or L are selected from the group consisting of

##STR00006## [0085] —S.sup.1-F, —O—S.sup.1-F, —O—S.sup.1—O—S.sup.2, wherein S.sup.1 is C.sub.1-12-alkylene or C.sub.2-12-alkenylene and S.sup.2 is H, C.sub.1-12-alkyl or C.sub.2-12-alkenyl, and very preferably are selected from the group consisting of

##STR00007## [0086] —O(CH.sub.2).sub.3OCH.sub.3, —O(CH.sub.2).sub.4OCH.sub.3, —O(CH.sub.2).sub.2F, —O(CH.sub.2).sub.3F, —O(CH.sub.2).sub.4F.

[0087] Halogen is preferably F or Cl, very preferably F.

[0088] The group —CR.sup.0=CR.sup.00— is preferably —CH=CH—.

[0089] —CO—, —C(=O)— and —C(O)— denote a carbonyl group, i.e.

##STR00008##

[0090] Preferred substituents L, are, for example, F, Cl, Br, I, —CN, —NO.sub.2, —NCO, —NCS, —OCN, —SCN, —C(=O)N(R.sup.x).sub.2, —C(=O)Y.sup.1, —C(=O)R.sup.x, —N(R.sup.x).sub.2, straight-chain or branched alkyl, alkoxy, alkylcarbonyl, alkoxycarbonyl, alkylcarbonyloxy or alkoxycarbonyloxy each having 1 to 25 C atoms, in which one or more H atoms may optionally be replaced by F or Cl, optionally substituted silyl having 1 to 20 Si atoms, or optionally substituted aryl having 6 to 25, preferably 6 to 15, C atoms, wherein Rx denotes H, F, Cl, CN, or straight chain, branched or cyclic alkyl having 1 to 25 C atoms, wherein one or more non-adjacent CH.sub.2-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F, Cl, P- or P-Sp-, and

[0091] Y.sup.1 denotes halogen.

[0092] Particularly preferred substituents L are, for example, F, Cl, CN, NO.sub.2, CH.sub.3, C.sub.2H.sub.5, OCH.sub.3, OC.sub.2H.sub.5, COCH.sub.3, COC.sub.2H.sub.5, COOCH.sub.3, COOC.sub.2H.sub.5, CF.sub.3, OCF.sub.3, OCHF.sub.2, OC.sub.2F.sub.5, furthermore phenyl.

##STR00009## [0093] in which L has one of the meanings indicated above.

[0094] In the compounds of formula L1 and its subformulae, R.sup.L1 preferably denotes alkyl with 1, 2, 3, 4, 5 or 6 C atoms in which a CH.sub.2 group is optionally replaced by cyclopropane-1,2-diyl or cyclopentane-1,2-diyl, very preferably propyl, pentyl, cyclopropyl, cyclopentyl, cyclopropylmethyl or cyclopentylmethyl, most preferably propyl or pentyl.

[0095] In the compounds of formula L1 and its subformulae, X.sup.L preferably denotes F, CF.sub.3 or CH.sub.3, very preferably F.

[0096] In the compounds of formula L1 and its subformulae, Y.sup.0 preferably denotes H or CH.sub.3, very preferably H.

[0097] Preferred compounds of formula L1 are selected from the following subformulae:

##STR00010## [0098] wherein R.sup.L1 has the meanings given in formula L1 above and preferably denotes alkyl having 1 to 6 C atoms in which a CH.sub.2 group is optionally replaced by cyclopropane-1,2-diyl or cyclopentane-1,2-diyl, very preferably propyl, pentyl, cyclopropyl, cyclopentyl, cyclopropylmethyl or cyclopentylmethyl, most preferably propyl or pentyl.

[0099] Very preferred are compounds of formula L1-1 and L1-2, especially wherein R.sup.L1 is n-propyl. Further preferred are compounds of formula L1-4 and L1-5, especially wherein R.sup.L1 is n-propyl or n-pentyl. Most preferred are compounds of formula L1-1, especially wherein R.sup.L1 is n-propyl or n-pentyl.

[0100] Preferably the total proportion of the compounds of formula L1 or its subformulae in the LC medium is from 0.1 to 10%, very preferably from 0.2 to 5%, most preferably from 0.3 to 3% by weight.

[0101] Preferably the LC medium, in addition to the compounds of formula L1 or its subformulae, comprises one, two or more polymerizable compounds, preferably selected of formula M:

R.sup.a—B.sup.1—(Z.sup.m—B.sup.2).sub.m—R.sup.b M [0102] wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings: [0103] P a polymerizable group, [0104] Sp a spacer group or a single bond, which is optionally substituted by L.sup.a, [0105] R.sup.a, R.sup.b P, P-Sp-, H, F, Cl, Br, I, —CN, —NO.sub.2, —NCO, —NCS, —OCN, —SCN, SF.sub.5 or straight-chain or branched alkyl having 1 to 25 C atoms, in which, in addition, one or more non-adjacent CH.sub.2 groups may each be replaced, independently of one another, by —C(R.sup.0)=C(R.sup.00)—, —C≡C—, —N(R.sup.00)—, —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, Br, I, CN, P or P-Sp-, where, if B.sup.1 and/or B.sup.2 contain a saturated C atom, R.sup.a and/or R.sup.b may also denote a radical which is spiro-linked to this saturated C atom, [0106] wherein at least one of the radicals R.sup.a and R.sup.b denotes or contains a group P or P-Sp-, [0107] B.sup.1, B.sup.2 an aromatic, heteroaromatic, alicyclic or heterocyclic group, preferably having 4 to 25 ring atoms, which may also contain fused rings, and which is unsubstituted, or mono- or polysubstituted by L, [0108] Z.sup.m —O—, —S—, —CO—, —CO—O—, —OCO—, —O—CO—O—, —OCH.sub.2—, —CH.sub.2O—, —SCH.sub.2—, —CH.sub.2S—, —CF.sub.2O—, —OCF.sub.2—, —CF.sub.2S—, —SCF.sub.2—, —(CH.sub.2).sub.n1—, —CF.sub.2CH.sub.2—, —CH.sub.2CF.sub.2—, —(CF.sub.2).sub.n1—, —CH=CH—, —CF=CF—, —CH=CF—, —CF=CH—, —C≡C—, —CH=CH—COO—, —OCO—CH=CH—, —CH2CH2—CO—O—, O—CO—CH.sub.2—CH.sub.2—, —CR.sup.0R.sup.00— or a single bond, [0109] R.sup.0, R.sup.00 H or alkyl having 1 to 12 C atoms, [0110] m 0, 1, 2, 3 or 4, [0111] n1 1, 2, 3 or 4, [0112] L P, P-Sp-, OH, CH.sub.2OH, F, Cl, Br, I, —CN, —NO.sub.2, —NCO, —NCS, —OCN, —SCN, —C(=O)N(R.sup.x).sub.2, —C(=O)Y.sup.1, —C(=O)R.sup.x, —N(R.sup.x).sub.2, optionally substituted silyl, optionally substituted aryl having 6 to 20 C atoms, or straight-chain or branched alkyl, alkoxy, alkylcarbonyl, alkoxycarbonyl, alkylcarbonyloxy or alkoxycarbonyloxy having 1 to 25 C atoms, in which, in addition, one or more H atoms may be replaced by F, Cl, P or P-Sp-, [0113] Y.sup.1 halogen, [0114] R.sup.x P, P-Sp-, H, halogen, straight-chain, branched or cyclic alkyl having 1 to 25 C atoms, in which, in addition, one or more non-adjacent CH.sub.2 groups may be replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, P or P-Sp-, an optionally substituted aryl or aryloxy group having 6 to 40 C atoms, or an optionally substituted heteroaryl or heteroaryloxy group having 2 to 40 C atoms.

[0115] Further preferred embodiments of the LC medium according to the present invention are listed below, including any combination thereof:

[0116] Preferably the LC medium further comprises one or more compounds of formula II,

##STR00011## [0117] wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings [0118] R.sup.21 and R.sup.22H, straight chain, branched or cyclic alkyl or alkoxy having 1 to 20 C atoms, wherein one or more non-adjacent CH.sub.2-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, CR.sup.0=CR.sup.00—, —C≡C—, ##STR00012## [0119] in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F, Cl, CN or CF.sub.3, preferably alkyl or alkoxy having 1 to 6 C atoms, [0120] R.sup.0, R.sup.00 H or alkyl having 1 to 12 C atoms, [0121] A.sup.1 and A.sup.2 a group selected from the following formulae ##STR00013## ##STR00014## [0122] preferably from formulae A1, A2, A3, A4, A5, A6, A9 and A10, very [0123] preferably from formulae A1, A2, A3, A4, A5, A9 and A10, [0124] Z.sup.1 and Z.sup.2 —CH.sub.2CH.sub.2—, —CH=CH—, —CF.sub.2O—, —OCF.sub.2—, —CH.sub.2O—, —OCH.sub.2—, —CO—O—, —O—CO—, —C.sub.2F.sub.4—, —CF=CF—, —CH=CH—CH.sub.2O— or a single bond, preferably a single bond, [0125] L.sup.1, L.sup.2, L.sup.3 and L.sup.4 F, Cl, OCF.sub.3, CF.sub.3, CH.sub.3, CH.sub.2F or CHF.sub.2, preferably F or Cl, very preferably F, [0126] Y H, F, Cl, CF.sub.3, CHF.sub.2 or CH.sub.3, preferably H or CH.sub.3, very preferably H, [0127] L.sup.C CH.sub.3 or OCH.sub.3, preferably CH.sub.3, [0128] a1 0, 1 or 2, [0129] a2 0 or 1.

[0130] Preferably the LC medium comprises one or more compounds of formula II selected from the group consisting of compounds of the formulae IIA, IIB, IIC, IID, IIE and IIF:

##STR00015## [0131] in which the individual radicals, on each occurrence identically or differently, and each, independently of one another, have the following meaning: [0132] R.sup.21, R.sup.22 H, an alkyl, alkoxy or alkenyl radical having up to 15, preferably up to 6, C atoms which is unsubstituted or monosubstituted by F, Cl, CN or CF.sub.3 and where, in addition, one or more CH.sub.2 groups in these radicals may be replaced by —O—, —S—, —C≡C—, —CF.sub.2O—, —OCF.sub.2—, —OC—O—, —O—CO—

##STR00016## [0133] in such a way that O- and/or S-atoms are not linked directly to one another, [0134] L.sup.1 to L.sup.5 F, Cl, CF.sub.3 or CHF.sub.2, [0135] Y H, F, Cl, CF.sub.3, CHF.sub.2 or CH.sub.3, preferably H or CH.sub.3, particularly preferably H, [0136] Z.sup.1, Z.sup.2 a single bond, —CH.sub.2CH.sub.2—, —CH=CH—, —CF.sub.2O—, —OCF.sub.2—, —CH.sub.2O—, —OCH.sub.2—, —COO—, —OCO—, —C.sub.2F.sub.4—, —CF=CF—, —CH=CHCH.sub.2O, [0137] p 0, 1 or 2, and [0138] q 0 or 1.

[0139] Preferred compounds of the formulae IIA, IIB, IIC, IID, IIE and IIF are those wherein R.sup.22 denotes an alkyl or alkoxy radical having up to 15 C atoms, and very preferably denotes (O)C.sub.vH.sub.2v+1 wherein (O) is an oxygen atom or a single bond and v is 1, 2, 3, 4, 5 or 6.

[0140] Further preferred compounds of the formulae IIA, IIB, IIC, IID, IIE and IIF are those wherein R.sup.21 or R.sup.22 denotes or contains a cycloalkyl or cycloalkoxy radical, preferably selected from the group consisting of

##STR00017## [0141] wherein S.sup.1 is C.sub.1-12-alkylene or C.sub.2-12-alkenylene and S.sub.2 is H, C.sub.1-12-alkyl or C.sub.2-12-alkenyl, and very preferably are selected from the group consisting of

##STR00018##

[0142] Further preferred compounds of the formulae IIA, IIB, IIC, IID, IIE and IIF are indicated below.

[0143] In a preferred embodiment the LC medium comprises one or more compounds of the formula IIA selected from the group consisting of the following formulae:

##STR00019## ##STR00020## ##STR00021##

##STR00022## ##STR00023## ##STR00024## ##STR00025## ##STR00026## [0144] in which the index a denotes 1 or 2, alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, alkenyl denotes a straight-chain alkenyl radical having 2-6

C atoms, and (O) denotes an oxygen atom or a single bond, and alkenyl preferably denotes CH.sub.2=CH—, CH.sub.2=CHCH.sub.2CH.sub.2—, CH.sub.3—CH=CH—, CH.sub.3—CH.sub.2—CH=CH—, CH.sub.3—(CH.sub.2).sub.2—CH=CH—, CH.sub.3—(CH.sub.2).sub.3—CH=CH— or CH.sub.3—CH=CH—(CH.sub.2).sub.2—.

[0145] Particularly preferred LC media according to the invention comprise one or more compounds selected from the group consisting of formulae IIA-2, IIA-8, IIA-10, IIA-16, II-18, IIA-40, IIA-41, IIA-42, IIA-43, IIA-77 and IIA-78.

[0146] Preferably, the LC medium comprises one or more compounds of the formula IIA-2 selected from the following subformulae:

##STR00027##

[0147] Alternatively, preferably in addition to the compounds of the formulae IIA-2-1 to IIA-2-5, the LC medium comprises one or more compounds of the following formulae:

##STR00028##

[0148] Further preferably, the LC medium comprises one or more compounds of the formula IIA-10 selected from the following sub-formulae:

##STR00029##

[0149] Alternatively, preferably in addition to the compounds of the formulae IIA-10-1 to IIA-10-5, the LC medium comprises one or more compounds of the following formulae:

##STR00030##

[0150] Further preferably, the LC medium comprises one or more compounds of the formula IIA-40 selected from the following sub-formulae:

##STR00031##

[0151] Alternatively, preferably in addition to the compounds of the formulae IIA-40-1 to IIA-40-5, the LC medium comprises one or more compounds of the following formulae:

##STR00032##

[0152] Further preferably, the LC medium comprises one or more compounds of the formula IIA-42 selected from the following sub-formulae:

##STR00033##

[0153] Further preferably, the LC medium comprises one or more compounds of the formula IIA-66 selected from the following subformulae:

##STR00034##

[0154] Further preferably, the LC medium comprises one or more compounds of the formula IIA-67 selected from the following sub-formulae:

##STR00035##

[0155] Further preferably, the LC medium comprises one or more compounds of the formula IIA 77 selected from the following with formulae:

##STR00036##

[0156] In another preferred embodiment the LC medium comprises one or more compounds of the formula IIB selected from the group consisting of formulae IIB-1 to IIB-26,

##STR00037## ##STR00038## ##STR00039## ##STR00040##

[0157] in which alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, alkenyl denotes a straight-chain alkenyl radical having 2-6 C atoms, and (O) denotes an oxygen atom or a single bond, and alkenyl preferably denotes CH.sub.2=CH—, CH.sub.2=CHCH.sub.2CH.sub.2—, CH.sub.3—CH=CH—, CH.sub.3—CH.sub.2—CH=CH—, CH.sub.3—(CH.sub.2).sub.2—CH=CH—, CH.sub.3—(CH.sub.2).sub.3—CH=CH— or CH.sub.3—CH=CH—(CH.sub.2).sub.2—.

[0158] Particularly preferred LC media according to the invention comprise one or more compounds selected from the group consisting of formulae IIB-2, IIB-11 and IIB-17.

[0159] Preferably, the LC medium comprises one or more compounds of the formula IIB-11 selected from the following sub-formulae:

##STR00041##

[0160] Alternatively, preferably in addition to the compounds of the formulae IIB-11-1 to IIB-11-5, the LC medium comprises one or more compounds of the following formulae:

##STR00042##

[0161] Preferably, the LC medium comprises one or more compounds of the formula IIB-17 selected from the following sub-formulae:

##STR00043##

[0162] The proportion of compounds of the formulae IIA and/or IIB in the mixture as a whole is preferably at least 20% by weight.

[0163] In another preferred embodiment the LC medium comprises one or more compounds of the formula IIC selected from the formula IIC-1,

##STR00044## [0164] in which alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, preferably in amounts of 0.5% to 5% by weight, in particular 1% to 3% by weight.

[0165] In another preferred embodiment the LC medium comprises one or more compounds of the formula IID selected from the group consisting of the following formulae,

##STR00045## ##STR00046## ##STR00047## ##STR00048## [0166] in which alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, alkenyl denotes a straight-chain alkenyl radical having 2-6 C atoms, (O) denotes an oxygen atom or a single bond, Y denotes H or CH₃ and alkenyl preferably denotes CH₂=CH—, CH₂=CHCH₂CH₂—, CH₃—CH=CH—, CH₃—CH₂—CH=CH—, CH₃—(CH₂)₂—CH=CH—, CH₃—(CH₂)₃—CH=CH— or CH₃—CH=CH—(CH₂)₂—.

[0167] Further preferred LC media according to the invention comprise one or more compounds of the formula IID-1 and/or IID-4.

[0168] Further preferred LC media according to the invention comprise one or more compounds of the formula IID, wherein q is 0, Y is H and R²² denotes an alkyl or alkoxy radical having 1 to 6 C atoms wherein one CH₂ group is optionally replaced by a cyclopropyl, cyclobutyl, cyclopentyl or cyclopentenyl group, preferably selected from formulae IID-1, IID-15, IID-16, IID-17, IID-18, IID-19, IID-20, IID-21, IID-22 and IID-23, very preferably selected from formulae IID-1, IID-16, IID-21, IID-22 and IID-23.

[0169] Very preferred compounds of the formula IID are compounds selected from the following subformulae,

##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053## [0170] wherein v is 1, 2, 3, 4, 5 or 6.

[0171] In a preferred embodiment, the LC medium comprises one or more compounds of formula IID-10a

##STR00054## [0172] in which R²¹, Y and q have the meanings given in formula IID, and R²³ is

##STR00055##

in which r is u, 1, 2, 3, 4, 5 or 6 and s is 1, 2 or 3.

[0173] Preferred compounds of formula IID-10a are selected from the following subformulae:

##STR00056## ##STR00057## ##STR00058## ##STR00059##

[0174] In a preferred embodiment the LC medium comprises one or more compounds of the formula IIE selected from the group consisting of the following formulae:

##STR00060## ##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065##
##STR00066## ##STR00067##

##STR00068## ##STR00069## ##STR00070## ##STR00071## [0175] in which the index a denotes 1 or 2, alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, alkenyl denotes a straight-chain alkenyl radical having 2-6 C atoms,

and (O) denotes an oxygen atom or a single bond, and alkenyl preferably denotes $\text{CH}_2=\text{CH}-$, $\text{CH}_2=\text{CHCH}_2-$, $\text{CH}_3-\text{CH}=\text{CH}-$, $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}-$, $\text{CH}_3-(\text{CH}_2)_2-\text{CH}=\text{CH}-$, $\text{CH}_3-(\text{CH}_2)_3-\text{CH}=\text{CH}-$ or $\text{CH}_3-\text{CH}=\text{CH}-(\text{CH}_2)_2-$.

[0176] Particularly preferred LC media according to the invention comprise one or more compounds selected from the group consisting of formulae IIE-2, IIE-8, IIE-10, IIE-16, IIE-18, IIE-37, IIE-38, IIE-39 and IIE-40.

[0177] Preferably, the LC medium comprises one or more compounds of the formula IIE-2 selected from the following sub-formulae:

##STR00072##

[0178] Preferably, the LC medium comprises one or more compounds of the formula IIE-10 selected from the following sub-formulae:

##STR00073## [0179] In another preferred embodiment the LC medium comprises one or more compounds of the formula IIF selected from the group consisting of the following formulae,

##STR00074## [0180] in which alkyl and alkyl* each, independently of one another, denote a straight-chain or branched alkyl radical having 1-6 C atoms.

[0181] Preferably, the LC medium comprises one or more compounds of the formula IIF-2 selected from the following sub-formulae:

##STR00075## [0182] Particularly preferred LC media according to the invention comprise one or more compounds selected from the formulae IIA-2, IIA-8, IIA-10, IIA-16, IIA-18, IIA-40, IIA-41, IIA-42, IIA-43, IIA-66, IIA-67, IIA-77, IIB-2, IIB-11, IIB-16, IIC-1, IID-1, IID-4, IID-10, IID-16, IIE-2, IIE-8, IIE-10, IIE-16, IIE-18, IIE-37, IIE-38, IIE-39, IIE-40 and IIF-2 or their subformulae.

[0183] In another preferred embodiment the LC medium comprises one or more compounds of formula III,

##STR00076## [0184] in which [0185] R^{sup.31} and R^{sup.32} each, independently of one another, denote H, an alkyl, alkoxy or alkenyl radical having up to 15 C atoms which is unsubstituted, monosubstituted by F, Cl, CN or CF₃ or at least monosubstituted by halogen, where, in addition, one or more CH₂ groups in these radicals may be replaced by —O—, —S—.

##STR00077## [0186] —C≡C—, —CF₂O—, —OCF₂—, —OC—O— or —O—CO— in such a way that O atoms are not linked directly to one another, [0187] Y^{sup.1}, Y^{sup.2} H, F, Cl, CF₃, CHF₂, CH₃ or OCH₃, preferably H, CH₃ or OCH₃, very preferably H, [0188] A^{sup.3} on each occurrence, independently of one another, denotes [0189] a) 1,4-cyclohexenylene or 1,4-cyclohexylene radical, in which one or two non-adjacent CH₂ groups may be replaced by —O— or —S—, [0190] b) a 1,4-phenylene radical, in which one or two CH groups may be replaced by N, or [0191] c) a radical selected from the group consisting of spiro[3.3]heptane-2,6-diyl, 1,4-bicyclo[2.2.2]octylene, naphthalene-2,6-diyl, decahydronaphthalene-2,6-diyl, 1,2,3,4-tetrahydronaphthalene-2,6-diyl, phenanthrene-2,7-diyl and fluorene-2,7-diyl, [0192] wherein the radicals a), b) and c) may be mono- or polysubstituted by halogen atoms, [0193] n denotes 0, 1 or 2, preferably 0 or 1, [0194] Z^{sup.1} on each occurrence independently of one another denotes —CO—O—, —O—CO—, —CF₂O—, —OCF₂—, —CH₂O—, —OCH₂—, —CH₂—, —CH₂CH₂—, —(CH₂)₄—, —CH=CH—CH₂O—, —C₂F₄—, —CH₂CF₂—, —CF₂CH₂—, —CF=CF—, —CH=CF—, —CF=CH—, —CH=CH—, —C≡C— or a single bond, and [0195] L^{sup.11} and L^{sup.12} each, independently of one another, denote F, Cl, CF₃ or CHF₂, preferably H or F, most preferably F, and [0196] W denotes O or S.

[0197] In the compounds of formula III R^{sup.31} and R^{sup.32} are preferably selected from straight-chain alkyl or alkoxy with 1 to 12, preferably 1 to 7 C atoms, straight-chain alkenyl with 2 to 12, preferably 2 to 7 C atoms and cyclic alkyl or alkoxy with 3 to 12, preferably 3 to 8 C atoms.

[0198] In a preferred embodiment of the present invention the LC medium comprises one or more compounds of formula III selected from the subformulae III-1 and III-2,

##STR00078## [0199] in which the occurring groups have the same meanings as given under formula III above and preferably [0200] R.sup.31 and R.sup.32 each, independently of one another, an alkyl, alkenyl or alkoxy radical having up to 15 C atoms, more preferably one or both of them denote an alkoxy radical and [0201] L.sup.11 and L.sup.12 each preferably denote F.

[0202] In another preferred embodiment the LC medium comprises one or more compounds of the formula III-1 selected from the group of compounds of subformulae III-1-1 to III-1-10, preferably of formula III-1-6,

##STR00079## [0203] in which alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, alkenyl and alkenyl* each, independently of one another, denote a straight-chain alkenyl radical having 2-6 C atoms, alkoxy and alkoxy* each, independently of one another, denote a straight-chain alkoxy radical having 1-6 C atoms, and L11 and L12 each, independently of one another, denote F or Cl, preferably both F.

[0204] In another preferred embodiment the LC medium comprises one or more compounds of the formula III-2 selected from the group of compounds of subformulae III-2-1 to III-2-10, preferably of subformula III-2-1,

##STR00080## [0205] in which alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, alkenyl and alkenyl* each, independently of one another, denote a straight-chain alkenyl radical having 2-6 C atoms, alkoxy and alkoxy* each, independently of one another, denote a straight-chain alkoxy radical having 1-6 C atoms, and L11 and L12 each, independently of one another, denote F or Cl, preferably both F.

[0206] Very preferred compounds of formula III-2 are selected from the group consisting of the following subformulae,

##STR00081## [0207] in which alkoxy denotes a straight-chain alkoxy radical having 1-6 C atoms, preferably ethoxy, propoxy, butoxy or pentoxy, very preferably ethoxy or propoxy.

[0208] Very preferred are the compounds of formula III-2-1-3, III-2-1-4 and III-2-1-5.

[0209] In another preferred embodiment of the present invention the LC medium comprises one or more compounds of formula III selected from the formulae III-3-1 and III-3-2

##STR00082## [0210] in which L.sup.11 and L.sup.12 have the same meanings as given under formula III, (O) denotes O or a single bond, [0211] R.sup.33 denotes alkyl or alkenyl having up to 7 C atoms or a group Cy-C.sub.mH.sub.2m+1—, [0212] m and n are, identically or differently, 0, 1, 2, 3, 4, 5 or 6, preferably 1, 2 or 3, very preferably 1, and [0213] Cy denotes a cycloaliphatic group having 3, 4 or 5 ring atoms, which is optionally substituted with alkyl or alkenyl each having up to 3 C atoms, or with halogen or CN, and preferably denotes cyclopropyl, cyclobutyl or cyclopentyl.

[0214] The compounds of formula III-3-1 and/or III-3-2 are contained in the LC medium either alternatively or additionally to the compounds of formula III-1 and/or III-2, preferably additionally.

[0215] Very preferred compounds of the formula III-3-1 are the following,

##STR00083## [0216] in which alkoxy denotes a straight-chain alkoxy radical having 1-6 C atoms.

[0217] Very preferred compounds of the formula III-3-2 are the following,

##STR00084## [0218] in which alkoxy denotes a straight-chain alkoxy radical having 1-6 C atoms, preferably ethoxy, propoxy, butoxy or pentoxy, very preferably ethoxy or propoxy.

[0219] In another preferred embodiment of the present invention, the LC medium comprises one or more compounds of the formulae III-4 to III-6, preferably of formula III-5,

##STR00085## [0220] in which the parameters have the meanings given above, R.sup.11 preferably denotes straight-chain alkyl and R.sup.12 preferably denotes alkoxy, each having 1 to 7 C atoms.

[0221] In another preferred embodiment of the present invention, the LC medium comprises one or more compounds of the formula III selected from the group of compounds of subformulae III-7 to III-9, preferably of subformula III-8,

##STR00086## [0222] in which the parameters have the meanings given above, R.sup.11 preferably denotes straight-chain alkyl and R.sup.12 preferably denotes alkoxy each having 1 to 7 C atoms.

[0223] In another preferred embodiment of the present invention, the LC medium comprises one or more compounds of formula III selected from the subformula III-10,

##STR00087## [0224] in which R.sup.31 and R.sup.32 have the meanings given above.

[0225] Very preferred compounds of formula III-10 are selected from the group consisting of the following formulae.

##STR00088## [0226] in which R.sup.32 denotes alkyl having 1 to 7 C-atoms, preferably ethyl, n-propyl or n-butyl, or alternatively cyclopropylmethyl, cyclobutylmethyl or cyclopentylmethyl or alternatively —(CH.sub.2).sub.nF in which n is 2,3,4, or 5, preferably C.sub.2H.sub.4F.

[0227] In another preferred embodiment of the present invention, the LC medium comprises one or more compounds of the formula III selected of the subformula III-11,

##STR00089## [0228] in which R.sup.31 and R.sup.32 have the meanings given above.

[0229] Very preferred compounds of formula III-11 are selected from the group consisting of the following formulae,

##STR00090## [0230] in which R.sup.32 denotes alkyl having 1 to 7 C-atoms, preferably ethyl, n-propyl or n-butyl, or alternatively cyclopropylmethyl, cyclobutylmethyl or cyclopentylmethyl or alternatively —(CH.sub.2).sub.nF in which n is 2,3,4, or 5, preferably C.sub.2H.sub.4F.

[0231] In a preferred embodiment, the LC medium comprises one or more compounds of the formula IV,

##STR00091## [0232] in which [0233] R.sup.41 denotes an unsubstituted alkyl radical having 1 to 7 C atoms where, in addition, one or more CH.sub.2 groups may be replaced by

##STR00092## [0234] or an unsubstituted alkenyl radical having 2 to 7 C atoms, preferably an n-alkyl radical, particularly preferably having 2, 3, 4 or 5 C atoms, and [0235] R.sup.42 denotes an unsubstituted alkyl radical having 1 to 7 C atoms or an unsubstituted alkoxy radical having 1 to 6 C atoms, both preferably having 2 to 5 C atoms, or an unsubstituted alkenyl radical having 2 to 7 C atoms, preferably having 2, 3 or 4 C atoms, more preferably a vinyl radical or a 1-propenyl radical and in particular a vinyl radical.

[0236] The compounds of the formula IV are preferably selected from the group of the compounds of the formulae IV-1 to IV-4,

##STR00093## [0237] in which [0238] alkyl and alkyl', independently of one another, denote alkyl having 1 to 7 C atoms, preferably having 2 to 5 C atoms, [0239] alkenyl denotes an alkenyl radical having 2 to 5 C atoms, preferably having 2 to 4 C atoms, particularly preferably 2 C atoms, [0240] alkenyl' denotes an alkenyl radical having 2 to 5 C atoms, preferably having 2 to 4 C atoms, particularly preferably having 2 to 3 C atoms, and [0241] alkoxy denotes alkoxy having 1 to 5 C atoms, preferably having 2 to 4 C atoms.

[0242] Preferably, the LC medium comprises one or more compounds selected from the compounds of the formulae IV-1-1 to IV-1-6:

##STR00094##

[0243] Very preferably, the LC medium according to the invention comprises one or more compounds of the formulae IV-2-1 and/or IV-2-2

##STR00095##

[0244] Very preferably, the LC medium according to the invention comprises a compound of formula IV-3, in particular selected from the compounds of the following subformulae:

##STR00096##

[0245] In another preferred embodiment, the LC medium according to the invention comprises one or more compounds of formula IV-3 selected from the compounds of the following subformulae:

##STR00097## ##STR00098## ##STR00099## [0246] in which alkyl denotes methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, or n-pentyl.

[0247] Very preferably, the LC medium according to the invention comprises a compound of formula IV-4, in particular selected from the compounds of the following formulae:

##STR00100##

[0248] In another preferred embodiment the LC medium comprises one or more compounds of formula IV-4 and its subformulae in which one or both of “alkenyl” and “alkenyl” denote

##STR00101##

in which [0249] m is 0, 1 or 2, and n is 0, 1 or 2, very preferably selected from compounds of formulae IV-4-3 to IV-4-6.

[0250] Very preferably, the LC medium according to the invention comprises one or more compounds of the formula IV-1 or its subformulae and/or one or more compounds of the formula IV-3 or its subformulae and/or one or more compounds of the formula IV-4 or its subformulae, where the total concentration of these compounds of the formula IV-1 is in the range from 1% to 30%.

[0251] The LC medium according to the invention preferably additionally comprises one or more compounds of the formula IVa,

##STR00102## [0252] in which [0253] R^{sup.41} and R^{sup.42} each, independently of one another, denote a straight-chain alkyl, alkoxy, alkenyl, alkoxyalkyl or alkoxy radical having up to 12 C atoms,

##STR00103## [0254] Z^{sup.4} denotes a single bond, —CH₂CH₂—, —CH=CH—, —CF₂O—, —OCF₂—, —CH₂O—, —OCH₂—, —COO—, —OCO—, —C₂F₄—, —C₄H₈— or —CF=CF—.

[0255] Preferred compounds of the formula IVa are indicated below:

##STR00104## [0256] in which alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1 to 6 C atoms.

[0257] The LC medium according to the invention preferably comprises at least one compound of the formula IVa-1 and/or formula IVa-2.

[0258] The proportion of compounds of the formula IVa in the mixture as a whole is preferably at least 5% by weight

[0259] Preferably, the LC medium comprises one or more compounds of formula IVb-1 to IVb-5

##STR00105## [0260] in which [0261] alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1 to 6 C atoms, and [0262] alkenyl and alkenyl* each, independently of one another, denote a straight-chain alkenyl radical having 2 to 6 C atoms.

[0263] The proportion of the compounds of the formulae IV-1 to IV-5 in the mixture as a whole is preferably at least 3% by weight, in particular ≥5% by weight.

[0264] Of the compounds of the formulae IVb-1 to IVb-5, the compounds of the formula IVb-2 are particularly preferred.

[0265] Particularly preferred compounds of the formulae IVb-1 to IVb-5 are selected from the group consisting of the following formulae

##STR00106## [0266] in which alkyl* denotes an alkyl radical having 1 to 6 C atoms and preferably denotes n-propyl.

[0267] The LC medium according to the invention particularly preferably comprises one or more compounds of the formulae IVb-1-1 and/or IVb-2-3.

[0268] In another preferred embodiment, the LC medium according to the invention comprises one or more compounds of formula V

##STR00107## [0269] in which [0270] R^{sup.51} and R^{sup.52} independently of one another, denote H, an alkyl, alkoxy or alkenyl radical having up to 15 C atoms which is unsubstituted, monosubstituted by F, Cl, CN or CF₃ or at least monosubstituted by halogen, where, in addition, one or more CH₂ groups in these radicals may be replaced by —O—, —S—, —C≡C—, —CF₂O—, —OCF₂—, —OC—O—, —O—CO—

##STR00108## [0271] in such a way that O atoms are not linked directly to one another, and

preferably denote alkyl having 1 to 7 C atoms, preferably n-alkyl, particularly preferably n-alkyl having 1 to 5 C atoms, alkoxy having 1 to 6 C atoms, preferably n-alkoxy, particularly preferably n-alkoxy having 2 to 5 C atoms, alkoxyalkyl, alkenyl or alkenyloxy having 2 to 7 C atoms, preferably having 2 to 4 C atoms, preferably alkenyloxy,

##STR00109##

identically or differently, denote

##STR00110## [0272] Z^{sup.51}, Z^{sup.52} each, independently of one another, denote —CH_{sub.2}—, —CH_{sub.2}—O—, —CH=CH—, —C≡C—, —COO— or a single bond, preferably —CH_{sub.2}—CH_{sub.2}—, —CH_{sub.2}—O— or a single bond and particularly preferably a single bond, and [0273] n is 1 or 2.

[0274] The compounds of formula V are preferably selected from the compounds of the formulae V-1 to V-17:

##STR00111## ##STR00112## [0275] in which R^{sup.1} and R^{sup.2} have the meanings indicated for R^{sup.51} and R^{sup.52} above. [0276] R^{sup.1} and R^{sup.2} preferably each, independently of one another, denote straight-chain alkyl or alkenyl.

[0277] Preferred LC media comprise one or more compounds of the formulae V-1, V-3, V-4, V-6, V-7, V-10, V-11, V-12, V-14, V-15, and/or V-16.

[0278] LC media according to the invention very particularly preferably comprise the compounds of the formula V-10 and/or IV-1, in particular in amounts of 5 to 30%.

[0279] Preferred compounds of the formulae V-10 are indicated below:

##STR00113##

[0280] The LC medium according to the invention particularly preferably comprises the tricyclic compounds of the formula V-10a and/or of the formula V-10b in combination with one or more bicyclic compounds of the formulae IV-1. The total proportion of the compounds of the formulae V-10a and/or V-10b in combination with one or more compounds selected from the bicyclohexyl compounds of the formula IV-1 is 5 to 40%, very particularly preferably 15 to 35%.

[0281] Particularly preferred LC media comprise the compounds V-10a and/or IV-1-1:

##STR00114##

[0282] The compounds V-10a and IV-1-1 are preferably present in the mixture in a concentration of 5 to 30%, very preferably 10 to 25%, based on the mixture as a whole.

[0283] Further preferred LC media comprise at least one compound selected from the group of the following compounds:

##STR00115## [0284] in which R^{sup.1}, R^{sup.2}, R^{sup.41} and R^{sup.42} have the meanings indicated above. Preferably in the compounds V-6, V-7 and IV, R^{sup.1} and R^{sup.41} denotes alkyl or alkenyl having 1 to 6 or 2 to 6 C atoms, respectively, and R^{sup.2} and R^{sup.42} denotes alkenyl having 2 to 6 C atoms. Preferably in the compounds V-14, R^{sup.1} denotes alkyl or alkenyl having 1 to 6 or 2 to 6 C atoms and R^{sup.2} denotes alkyl having 1 to 6 C atoms.

[0285] In another preferred embodiment the LC medium according to the invention comprises one or more compounds of the formula V-7, preferably selected from the compounds of the formulae V-7a to V-7e:

##STR00116## [0286] in which alkyl denotes an alkyl group having 1 to 7 C atoms, alkenyl denotes an alkenyl group having 2 to 7 C atoms, and cycloalkyl denotes a cyclic alkyl group having 3 to 12 C atoms, preferably cyclopropyl, cyclobutyl, cyclopentyl, cyclopropylalkyl, cyclobutylalkyl or cyclopentylalkyl.

[0287] Very preferred compounds of the formulae V-7a to V-7e are selected from the compounds of the following subformulae:

##STR00117## ##STR00118## ##STR00119## [0288] in which alkyl denotes ethyl, n-propyl, n-butyl or n-pentyl, preferably n-propyl.

[0289] Further preferred are compounds of formula V, wherein R^{sup.51} and R^{sup.52} independently of one another denote straight-chain alkyl having 1 to 7 C atoms or alkenyl having 2

to 7 C atoms.

[0290] In a preferred embodiment of the present invention the LC medium additionally comprises one or more compounds of the formulae VI-1 to VI-24,

##STR00120## ##STR00121## ##STR00122## [0291] in which [0292] R denotes a straight-chain alkyl or alkoxy radical having 1 to 6 C atoms, (O) denotes-O— or a single bond, X denotes F, Cl, OCF.sub.3 or OCHF.sub.2, Lx denotes H or F, m is 0, 1, 2, 3, 4, 5 or 6 and n is 0, 1, 2, 3 or 4.

[0293] R preferably denotes methyl, ethyl, propyl, butyl, pentyl, hexyl, methoxy, ethoxy, propoxy, butoxy, pentoxy. [0294] X preferably denotes F or OCH.sub.3, very preferably F.

[0295] The LC medium according to the invention preferably comprises the terphenyls of the formulae VI-1 to VI-25 in amounts of 2 to 30% by weight, in particular 5 to 20% by weight.

[0296] Particular preference is given to compounds of the formulae VI-1, VI-3, VI-19, VI-20, and VI-21 wherein X denotes F. In these compounds, R preferably denotes alkyl, furthermore alkoxy, each having 1 to 5 C atoms. In the compounds of the formula VI-19, R preferably denotes alkyl or alkenyl, in particular alkyl. In the compounds of the formula VI-20, R preferably denotes alkyl. In the compounds of the formulae VI-21 to VI-24, X preferably denotes F.

[0297] The terphenyls of formula VI-1 to VI-24 are preferably employed in the LC media according to the invention if the Δn value of the mixture is to be >0.1 . Preferred LC media comprise 2 to 20% by weight of one or more terphenyl compounds selected from the group of the compounds of formulae VI-1 to VI-24.

[0298] In another preferred embodiment of the present invention the LC medium additionally comprises one or more compounds of the formulae VII-1 to VII-9

##STR00123## [0299] in which [0300] R.sup.1 each, independently of one another, have one of the meanings indicated for R.sup.21 in formula IIA, and [0301] w and x each, independently of one another, denote 1 to 6.

[0302] Particular preference is given to LC media comprising at least one compound of the formula VII-9.

[0303] In another preferred embodiment of the present invention the LC medium additionally comprises one or more compounds which contain a tetrahydronaphthyl or naphthyl unit, such as, for example, the compounds of the formulae N-1 to N-5,

##STR00124## [0304] in which R.sup.61 and R.sup.62 each, independently of one another, have the meanings indicated for R.sup.21, preferably denote straight-chain alkyl, straight-chain alkoxy or straight-chain alkenyl, and [0305] Z.sup.61 and Z.sup.62 each, independently of one another, denote —C.sub.2H.sub.4—, —CH=CH—, —(CH.sub.2).sub.4—, —(CH.sub.2).sub.3O—, —O(CH.sub.2).sub.3—, —CH=CHCH.sub.2CH.sub.2—, —CH.sub.2CH.sub.2CH=CH—, —CH.sub.2O—, —OCH.sub.2—, —COO—, —OCO—, —C.sub.2F.sub.4—, —CF=CF—, —CF=CH—, —CH=CF—, —CF.sub.2O—, —OCF.sub.2—, —CH.sub.2— or a single bond.

[0306] In another preferred embodiment of the present invention the LC medium additionally comprises one or more compounds selected from the group of the difluorodibenzochroman compounds of the formula BC, chromans of the formula CR, and fluorinated phenanthrenes of the formulae PH-1 and PH-2,

##STR00125## [0307] in which [0308] R.sup.71 and R.sup.72 each, independently of one another, have the meaning of R.sup.21 and c is 0, 1 or 2. R.sup.71 and R.sup.72 preferably, independently of one another, denote alkyl or alkoxy having 1 to 6 C atoms.

[0309] The LC medium according to the invention preferably comprises the compounds of the formulae BC, CR, PH-1, PH-2 in amounts of 3 to 20% by weight, in particular in amounts of 3 to 15% by weight.

[0310] Particularly preferred compounds of the formulae BC and CR are the compounds BC-1 to BC-7 and CR-1 to CR-5,

##STR00126## [0311] in which [0312] alkyl and alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1 to 6 C atoms, and [0313] alkenyl and alkenyl* each,

independently of one another, denote a straight-chain alkenyl radical having 2 to 6 C atoms.

[0314] Very particular preference is given to an LC medium comprising one, two or three compounds of the formula BC-2, BF-1 and/or BF-2.

[0315] In another preferred embodiment of the present invention the LC medium additionally comprises one or more indane compounds of the formula In,

##STR00127## [0316] in which [0317] R.sup.81, R.sup.82, R.sup.83 each, independently of one another, denote a straight-chain alkyl, alkoxy, alkoxyalkyl or alkenyl radical having 1 to 6 C atoms, [0318] R.sup.82 and R.sup.83 may also denote halogen, preferably F,

##STR00128##

denotes 0, 1 or 2.

[0319] Preferred compounds of the formula In are the compounds of the formulae In-1 to In-16 indicated below:

##STR00129## ##STR00130##

[0320] Particular preference is given to the compounds of the formulae In-1, In-2, In-3 and In-4.

[0321] The compounds of the formula In and the sub-formulae In-1 to In-16 are preferably employed in the LC media according to the invention in concentrations $\geq 5\%$ by weight, in particular 5 to 30% by weight and very particularly preferably 5 to 25% by weight.

[0322] In another preferred embodiment of the present invention the LC medium additionally comprises one or more compounds of the formulae L-1 to L-8,

##STR00131## [0323] in which [0324] R.sup.L1 and R.sup.L2 each, independently of one another, have the meanings indicated for R.sup.21 in formula IIA above, alkyl denotes an alkyl radical having 1 to 6 C atoms, and s denotes 1 or 2.

[0325] The compounds of the formulae L-1 to L8 are preferably employed in concentrations of 5 to 15% by weight, in particular 5 to 12% by weight and very particularly preferably 8 to 10% by weight.

[0326] In another preferred embodiment of the present invention the LC medium additionally comprises one or more compounds of formula IIA-Y,

##STR00132## [0327] in which R.sup.11 and R.sup.12 have one of the meanings given for R.sup.21 in formula IIA above, and L.sup.1 and L.sup.2, identically or differently, denote F or Cl.

[0328] Preferred compounds of the formula IIA-Y are selected from the group consisting of the following subformulae:

##STR00133## [0329] in which Alkyl and Alkyl* each, independently of one another, denote a straight-chain alkyl radical having 1-6 C atoms, Alkoxy denotes a straight-chain alkoxy radical having 1-6 C atoms, Alkenyl and Alkenyl* each, independently of one another, denote a straight-chain alkenyl radical having 2-6 C atoms, and O denotes an oxygen atom or a single bond. Alkenyl and Alkenyl* preferably denote CH.sub.2=CH—, CH.sub.2=CHCH.sub.2CH.sub.2—, CH.sub.3—CH=CH—, CH.sub.3—CH.sub.2—CH=CH—, CH.sub.3—(CH.sub.2).sub.2—CH=CH—, CH.sub.3—(CH.sub.2).sub.3—CH=CH— or CH.sub.3—CH=CH—(CH.sub.2).sub.2—.

[0330] Particularly preferred compounds of the formula IIA-Y are selected from the group consisting of following subformulae:

##STR00134## [0331] in which Alkoxy and Alkoxy* have the meanings defined above and preferably denote methoxy, ethoxy, n-propyloxy, n-butyloxy or n-pentyloxy.

[0332] In another preferred embodiment of the present invention the LC medium additionally comprises one or more quaterphenyl compounds selected from the following formula:

##STR00135## [0333] in which [0334] R.sup.Q is alkyl, alkoxy, oxaalkyl or alkoxyalkyl having 1 to 9 C atoms or alkenyl or alkenyloxy having 2 to 9 C atoms, all of which are optionally fluorinated, [0335] X.sup.Q is F, Cl, halogenated alkyl or alkoxy having 1 to 6 C atoms or halogenated alkenyl or alkenyloxy having 2 to 6 C atoms, [0336] L.sup.Q1 to L.sup.Q6 independently of each other are H or F, with at least one of L.sup.Q1 to L.sup.Q6 being F.

[0337] Preferred compounds of formula Q are those wherein R.sup.O denotes straight-chain alkyl

with 2 to 6 C-atoms, very preferably ethyl, n-propyl or n-butyl.

[0338] Preferred compounds of formula Q are those wherein L.sup.Q3 and L.sup.Q4 are F. Further preferred compounds of formula Q are those wherein L.sup.Q3, L.sup.Q4 and one or two of L.sup.Q1 and L.sup.Q2 are F.

[0339] Preferred compounds of formula Q are those wherein X.sup.Q denotes F or OCF.sub.3, very preferably F.

[0340] The compounds of formula Q are preferably selected from the following subformulae:

##STR00136## [0341] in which R.sup.Q has one of the meanings of formula Q or one of its preferred meanings given above and below, and is preferably ethyl, n-propyl or n-butyl.

[0342] Especially preferred are compounds of formula Q1, in particular those wherein R.sup.Q is n-propyl.

[0343] Preferably the proportion of compounds of formula Q in the LC medium is from >0 to ≤5% by weight, very preferably from 0.05 to 2% by weight, more preferably from 0.1 to 1% by weight, most preferably from 0.1 to 0.8% by weight.

[0344] Preferably the LC medium contains 1 to 5, preferably 1 or 2 compounds of formula Q.

[0345] The addition of quaterphenyl compounds of formula Q to the LC mixture of a polymerizable LC medium enables to reduce ODF mura, whilst maintaining high UV absorption, enabling quick and complete polymerization, enabling strong and quick tilt angle generation, and increasing the UV stability of the LC medium.

[0346] Besides, the addition of compounds of formula Q, which have positive dielectric anisotropy, to the LC medium with negative dielectric anisotropy allows a better control of the values of the dielectric constants $\epsilon_{\parallel}$ and $\epsilon_{\perp}$, and in particular enables to achieve a high value of the dielectric constant $\epsilon_{\parallel}$ while keeping the dielectric anisotropy $\Delta\epsilon$ constant, thereby reducing the kick-back voltage and reducing image sticking.

[0347] The LC medium according to the invention preferably comprises: [0348] one or more compounds of formula L1 or its subformulae, preferably selected from subformula L1-1 to L1-3, very preferably from subformula L1-2, preferably in a proportion from 0.1 to 10%, very preferably from 0.2 to 5%, most preferably from 0.3 to 3% by weight,

[0349] and/or [0350] one or more compounds of formula IV, preferably in a total concentration in the range of 10% to 60%, more preferably 15% to 55%, particularly preferably 20% to 45%,

[0351] and/or [0352] one or more compounds of formula III, preferably of formula III-2, very preferably of formula III-2-1, preferably in a total concentration in the range of 2% to 25%, very preferably 4% to 15%,

[0353] and/or [0354] one or more compounds of formula III-3-2, preferably of formula III-3-2-3 or III-3-2-4, preferably in a total concentration in the range of 2% to 25%, very preferably 4% to 15%,

[0355] and/or [0356] one or more compounds of formula III-10, preferably selected from the group consisting of the subformulae III-10-1 to III-10-11, preferably in a total concentration in the range of 2% to 25%, very preferably 4% to 15%,

[0357] and/or [0358] one or more compounds of formula IID-1 and/or IID-4, preferably in a total concentration in the range of 0.5% to 15%, very preferably 0.5% to 8%,

[0359] and/or [0360] one or more compounds selected from formulae IID-1, IID-10, IID-15, IID-17, IID-18, IID-19, IID-20, IID-21, IID-22 and IID-23, very preferably from formulae IID-1, IID-21, IID-22 and IID-23, preferably in a total concentration in the range of 0.5% to 15%, very preferably 0.5% to 8%.

[0361] In particular, the LC medium comprises [0362] one or more compounds CY-n-Om, in particular CY-3-O4, CY-5-O4 and/or CY-3-O2, preferably in a total concentration in the range of from 5% to 30%, preferably 10% to 20%, based on the mixture as a whole, [0363] one or more compounds PY-n-Om, in particular PY-1-O2, PY-2-O2 and/or PY-3-O2, preferably in a total concentration in the range of from 5% to 40%, preferably 10% to 30%, based on the mixture as a whole, and/or [0364] one or more compounds CPY-n-Om, in particular CPY-2-O2, CPY-3-O2

and/or CPY-5-O2, preferably in concentrations >5%, in particular 7% to 20%, based on the mixture as a whole,

[0365] and/or [0366] one or more compounds CCY-n-Om, preferably CCY-4-O2, CCY-3-O2, CCY-3-O3, CCY-3-O1 and/or CCY-5-O2, preferably in concentrations >3%, in particular 5 to 15%, based on the mixture as a whole,

[0367] and/or [0368] one or more compounds CPY-n-Om and CY-n-Om, preferably in concentrations of 10 to 80%, based on the mixture as a whole,

[0369] and/or [0370] one or more compounds CPY-n-Om and PY-n-Om, preferably CPY-2-O2 and/or CPY-3-O2 and PY-3-O2 or PY-1-O2, preferably in concentrations of 5 to 20%, more preferably 10 to 15%, based on the mixture as a whole,

[0371] and/or [0372] one or more compound(s) selected from the group consisting of CCH-13, CCH-23, CCH-34, CCH-35, CCH-301 and CCH-303, preferably in a total concentration of 3 to 40%, preferably 3 to 25% based on the mixture as a whole,

[0373] and/or [0374] one or more compounds selected from the group consisting of CC-2-V1, CC-3-V1, CC-3-V2, CC-4-V1, CC-3-2V1, CC-3-V, CC-4-V, CC-5-V, CC-V-V, CC-V-V1, CC-1V-V1 and CC-2V-V2, preferably in a total concentration of 3 to 60%, more preferably 3 to 40%, very preferably from 5% to 30% based on the mixture as a whole,

[0375] and/or [0376] one or more compounds CC-3-V, preferably in a total concentration of $\geq 10\%$, more preferably $\geq 15\%$, very preferably from 20% to 60%, most preferably from 20% to 40%, based on the mixture as a whole,

[0377] and/or [0378] one or more compound(s) CCP-n-m and/or CCP-Vn-m and/or CPP-n-m, preferably selected from the group consisting of CCP-3-1, CCP-V-1, CCP-V2-1 and CPP-3-2, preferably in a total concentration of 4 to 35%, preferably 5 to 25% based on the mixture as a whole,

[0379] and/or [0380] one or more compound(s) CLP-n-m and/or CLP-Vn-m, preferably selected from the group consisting of CLP-3-1, CLP-3-2 and CLP-V-1, preferably in a total concentration of 1 to 25%, preferably 2 to 15% based on the mixture as a whole,

[0381] and/or [0382] one or more compounds PGP-n-m, preferably selected from the group consisting of PGP-2-3, PGP-2-4, PGP-2-5, PGP-3-5, preferably in a total concentration of 2 to 20%, more preferably 2% to 15%, most preferably 2 to 10%, based on the mixture as a whole,

[0383] and/or [0384] one or more compounds selected from the group consisting of PYP-n-m, PGIY-n-Om and PGP-n-2V, preferably selected from the group consisting of PGP-1-2V, PGP-2-2V and PGP-3-2V, preferably in a total concentration of 2 to 20%, more preferably 2% to 15%, most preferably 2 to 10%, based on the mixture as a whole,

[0385] and/or [0386] one or more compound(s) PP-n-m and/or PP-n-nVm, preferably selected from the group consisting of PP-1-3, PP-1-4, PP-1-5, PP-1-2V and PP-1-2V1, preferably in a total concentration of 1 to 15%, preferably 2 to 10% based on the mixture as a whole,

[0387] and/or [0388] the compound CCG-V-F, preferably in a concentration of 0.5% to 10% based on the mixture as a whole,

[0389] and/or [0390] the compound PPGU-3-F, preferably in a concentration of 0.1% to 3% based on the mixture as a whole.

[0391] It is advantageous for the liquid-crystalline medium according to the invention to preferably have a nematic phase from $\leq -20^\circ \text{C.}$ to $\geq 70^\circ \text{C.}$, particularly preferably from $\leq -30^\circ \text{C.}$ to $> 80^\circ \text{C.}$, very particularly preferably from $\leq -40^\circ \text{C.}$ to $\geq 90^\circ \text{C.}$

[0392] The LC medium according to the invention preferably has a clearing temperature of 70°C. or more, preferably of 74°C. or more.

[0393] The expression “have a nematic phase” here means on the one hand that no smectic phase and no crystallisation are observed at low temperatures at the corresponding temperature and on the other hand that clearing still does not occur on heating from the nematic phase. The investigation at low temperatures is carried out in a flow viscometer at the corresponding temperature and checked

by storage in test cells having a layer thickness corresponding to the electro-optical use for at least 100 hours. If the storage stability at a temperature of -20°C . in a corresponding test cell is 1000 h or more, the LC medium is referred to as stable at this temperature. At temperatures of -30°C . and -40°C ., the corresponding times are 500 h and 250 h respectively. At high temperatures, the clearing point is measured by conventional methods in capillaries.

[0394] The liquid-crystal mixture preferably has a nematic phase range of at least 60 K and a flow viscosity $\eta_{\text{sub.20}}$ of at most 30 mPa.s at 20°C .

[0395] The mixture is nematic at a temperature of -20°C . or less, preferably at -30°C . or less, very preferably at -40°C . or less.

[0396] The values of the birefringence Δn in the liquid-crystal mixture are generally between 0.08 and 0.17, preferably between 0.09 and 0.165, very preferably between 0.10 and 0.16. In a preferred embodiment of the present invention, the LC medium has a birefringence in the range of from 0.11 to 0.155.

[0397] The liquid-crystal mixture according to the invention has a dielectric anisotropy $\Delta\epsilon$ of -1.5 to -8.0 , preferably of -2.0 to -5.0 , in particular -2.5 to -4.8 ,

[0398] The rotational viscosity $\gamma_{\text{sub.1}}$ at 20°C . is preferably ≤ 200 mPa.s, more preferably ≤ 180 mPa.s, very preferably ≤ 125 mPa.s, most preferably ≤ 115 mPa.s. In another preferred embodiment, the rotational viscosity $\gamma_{\text{sub.1}}$ at 20°C . is ≤ 110 mPa.s, in particular ≤ 105 mPa.s.

[0399] The liquid-crystal media according to the invention have relatively low values for the threshold voltage ($V_{\text{sub.0}}$). They are preferably in the range from 1.7 V to 3.0 V, particularly preferably ≤ 2.7 V and very particularly preferably ≤ 2.5 V.

[0400] For the present invention, the term “threshold voltage” relates to the capacitive threshold ($V_{\text{sub.0}}$), also called the Fredericks threshold, unless explicitly indicated otherwise.

[0401] In addition, the liquid-crystal media according to the invention have high values for the voltage holding ratio in liquid-crystal cells.

[0402] In general, liquid-crystal media having a low addressing voltage or threshold voltage exhibit a lower voltage holding ratio than those having a higher addressing voltage or threshold voltage and vice versa.

[0403] For the present invention, the term “dielectrically positive compounds” denotes compounds having a $\Delta\epsilon > 1.5$, the term “dielectrically neutral compounds” denotes those having $-1.5 \leq \Delta\epsilon \leq 1.5$ and the term “dielectrically negative compounds” denotes those having $\Delta\epsilon < -1.5$. The dielectric anisotropy of the compounds is determined here by dissolving 10% of the compounds in a liquid-crystalline host and determining the capacitance of the resultant mixture in at least one test cell in each case having a layer thickness of 20 μm with homeotropic and with homogeneous surface alignment at 1 KHz. The measurement voltage is typically 0.5 V to 1.0 V, but is always lower than the capacitive threshold of the respective liquid-crystal mixture investigated.

[0404] All temperature values indicated for the present invention are in $^{\circ}\text{C}$.

[0405] The LC media according to the invention are suitable for all VA-TFT (vertical alignment-thin film transistor) applications, such as, for example, VAN (vertically aligned nematic), MVA (multidomain VA), (S)-PVA (super patterned VA), ASV (advanced super view, or axially symmetric VA), PSA (polymer sustained VA) and PS-VA (polymer stabilised VA). They are furthermore suitable for IPS (in-plane switching) and FFS (fringe field switching) applications having negative $\Delta\epsilon$.

[0406] The nematic LC media in the displays according to the invention generally comprise two components A and B, which themselves consist of one or more individual compounds.

[0407] Component A has significantly negative dielectric anisotropy and gives the nematic phase a dielectric anisotropy of ≤ -0.5 . It preferably comprises one or more of the compounds of the formulae IIA, IIB, IIC and/or IID, furthermore one or more compounds of the formula III.

[0408] The proportion of component A is preferably between 45 and 100%, in particular between

60 and 85%.

[0409] For component A, one (or more) individual compound(s) which has (have) a value of $\Delta\epsilon \leq -1.5$ is (are) preferably selected. This value must be more negative, the smaller the proportion A in the mixture as a whole.

[0410] Component B has pronounced nematogeneity and a flow viscosity of not greater than $30 \text{ mm} \cdot \text{sup.2} \cdot \text{Math.s.sup.} -1$, preferably not greater than $25 \text{ mm} \cdot \text{sup.2} \cdot \text{Math.s.sup.} -1$, at 20° C .

[0411] A multiplicity of suitable materials is known to the person skilled in the art from the literature. Particular preference is given to compounds of the formula IV.

[0412] Particularly preferred individual compounds in component B are extremely low-viscosity nematic liquid crystals having a flow viscosity of not greater than $18 \text{ mm} \cdot \text{sup.2} \cdot \text{Math.s.sup.} -1$, preferably not greater than $12 \text{ mm} \cdot \text{sup.2} \cdot \text{Math.s.sup.} -1$, at 20° C .

[0413] Component B is monotropically or enantiotropically nematic, has no smectic phases and is able to prevent the occurrence of smectic phases down to very low temperatures in LC media. For example, if various materials of high nematogeneity are added to a smectic liquid-crystal mixture, the nematogeneity of these materials can be compared through the degree of suppression of smectic phases that is achieved.

[0414] The mixture may optionally also comprise a component C, comprising compounds having a dielectric anisotropy of $\Delta\epsilon \geq 1.5$. These so-called positive compounds are generally present in a mixture of negative dielectric anisotropy in amounts of $\leq 20\%$ by weight, based on the mixture as a whole.

[0415] Besides the polymerizable compounds and the compounds of formula L1, the LC medium preferably comprises 4 to 15, in particular 5 to 12, and particularly preferably < 10 , compounds selected from the formulae IIA, IIB, IIC, IID, IIE, IIF, III and V and one or more compounds of the formula IV.

[0416] Besides the polymerizable compounds and the compounds of formula L1 and the compounds selected from the formulae IIA, IIB, IIC, IID, IIE, IIF, III, IV and V, other constituents may also be present, for example in an amount of up to 45% of the mixture as a whole, but preferably up to 35%, in particular up to 10%.

[0417] The other constituents are preferably selected from nematic or nematogenic substances, in particular known substances, from the classes of the azoxybenzenes, benzylideneanilines, biphenyls, terphenyls, phenyl or cyclohexyl benzoates, phenyl or cyclohexyl cyclohexanecarboxylates, phenylcyclohexanes, cyclohexylbiphenyls, cyclohexylcyclohexanes, cyclohexylnaphthalenes, 1,4-biscyclohexylbiphenyls or cyclohexylpyrimidines, phenyl- or cyclohexyldioxanes, optionally halogenated stilbenes, benzyl phenyl ethers, tolans and substituted cinnamic acid esters.

[0418] The most important compounds which are suitable as constituents of liquid-crystal phases of this type can be characterised by the formula R

R^{sup.}R1-L-G-E-R^{sup.}R2 R [0419] in which L and E each denote a carbo- or heterocyclic ring system from the group formed by 1,4-disubstituted benzene and cyclohexane rings, 4,4'-disubstituted biphenyl, phenylcyclohexane and cyclohexylcyclohexane systems, 2,5-disubstituted pyrimidine and 1,3-dioxane rings, 2,6-disubstituted naphthalene, di- and tetrahydronaphthalene, quinazoline and tetrahydroquinazoline, [0420] G denotes —CH=CH— —N(O)=N— [0421] —CH=CQ— —CH=N(O)— [0422] $\text{—C}\equiv\text{C—}$ $\text{—CH.sub.2—CH.sub.2—}$ [0423] —CO—O— —CH.sub.2—O— [0424] —CO—S— —CH.sub.2—S— [0425] —CH=N— —COO-Phe-COO— [0426] —CF.sub.2O— —CF=CF— [0427] —OCF.sub.2— —OCH.sub.2— [0428] $\text{—(CH.sub.2).sub.4—}$ $\text{—(CH.sub.2).sub.3O—}$ [0429] or a C—C single bond, Q denotes halogen, preferably chlorine, or —CN , and R^{sup.}R1 and R^{sup.}R2 each denote alkyl, alkenyl, alkoxy, alkoxyalkyl or alkoxycarbonyloxy having up to 18, preferably up to 8, carbon atoms, or one of these radicals alternatively denotes CN, NC, NO₂, NCS, CF₃, SF₅, OCF₃, F, Cl

or Br.

[0430] In most of these compounds, R.sup.R1 and R.sup.R2 are different from one another, one of these radicals usually being an alkyl or alkoxy group. Other variants of the proposed substituents are also common. Many such substances or also mixtures thereof are commercially available. All these substances can be prepared by methods known from the literature.

[0431] It goes without saying for the person skilled in the art that the VA, IPS or FFS mixture according to the invention may also comprise compounds in which, for example, H, N, O, Cl and F have been replaced by the corresponding isotopes.

[0432] The LC medium has preferably a nematic LC phase.

[0433] In a preferred embodiment the LC medium further comprises one or more, very preferably one, two or three, polymerizable compounds, preferably selected from polymerizable mesogenic compounds, also known as "reactive mesogens" or RMs, very preferably from formula M as described above.

[0434] In the polymerizable compounds, the polymerizable group P is a group which is suitable for a polymerization reaction, such as, for example, free-radical or ionic chain polymerization, polyaddition or polycondensation, or for a polymer-analogous reaction, for example addition or condensation onto a main polymer chain. Particular preference is given to groups for chain polymerization, in particular those containing a C=C double bond or —C≡C— triple bond, and groups which are suitable for polymerization with ring opening, such as, for example, oxetane or epoxide groups.

[0435] Preferred groups P are selected from the group consisting of CH.sub.2=CW.sup.1—CO—O—, CH.sub.2=CW.sup.1—CO—,

##STR00137##

CH.sub.2=CW.sup.2—(O).sub.k3—, CW.sup.1=CH—CO—(O).sub.k3—, CW.sup.1=CH—CO—NH—, CH.sub.2=CW.sup.1—CO—NH—, CH.sub.3—CH=CH—O—, (CH.sub.2=CH).sub.2CH—OCO—, (CH.sub.2=CH—CH.sub.2).sub.2CH—OCO—, (CH.sub.2=CH).sub.2CH—O—, (CH.sub.2=CH—CH.sub.2).sub.2N—, (CH.sub.2=CH—CH.sub.2).sub.2N—CO—, HO—CW.sup.2W.sup.3—, HS—CW.sup.2W.sup.3—, HW.sup.2N—, HO—CW.sup.2W.sup.3—NH—, CH.sub.2=CW.sup.1—CO—NH—, CH.sub.2=CH—(COO).sub.k1-Phe-(O).sub.k2—, CH.sub.2=CH—(CO).sub.k1-Phe-(O).sub.k2—, Phe-CH=CH—, HOOC—, OCN— and W.sup.4W.sup.5W.sup.6Si—, in which W.sup.1 denotes H, F, Cl, CN, CF.sub.3, phenyl or alkyl having 1 to 5 C atoms, in particular H, F, Cl or CH.sub.3, W.sup.2 and W.sup.3 each, independently of one another, denote H or alkyl having 1 to 5 C atoms, in particular H, methyl, ethyl or n-propyl, W.sup.4, W.sup.5 and W.sup.6 each, independently of one another, denote Cl, oxaalkyl or oxacarbonylalkyl having 1 to 5 C atoms, W.sup.7 and W.sup.8 each, independently of one another, denote H, Cl or alkyl having 1 to 5 C atoms, Phe denotes 1,4-phenylene, which is optionally substituted by one or more radicals L as defined above which are other than P-Sp-, k.sub.1, k.sub.2 and k.sub.3 each, independently of one another, denote 0 or 1, k.sub.3 preferably denotes 1, and k.sub.4 denotes an integer from 1 to 10.

[0436] Very preferred groups P are selected from the group consisting of CH.sub.2=CW.sup.1—CO—O—, CH.sub.2=CW.sup.1—CO—,

##STR00138##

CH.sub.2=CW.sup.2—O—, CH.sub.2=CW.sup.2—, CW.sup.1=CH—CO—(O).sub.k3—, CW.sup.1=CH—CO—NH—, CH.sub.2=CW.sup.1—CO—NH—, (CH.sub.2=CH).sub.2CH—OCO—, (CH.sub.2=CH—CH.sub.2).sub.2CH—OCO—, (CH.sub.2=CH).sub.2CH—O—, (CH.sub.2=CH—CH.sub.2).sub.2N—, (CH.sub.2=CH—CH.sub.2).sub.2N—CO—, CH.sub.2=CW.sup.1—CO—NH—, CH.sub.2=CH—(COO).sub.k1-Phe-(O).sub.k2—, CH.sub.2=CH—(CO).sub.k1-Phe-(O).sub.k2—, Phe-CH=CH— and W.sup.4W.sup.5W.sup.6Si—, in which W.sup.1 denotes H, F, Cl, CN, CF.sub.3, phenyl or alkyl having 1 to 5 C atoms, in particular H, F, Cl or CH.sub.3, W.sup.2 and W.sup.3 each, independently of one another, denote H

or alkyl having 1 to 5 C atoms, in particular H, methyl, ethyl or n-propyl, W^{sup.4}, W^{sup.5} and W^{sup.6} each, independently of one another, denote Cl, oxaalkyl or oxacarbonylalkyl having 1 to 5 C atoms, W^{sup.7} and W^{sup.8} each, independently of one another, denote H, Cl or alkyl having 1 to 5 C atoms, Phe denotes 1,4-phenylene, k_{sub.1}, k_{sub.2} and k_{sub.5} each, independently of one another, denote 0 or 1, k_{sub.3} preferably denotes 1, and k_{sub.4} denotes an integer from 1 to 10. [0437] Very particularly preferred groups P are selected from the group consisting of CH_{sub.2}=CW^{sup.1}—CO—O—, in particular CH_{sub.2}=CH—CO—O—, CH_{sub.2}=C(CH_{sub.3})—CO—O— and CH_{sub.2}=CF—CO—O—, furthermore CH_{sub.2}=CH—O—, (CH_{sub.2}=CH).sub.2CH—O—CO—, (CH_{sub.2}=CH).sub.2CH—O—, ##STR00139##

[0438] Further preferred polymerizable groups P are selected from the group consisting of vinyloxy, acrylate, methacrylate, fluoroacrylate, chloroacrylate, oxetane and epoxide, most preferably from acrylate and methacrylate.

[0439] Very preferably all polymerizable groups in the polymerizable compound have the same meaning.

[0440] If the spacer group Sp is different from a single bond, it is preferably of the formula Sp''-X'', so that the respective radical P-Sp-conforms to the formula P-Sp''-X'', wherein [0441] Sp'' denotes linear or branched alkylene having 1 to 20, preferably 1 to 12, C atoms, which is optionally mono- or polysubstituted by F, Cl, Br, I or CN and in which, in addition, one or more non-adjacent CH_{sub.2} groups may each be replaced, independently of one another, by —O—, —S—, —NH—, —N(R^{sup.0})—, —Si(R^{sup.0}R^{sup.00})—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, —S—CO—, —CO—S—, —N(R^{sup.00})—CO—O—, —O—CO—N(R^{sup.0})—, —N(R^{sup.0})—CO—N(R^{sup.00})—, —CH=CH— or —C≡C— in such a way that O and/or S atoms are not linked directly to one another, [0442] X'' denotes —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, —CO—N(R^{sup.0})—, —N(R^{sup.0})—CO—, —N(R^{sup.0})—CO—N(R^{sup.00})—, —OCH_{sub.2}—, —CH_{sub.2}O—, —SCH_{sub.2}—, —CH_{sub.2}S—, —CF_{sub.2}O—, —OCF_{sub.2}—, —CF_{sub.2}S—, —SCF_{sub.2}—, —CF_{sub.2}CH_{sub.2}—, —CH_{sub.2}CF_{sub.2}—, —CF_{sub.2}CF_{sub.2}—, —CH=N—, —N=CH—, —N=N—, —CH=CR^{sup.0}—, —CY_{sub.2}=CY_{sub.3}—, —C≡C—, —CH=CH—CO—O—, —O—CO—CH=CH— or a single bond, [0443] R^{sup.0} and R^{sup.00} each, independently of one another, denote H or alkyl having 1 to 20 C atoms, and [0444] Y^{sup.2} and Y^{sup.3} each, independently of one another, denote H, F, Cl or CN. [0445] X'' is preferably —O—, —S—, —CO—, —COO—, —OCO—, —O—COO—, —CO—NR^{sup.0}—, —NR^{sup.0}—CO—, —NR^{sup.0}—CO—NR^{sup.00}— or a single bond. [0446] Typical spacer groups Sp and -Sp''-X''— are, for example, —(CH_{sub.2}).sub.p1—, —(CH_{sub.2}).sub.p1—O—, —(CH_{sub.2}).sub.p1—O—CO—, —(CH_{sub.2}).sub.p1—CO—O—, —(CH_{sub.2}).sub.p1—O—CO—O—, —(CH_{sub.2}CH_{sub.2}O).sub.q1—CH_{sub.2}CH_{sub.2}—, —CH_{sub.2}CH_{sub.2}—S—CH_{sub.2}CH_{sub.2}—, —CH_{sub.2}CH_{sub.2}—NH—CH_{sub.2}CH_{sub.2}— or —(SiR^{sup.0}R^{sup.00}—O).sub.p1—, in which p1 is an integer from 1 to 12, q1 is an integer from 1 to 3, and R^{sup.0} and R^{sup.00} have the meanings indicated above.

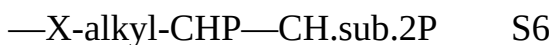
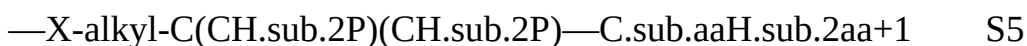
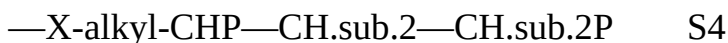
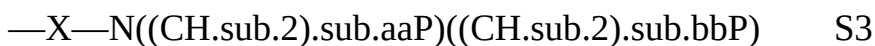
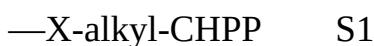
[0447] Particularly preferred groups Sp and -Sp''-X''— are —(CH_{sub.2}).sub.p1—, —(CH_{sub.2}).sub.p1—O—, —(CH_{sub.2}).sub.p1—O—CO—, —(CH_{sub.2}).sub.p1—CO—O—, —(CH_{sub.2}).sub.p1—O—CO—O—, in which p1 and q1 have the meanings indicated above.

[0448] Particularly preferred groups Sp'' are, in each case straight-chain, ethylene, propylene, butylene, pentylene, hexylene, heptylene, octylene, nonylene, decylene, undecylene, dodecylene, octadecylene, ethyleneoxyethylene, methyleneoxybutylene, ethylenethioethylene, ethylene-N-methyliminoethylene, 1-methylalkylene, ethenylene, propenylene and butenylene.

[0449] In a preferred embodiment of the invention the polymerizable compounds contain a spacer group Sp that is substituted by one or more polymerizable groups P, so that the group Sp-P corresponds to Sp (P) s, with s being ≥2 (branched polymerizable groups).

[0450] Preferred polymerizable compounds according to this preferred embodiment are those

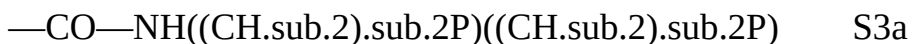
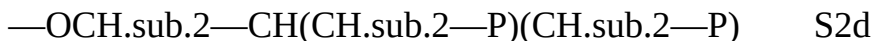
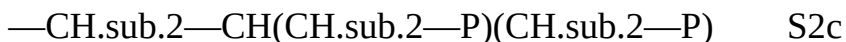
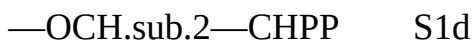
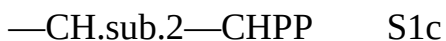
wherein s is 2, i.e. compounds which contain a group Sp (P) 2. Very preferred polymerizable compounds according to this preferred embodiment contain a group selected from the following formulae:



$\text{—X-alkyl-CHPCHP-C.sub.aaH.sub.2aa+1}$ S8 [0451] in which P is as defined in formula M, [0452] alkyl denotes a single bond or straight-chain or branched alkylene having 1 to 12 C atoms which is unsubstituted or mono- or polysubstituted by F, Cl or CN and in which one or more non-adjacent CH.sub.2 groups may each, independently of one another, be replaced by —C(R.sup.0)=C(R.sup.0)—, —C≡C—, —N(R.sup.0)—, —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, where R.sup.0 has the meaning indicated above, [0453] aa and bb each, independently of one another, denote 0, 1, 2, 3, 4, 5 or 6, [0454] X has one of the meanings indicated for X", and is preferably O, CO, SO.sub.2, O—CO—, CO—O or a single bond.

[0455] Preferred spacer groups Sp (P) 2 are selected from formulae S1, S2 and S3.

[0456] Very preferred spacer groups Sp (P) 2 are selected from the following subformulae:



[0457] P is preferably selected from the group consisting of vinyloxy, acrylate, methacrylate, fluoroacrylate, chloroacrylate, oxetane and epoxide, very preferably from acrylate and methacrylate, most preferably from methacrylate.

[0458] Further preferably all polymerizable groups P that are present in the same compound have the same meaning, and very preferably denote acrylate or methacrylate, most preferably methacrylate.

[0459] Sp preferably denotes a single bond or $-(CH_{sub.2})_{sub.p1}-$, $-(CH_{sub.2})_{sub.p2}-CH=CH-(CH_{sub.2})_{sub.p3}-$, $-O-(CH_{sub.2})_{sub.p1}-$, $-O-CO-(CH_{sub.2})_{sub.p1}$, or $-CO-O-(CH_{sub.2})_{sub.p1}$, wherein p1 is 2, 3, 4, 5 or 6, preferably 2 or 3, p2 and p3 are independently of each other 0, 1, 2 or 3 and, if Sp is $-O-(CH_{sub.2})_{sub.p1}-$, $-O-CO-(CH_{sub.2})_{sub.p1}$ or $-CO-O-(CH_{sub.2})_{sub.p1}$ the O-atom or CO-group, respectively, is linked to the benzene ring.

[0460] Further preferably at least one group Sp is a single bond.

[0461] Further preferably at least one group Sp is different from a single bond, and is preferably selected from $-(CH_{sub.2})_{sub.p1}-$, $-(CH_{sub.2})_{sub.p2}-CH=CH-(CH_{sub.2})_{sub.p3}-$, $-O-(CH_{sub.2})_{sub.p1}-$, $-O-CO-(CH_{sub.2})_{sub.p1}$, or $-CO-O-(CH_{sub.2})_{sub.p1}$, wherein p1 is 2, 3, 4, 5 or 6, preferably 2 or 3, p2 and p3 are independently of each other 0, 1, 2 or 3 and, if Sp is $-O-(CH_{sub.2})_{sub.p1}-$, $-O-CO-(CH_{sub.2})_{sub.p1}$ or $-CO-O-(CH_{sub.2})_{sub.p1}$ the O-atom or CO-group, respectively, is linked to the benzene ring.

[0462] Very preferably Sp is different from a single bond, and is selected from $-(CH_{sub.2})_{sub.2}-$, $-(CH_{sub.2})_{sub.3}-$, $-(CH_{sub.2})_{sub.4}-$, $-O-(CH_{sub.2})_{sub.2}-$, $-O-(CH_{sub.2})_{sub.3}-$, $-O-CO-(CH_{sub.2})_{sub.2}$ and $-CO-O-(CH_{sub.2})_{sub.2}-$, wherein the O atom or the CO group is attached to the benzene ring.

[0463] Particularly preferred compounds of the formula M are those in which B1 and B2 each, independently of one another, denote 1,4-phenylene, 1,3-phenylene, naphthalene-1,4-diyl, naphthalene-2,6-diyl, phenanthrene-2,7-diyl, 9,10-dihydro-phenanthrene-2,7-diyl, anthracene-2,7-diyl, fluorene-2,7-diyl, coumarine, flavone, where, in addition, one or more CH groups in these groups may be replaced by N, cyclohexane-1,4-diyl, in which, in addition, one or more non-adjacent CH_{sub.2} groups may be replaced by O and/or S, 1,4-cyclohexenylen, bicycle[1.1.1]pentane-1,3-diyl, bicyclo[2.2.2]octane-1,4-diyl, spiro[3.3]heptane-2,6-diyl, piperidine-1,4-diyl, decahydronaphthalene-2,6-diyl, 1,2,3,4-tetrahydronaphthalene-2,6-diyl, indane-2,5-diyl or octahydro-4,7-methanoindane-2,5-diyl, where all these groups may be unsubstituted or mono- or polysubstituted by L as defined above.

[0464] Particularly preferred compounds of the formula M are those in which B1 and B2 each, independently of one another, denote 1,4-phenylene, 1,3-phenylene, naphthalene-1,4-diyl or naphthalene-2,6-diyl.

[0465] Further preferred compounds of the formula M are those in which the group $-B_{sup.1}-$ ($Z_{sup.m}-B_{sup.2})_{sub.m}-$ in formula M is selected from the following formulae

##STR00140## [0466] wherein at least one benzene ring is substituted by at least one group L and the benzene rings are optionally further substituted by one or more groups L or P-Sp-

[0467] Preferred compounds of formula M and its sub-formulae are selected from the following preferred embodiments, including any combination thereof: [0468] All groups P in the compound have the same meaning, [0469] $B_{sup.1}-(Z_{sup.m}-B_{sup.2})_{sub.m}-$ is selected from formulae A1, A2 and A5, [0470] the compounds contain exactly two polymerizable groups (represented by the groups P), [0471] the compounds contain exactly three polymerizable groups (represented by the groups P), [0472] P is selected from the group consisting of acrylate, methacrylate and oxetane, very preferably acrylate or methacrylate, [0473] P is methacrylate, [0474] all groups Sp are a single bond, [0475] at least one of the groups Sp is a single bond and at least one of the groups Sp is different from a single bond, [0476] Sp, when being different from a single bond, is $-(CH_{sub.2})_{sub.p2}-$, $-(CH_{sub.2})_{sub.p2}-O-$, [0477] $-(CH_{sub.2})_{sub.p2}-CO-O-$, $-(CH_{sub.2})_{sub.p2}-O-CO-$, wherein p2 is 2, 3, 4, 5 or 6, and the O-atom or the CO-group, respectively, is connected to the benzene ring, [0478] Sp is a single bond or denotes $-(CH_{sub.2})_{sub.p2}-$, $-(CH_{sub.2})_{sub.p2}-O-$, $-(CH_{sub.2})_{sub.p2}-CO-O-$, $-$

(CH.sub.2).sub.p2—O—CO—, wherein p2 is 2, 3, 4, 5 or 6, and the O-atom or the CO-group, respectively, is connected to the benzene ring, [0479] R denotes P-Sp-, [0480] R does not denote or contain a polymerizable group, [0481] R does not denote or contain a polymerizable group and denotes straight chain, branched or cyclic alkyl having 1 to 25 C atoms, wherein one or more non-adjacent CH.sub.2-groups are optionally replaced by —O—, [0482] —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F, Cl or L.sup.a, [0483] L or L' denote F, Cl or CN, [0484] L is F.

[0485] Very preferred compounds of formula M are selected from the following formulae:

##STR00141## ##STR00142## ##STR00143## ##STR00144## ##STR00145## [0486] in which the individual radicals, on each occurrence identically or differently, and each, independently of one another, have the following meaning: [0487] P.sup.1, P.sup.2, P.sup.3 a polymerizable group, preferably selected from vinyloxy, acrylate, methacrylate, fluoroacrylate, chloroacrylate, oxetane and epoxy, very preferably acrylate or methacrylate, [0488] Sp.sup.1, Sp.sup.2, Sp.sup.3 a single bond or a spacer group where, in addition, one or more of the radicals P.sup.1-Sp.sup.1-, P.sup.2-Sp.sup.2- and P.sup.3-Sp.sup.3- may denote R.sup.M, with the proviso that at least one of the radicals P.sup.1-Sp.sup.1-, P.sup.2-Sp.sup.2- and P.sup.3-Sp.sup.3-present is different from R.sup.M, preferably having one of the preferred meanings of Sp as given above, very preferably —(CH.sub.2).sub.p1—, —(CH.sub.2).sub.p1—O—, —(CH.sub.2).sub.p1—CO—O— or —(CH.sub.2).sub.p1—O—CO—O—, wherein p1 is an integer from 1 to 12, [0489] R.sup.M H, F, Cl, CN or straight-chain or branched alkyl having 1 to 25 C atoms, in which, in addition, one or more non-adjacent CH.sub.2 groups may each be replaced, independently of one another, by —C(R.sup.0)=C(R.sup.00)—, —C≡C—, —N(R.sup.0)—, —O—, —S—, —CO—, [0490] —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, CN or P.sup.1-Sp.sup.1-, particularly preferably straight-chain or branched, optionally mono- or polyfluorinated alkyl, alkoxy, alkenyl, alkynyl, alkylcarbonyl, alkoxycarbonyl, alkylcarbonyloxy or alkoxycarbonyloxy having 1 to 12 C atoms (where the alkenyl and alkynyl radicals have at least two C atoms and the branched radicals have at least three C atoms), and wherein R.sup.M does not denote or contain a group P.sup.1, P.sup.2 or P.sup.3, [0491] R.sup.0, R.sup.00 H or alkyl having 1 to 12 C atoms, [0492] R.sup.y and R.sup.z H, F, CH.sub.3 or CF.sub.3, [0493] X.sup.1, X.sup.2, X.sup.3 —CO—O—, —O—CO— or a single bond, [0494] Z.sup.M1 —O—, —CO—, —C(R.sup.yR.sup.z)— or —CF.sub.2CF.sub.2—, [0495] Z.sup.M2, Z.sup.M3 —CO—O—, —O—CO—, —CH.sub.2O—, —OCH.sub.2—, —CF.sub.2O—, —OCF.sub.2— or —(CH.sub.2).sub.n—, where n is 2, 3 or 4, [0496] L F, Cl, CN or straight-chain or branched, optionally mono- or polyfluorinated alkyl, alkoxy, thioalkyl, alkenyl, alkynyl, alkylcarbonyl, alkoxycarbonyl, alkylcarbonyloxy or alkoxycarbonyloxy having 1 to 12 C atoms, [0497] L', L'' H, F or Cl, [0498] k 0 or 1, [0499] r 0, 1, 2, 3 or 4, [0500] s 0, 1, 2 or 3, [0501] t 0, 1 or 2, [0502] x 0 or 1.

[0503] Very preferred are compounds of formulae M2, M13 and M32, especially direactive compounds containing exactly two polymerizable groups P.sup.1 and P.sup.2.

[0504] Further preferred are compounds selected from formulae M17 to M31, in particular from formulae M20, M22, M26, M29 and M31, especially trireactive compounds containing exactly three polymerizable groups P.sup.1, P.sup.2 and P.sup.3.

[0505] In the compounds of formulae M1 to M32 the group

##STR00146## [0506] wherein L on each occurrence, identically or differently, has one of the meanings given above or below, and is preferably F, Cl, CN, NO.sub.2, CH.sub.3, C.sub.2H.sub.5, C(CH.sub.3).sub.3, CH(CH.sub.3).sub.2, CH.sub.2CH(CH.sub.3)C.sub.2H.sub.5, —CH=CH.sub.2, C(CH.sub.3)=CH.sub.2, SCH.sub.3, OCH.sub.3, OC.sub.2H.sub.5, COCH.sub.3, COC.sub.2H.sub.5, COOCH.sub.3, COOC.sub.2H.sub.5, CF.sub.3, OCF.sub.3, OCHF.sub.2, OC.sub.2F.sub.5 or P-Sp-, very preferably F, Cl, CN, CH.sub.3, C.sub.2H.sub.5, —CH=CH.sub.2,

C(CH.sub.3)=CH.sub.2, SCH.sub.3, COCH.sub.3, OCF.sub.3 or P-Sp-, more preferably F, Cl, CH.sub.3, —CH=CH.sub.2, C(CH.sub.3)=CH.sub.2, SCH.sub.3, OCH.sub.3, COCH.sub.3 or OCF.sub.3, most preferably F, SCH.sub.3 or OCH.sub.3.

[0507] Preferred compounds of formulae M1 to M32 are those wherein P.sup.1, P.sup.2 and P.sup.3 denote an acrylate, methacrylate, oxetane or epoxy group, very preferably an acrylate or methacrylate group, most preferably a methacrylate group.

[0508] Further preferred compounds of formulae M1 to M32 are those wherein Sp.sup.1, Sp.sup.2 and Sp.sup.3 are a single bond.

[0509] Further preferred compounds of formulae M1 to M32 are those wherein one of Sp.sup.1, Sp.sup.2 and Sp.sup.3 is a single bond and another one of Sp.sup.1, Sp.sup.2 and Sp.sup.3 is different from a single bond.

[0510] Further preferred compounds of formulae M1 to M32 are those wherein those groups Sp.sup.1, Sp.sup.2 and Sp.sup.3 that are different from a single bond denote —(CH.sub.2).sub.s1—X"—, wherein s1 is an integer from 1 to 6, preferably 2, 3, 4 or 5, and X" is X" is the linkage to the benzene ring and is —O—, —O—CO—, —CO—O—, —O—CO—O— or a single bond.

[0511] Further preferred polymerizable compounds are selected from Table E below, especially selected from the group consisting of formulae RM-1, RM-4, RM-8, RM-17, RM-19, RM-35, RM-37, RM-39, RM-40, RM-41, RM-48, RM-52, RM-54, RM-57, RM-58, RM-64, RM-74, RM-76, RM-88, RM-91, RM-102, RM-103, RM-109, RM-116, RM-117, RM-120, RM-121, RM-122, RM-139, RM-140, RM-142, RM-143, RM-145, RM-146, RM-147, RM-149, RM-156 to RM-163, RM-169, RM-170 and RM-171 to RM-183.

[0512] Particularly preferred are LC media comprising two, three or more polymerizable compounds of formula M.

[0513] Further preferred are LC media comprising two or more direactive polymerizable compounds of formula M, preferably selected from formulae M1 to M16 and M32, very preferably selected from formulae M2, M13 and M32.

[0514] Further preferred are LC media comprising one or more direactive polymerizable compounds of formula M, preferably selected from formulae M1 to M16 and M32, very preferably from formulae M2, M13 and M32, and one or more trireactive polymerizable compounds of formula M, preferably selected from formulae M17 to M32, very preferably from formulae M20, M22, M26, M29 and M31.

[0515] Further preferred are LC media comprising one or more polymerizable compounds of formula M wherein at least one r is not 0, or at least one of s and t is not 0, very preferably selected from formulae M2, M13, M22, M24, M27, M29, M31 and M32, and wherein L is selected from the preferred groups shown above, most preferably from F, OCH.sub.3 and SCH.sub.3.

[0516] Further preferred are LC media comprising one or more polymerizable compounds which show absorption in the wavelength range from 320 to 380 nm, preferably selected from formula M, very preferably from formulae M1 to M32.

[0517] Particular preference is given to LC media comprising one, two or more, very preferably two or three, polymerizable compounds selected from formula M or formulae M1 to M32.

[0518] The combination of compounds of the preferred embodiments mentioned above with the polymerized compounds described above and below causes low threshold voltages, low rotational viscosities and very good low-temperature stabilities in the LC media according to the invention at the same time as constantly high clearing points and high HR values, and allows the rapid establishment of a particularly low tilt angle (i.e. a large tilt) in PSA displays. In particular, the LC media exhibit significantly shortened response times, in particular also the grey-shade response times, in PSA displays compared with the LC media from the prior art.

[0519] For use in PSA displays the total proportion of the polymerizable compounds, like those of formula M or M1 to M32, in the LC medium is preferably from 0.01 to 2.0%, very preferably from 0.1 to 1.0%, most preferably from 0.2 to 0.8%.

[0520] For use in SA-VA displays the total proportion of the polymerizable compounds, like those of formula M or M1 to M32, in the LC medium is preferably from >0 to <3%, very preferably from >0 to <2%, more preferably from 0.05 to 2.0, most preferably from 0.05 to 1.0%.

[0521] In a preferred embodiment of the present invention the LC medium comprises one or more, preferably two or more, polymerizable compounds of formula M selected from formulae IA, IB and IC:

P-Sp-M.sup.1-Sp-P IA

P-Sp-M.sup.2-Sp-P IB

P-Sp-M.sup.3-Sp-P IC [0522] in which the individual radicals, on each occurrence identically or differently, and each, independently of one another, have the following meanings: [0523]

M.sup.1, M.sup.2, M.sup.3 a group each individually selected from the following formulae:

##STR00147## [0524] wherein the benzene rings are optionally substituted by one or more groups L or P-Sp-, [0525] L L.sup.a, L.sup.b, F, Cl, —CN, P-Sp-, or straight chain, branched or cyclic alkyl having 1 to 25 C atoms, wherein one or more non-adjacent CH.sub.2-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by P, F or Cl, [0526] L.sup.a —C(R.sup.aa)(R.sup.bb)OH, [0527] R.sup.aa, R.sup.bb straight-chain alkyl with 1 to 6 C atoms, [0528] L.sup.b straight-chain or branched alkenyl with 2 to 7 C atoms, preferably 3 or 4 C atoms, [0529] wherein in the compounds of formula IA the group M.sup.1 and/or at least one spacer group Sp is at least monosubstituted with L.sup.a, [0530] and wherein in the compounds of formula IC the group M.sup.3 is at least monosubstituted with L.sup.b.

[0531] In a preferred embodiment of the invention the compounds of formula IA and its subformulae contain a spacer group Sp that is substituted by one or more, preferably by one, group L.sup.a.

[0532] In the compounds of formula IA and its subformulae, R.sup.aa and R.sup.bb preferably denote straight chain alkyl with 1 to 6 C atoms or branched alkyl with 3 to 6 C atoms. More preferably R.sup.aa and R.sup.bb denote, independently of each other, methyl, ethyl, propyl and butyl, very preferably methyl or ethyl, most preferably methyl.

[0533] Further preferred are compounds of formula IA and its subformulae as described above and below, wherein R.sup.aa and R.sup.bb together with the C atom to which they are attached form a cyclic alkyl group with 3 to 12 C atoms, very preferably a cyclopropyl, cyclobutyl, cyclopentyl or cyclohexyl group.

[0534] Very preferably the compounds of formula IA contain a group L.sup.a selected from the following formulae:

##STR00148## [0535] wherein the asterisk denotes the linkage to the adjacent group in the compound of formula I.

[0536] In another preferred embodiment of the invention the compounds of formula IA and its subformulae contain a spacer group Sp that is a linear or branched alkylene with 1 to 12, preferably 1 to 7 C atoms which is substituted by one or more groups L.sup.a. Preferred compounds of formula IA according to this preferred embodiment contain a group P-Sp-selected from the following formulae:

P-CHL.sup.a- SL1

P—(CH.sub.2).sub.cc—O—CHL.sup.a- SL2

P—(CH.sub.2).sub.cc—CO—O—CHL.sup.a- SL3

P—(CH.sub.2).sub.cc—CHL.sup.a- SL4 [0537] in which P and L.sup.a are as defined in formula I or have one of the meanings given above and below, and cc is 1, 2, 3, 4, 5 or 6, preferably 1, 2 or 3.

[0538] Preferred compounds of formula I contain one or more groups P-Sp-selected from formulae SL1, SL2 and SL3, very preferably of formula SL1.

[0539] Preferably in the compounds of formula IA M1 is selected of formula 1 or 2.

[0540] Preferred compounds of formula IA are selected from the following subformulae:

##STR00149## [0541] wherein P, Sp and L have the meanings given in formula IA or one of the preferred meanings as given above and below, [0542] r1, r2, r3 are independently of each other 0, 1, 2, 3 or 4, [0543] r4 is 0, 1, 2 or 3, [0544] wherein the compounds contain at least one group Sp that is at least monosubstituted by L.sup.a and/or at least one group L that denotes L.sup.a.

[0545] Very preferred compounds of subformulae IA-1 to IA-10 are those wherein all groups P are identical and denote acrylate or methacrylate, preferably methacrylate.

[0546] Further preferred compounds of subformulae IA-1 to IA-10 are those wherein at least one group Sp denotes a single bond.

[0547] Further preferred compounds of subformulae IA-1 to IA-10 are those wherein one group Sp is selected from formulae SL1 to SL4, very preferably from formula SL1, and the other groups Sp denote a single bond or, if being different from a single bond, denote is, —(CH.sub.2).sub.p1—, —(CH.sub.2).sub.p1—O—, —(CH.sub.2).sub.p1—O—CO— or —(CH.sub.2).sub.p1—CO—O—, in which p1 is an integer from 1 to 12, preferably 1 to 6, and the O— or CO-group is connected to the benzene ring, very preferably wherein the other groups Sp denote single bonds.

[0548] Further preferred compounds of formula IA and its subformulae are selected from the following preferred embodiments, including any combination thereof: [0549] All groups P in the compound have the same meaning, [0550] M.sup.a is of formula 1 or 2, very preferably of formula 1, [0551] the compounds contain exactly two polymerizable groups (represented by the groups P), [0552] the compounds contain exactly three polymerizable groups (represented by the groups P), [0553] P is selected from the group consisting of acrylate, methacrylate and oxetane, very preferably acrylate or methacrylate, [0554] the compounds contain at least one, preferably exactly one, group Sp which is substituted by L.sup.a, and which is preferably selected from formulae SL1-SL4, very preferably from formulae SL1, SL2 and SL3, [0555] Sp, when being different from a single bond, is —(CH.sub.2).sub.p2—, —(CH.sub.2).sub.p2—O—, —(CH.sub.2).sub.p2—CO—O—, —(CH.sub.2).sub.p2—O—CO—, wherein p2 is 2, 3, 4, 5 or 6, and the O-atom or the CO-group, respectively, is connected to the benzene ring, [0556] Sp' is selected from formula SL1, [0557] L denotes F, Cl, CH.sub.3, C.sub.2H.sub.5, OCH.sub.3 or OC.sub.2H.sub.5, very preferably F, [0558] L' denotes F, Cl, CH.sub.3, C.sub.2H.sub.5, OCH.sub.3 or OC.sub.2H.sub.5, very preferably F, [0559] L.sup.a denotes —C(CH.sub.3).sub.2—OH, —C(C.sub.2H.sub.5).sub.2—OH or —C(CH.sub.3)(C.sub.2H.sub.5)OH, very preferably —C(CH.sub.3).sub.2—OH, [0560] r1, r2, r3 and r4 denote 0 or 1, [0561] r1+r2 is 0, [0562] r1+r2 is 1, [0563] r3 is 0, [0564] r4 is 0.

[0565] Very preferred compounds of formula IA and its subformulae are selected from the following list:

##STR00150## ##STR00151## ##STR00152## [0566] wherein "Me" is methyl and "Et" is ethyl.

[0567] In a preferred embodiment the LC medium comprises one or more compounds of formula IB having two polymerizable groups, which are preferably selected from formula IB-D

##STR00153## [0568] wherein P and Sp have the meanings given in formula IB, L has one of the meanings given in formula IB which is different from P-Sp-, r1, r2 and r3 are independently of each other 0, 1, 2, 3 or 4, preferably 0, 1 or 2, very preferably 0 or 1, and k is 0 or 1.

[0569] Preferred compounds of formula IB-D are selected from the following subformulae

##STR00154## [0570] wherein P and Sp, L, r1, r2 and r3 have independently of each other one of

the meanings given in formula IB-D or one of their preferred meanings as given above and below.
[0571] Especially preferred are the compounds of formula IB-D-1.

[0572] In the compounds of formula IB-D preferably at least one of r1, r2 and r3 is not 0. P is preferably acrylate or methacrylate, very preferably methacrylate. Preferably all groups P in the formulae IB-D, IB-D-1 and IB-D-2 have the same meaning, and very preferably denote methacrylate. Sp'' is preferably selected from —(CH.sub.2).sub.2—, —(CH.sub.2).sub.3—, —(CH.sub.2).sub.4—, —O—(CH.sub.2).sub.2—, —O—(CH.sub.2).sub.3—, —O—CO—(CH.sub.2).sub.2 and —CO—O—(CH).sub.2—, wherein the O atom or the CO group is attached to the benzene ring. L is preferably selected from F, CH.sub.3, OCH.sub.3, OC.sub.2H.sub.5 and C.sub.2H.sub.5, very preferably from F.

[0573] Very preferred compounds of formula IB-D are selected from the following subformulae:

##STR00155## ##STR00156## ##STR00157## ##STR00158##

[0574] Very preferred are the compounds of formula IBD1, IBD4, IBD6, IBD7, IBD19, IBD21 and IBD23.

[0575] Further preferred are the compounds of formulae IBD1 to IBD25 wherein one or two of the methacrylate groups are replaced by acrylate groups.

[0576] Further preferred compounds of formula IB-D are selected from Table D below, very preferably from the group consisting of RM-1, RM-2, RM-3, RM-7 to RM-49 and RM-58 to RM-77, very preferably selected from the group consisting of RM-1, RM-4, RM-8, RM-17, RM-19, RM-35, RM-37, RM-39, RM-40, RM-41, RM-48, RM-58, RM-64, RM-72 and RM-74.

[0577] In another preferred embodiment the LC medium comprises one or more polymerizable compounds of formula IB having three polymerizable groups, which are preferably selected from formula IB-T

##STR00159## [0578] wherein P, Sp, L, r1, r2 and k independently of each other have the meanings given in formula IB-D or one of their preferred meanings as given above and below, and r4 is 0, 1, 2 or 3, preferably 0, 1 or 2, very preferably 0 or 1.

[0579] Preferred compounds of formula IB-T are selected from the following subformulae

##STR00160## [0580] wherein P, Sp, L, r1, r2 and r4 have independently of each other one of the meanings given in formula IB-T or one of their preferred meanings as given above and below.

Preferably at least one of r1, r2 and r4 is not 0. P is preferably acrylate or methacrylate, very preferably methacrylate. L is preferably selected from F, CH.sub.3, OCH.sub.3, OC.sub.2H.sub.5 and C.sub.2H.sub.5, very preferably from OCH.sub.3 or F. Preferably all groups P in the formulae IB-T and IB-T-1 to IB-T-6 have the same meaning, and very preferably denote methacrylate.

[0581] Especially preferred are the compounds of formula IB-T-1, IB-T-4 and IB-T-5.

[0582] Very preferred compounds of formula IB-T are selected from the following subformulae:

##STR00161## ##STR00162## ##STR00163## ##STR00164## ##STR00165##

##STR00166## ##STR00167## ##STR00168## ##STR00169## ##STR00170## ##STR00171##
##STR00172## ##STR00173## ##STR00174##

[0583] Of the compounds of formula IBT1 to IB-T21 with two benzene rings very preferred are the compounds of formula IBT1, IBT2, IBT3, IBT8, IBT9, IBT10, IBT15, IBT16, IBT17. Most preferred are the compounds of formula IBT1, IBT2, IBT3, IBT8, IBT9 and IB10.

[0584] Of the compounds of formula IBT22 to IBT53 with three benzene rings very preferred are the compounds of formula IBT22 to IBT46. Most preferred are the compounds of formula IBT22, IBT28, IBT29, IBT35 and IBT36.

[0585] Further preferred are the compounds of formulae IBT1 to IBT53 wherein one or two of the methacrylate groups are replaced by acrylate groups.

[0586] Further preferred are the compounds of formulae IBT1 to IBT53 wherein all methacrylate groups are replaced by acrylate groups.

[0587] In another preferred embodiment the LC medium contains at least one compound of formula IC, which is at least monosubstituted by an alkenyl group L.sup.b. Preferred compounds of

formula IC of this preferred embodiment are selected from the following subformulae:
##STR00175## ##STR00176## [0588] wherein P, Sp, L, r1, r2, r3 and r4 have the meanings given in formula IB-D and IB-T or one of the preferred meanings as given above and below, with $r1+r2+r3+r4 \geq 1$, and wherein the compounds contain at least one group L that denotes L.sub.2, —CH=CH.sub.2, —CH.sub.2—CH=CH.sub.2, —CH=CH—CH.sub.3, —CH=CH—CH—CH.sub.2 or —C(CH.sub.3)=CH.sub.2, very preferably —CH=CH.sub.2 or C(CH.sub.3)=CH.sub.2.
[0589] In the compounds of formulae IC-1 to IC-9 preferably L' denotes —CH=CH.sub.2, —CH.sub.2—CH=CH.sub.2, —CH=CH—CH.sub.3, —CH=CH—CH—CH.sub.2 or —C(CH.sub.3)=CH.sub.2, very preferably —CH=CH.sub.2 or C(CH.sub.3)=CH.sub.2.
[0590] In the compounds of formulae IC-1 to IC-9 preferably P is acrylate or methacrylate, very preferably methacrylate. If Sp is different from a single bond, it is preferably selected from —(CH.sub.2).sub.2—, —(CH.sub.2).sub.3—, —(CH.sub.2).sub.4—, —O—(CH.sub.2).sub.2—, —O—(CH.sub.2).sub.3—, —O—CO—(CH.sub.2).sub.2 and —CO—O—(CH).sub.2—, wherein the O atom or the CO group is attached to the benzene ring. L is preferably selected from F, CH.sub.3, OCH.sub.3, OC.sub.2H.sub.5 or C.sub.2H.sub.5, very preferably from OCH.sub.3 or F. Preferably all groups P in the formulae IC-1 to IC-9 have the same meaning, and very preferably denote methacrylate.

[0591] Very preferred compounds of formulae IC-1 to IC-9 are selected from the following subformulae:

##STR00177## ##STR00178## ##STR00179## ##STR00180## ##STR00181## ##STR00182## ##STR00183##

[0592] Further preferred are compounds of formula IC1 to IC46 wherein one, two or all of the methacrylate groups are replaced by acrylate groups.

[0593] In a preferred embodiment the LC medium comprises at least one polymerizable compound of formula IA and/or formula IB and/or formula IC which have absorption in the range from 330 to 390 nm. Very preferably these compounds have an extinction coefficient of at least 0.5 at a wavelength in the range from 330 to 390 nm, more preferably in the range from 340 to 380 nm, very preferably in the range from 350 to 370 nm, most preferably in the range from 355 to 365 nm. The extinction coefficient and absorption wavelength are measured unless stated otherwise in a solution of the compound in DCM at a concentration of 3 g/L.

[0594] The total proportion of the polymerizable compounds of formulae IA, IB, IC and their subformulae in the LC medium according to the present invention is preferably from 0.05 to 3.0%, more preferably from 0.1 to 1.5%, very preferably from 0.1 to 0.9%.

[0595] In a first preferred embodiment of the present invention, the LC medium contains one or more, preferably exactly one, compound(s) of formula IA or its subformulae and one or more, preferably exactly one, compound(s) of formula IB or IC their subformulae, and preferably does not contain further polymerizable compounds.

[0596] Preferably, in the LC medium of this first preferred embodiment the proportion of the compound(s) of formula IA or its subformulae is from 0.01 to 1.0%, more preferably from 0.05 to 0.8%, very preferably from 0.1 to 0.6%, and the proportion of the compound(s) of formula IB or IC or their subformulae is from 0.01 to 1.0%, more preferably from 0.02 to 0.8%, very preferably from 0.05 to 0.5%.

[0597] In a second preferred embodiment of the present invention, the LC medium contains one or more, preferably exactly one, compound(s) of formula IA or its subformulae, one or more, preferably exactly one, compound(s) of formula IB or their subformulae, and one or more, preferably exactly one, compound(s) of formula IC or their subformulae, and preferably does not contain further polymerizable compounds.

[0598] More preferably the LC medium of this second preferred embodiment contains one or more, preferably exactly one, compound(s) of formula IA or its subformulae, one or more, preferably exactly one, compound(s) of formula IB-D or IB-T or their subformulae, and one or more, preferably exactly one, compound(s) selected from formulae IC-1 to IC-9 or their subformulae, and preferably does not contain further polymerizable compounds.

[0599] Preferably, in the LC medium of this second preferred embodiment the proportion of the compound(s) of formula IA or its subformulae is from 0.01 to 1.0%, more preferably from 0.05 to 0.8%, very preferably from 0.1 to 0.6%, the total proportion of the compound(s) of formula IB-D and IB-T or their subformulae is from 0.01 to 1.0%, more preferably from 0.05 to 0.8%, very preferably from 0.1 to 0.6%, and the total proportion of the compound(s) of the formulae IC-1 to IC-9 or their subformulae is from 0.01 to 1.0%, more preferably from 0.02 to 0.8%, very preferably from 0.05 to 0.5%.

[0600] In another preferred embodiment the LC medium comprises, in addition to the polymerizable compounds of formulae IA, IB, IC and their subformulae, at least one further polymerizable compound.

[0601] The compounds of the formulae M and its subformulae can be prepared analogously to processes known to the person skilled in the art and described in standard works of organic chemistry, such as, for example, in Houben-Weyl, Methoden der organischen Chemie [Methods of Organic Chemistry], Thieme-Verlag, Stuttgart.

[0602] For example, acrylic or methacrylic esters can be prepared by esterification of the corresponding alcohols with acid derivatives like, for example, (meth)acryloyl chloride or (meth)acrylic anhydride in the presence of a base like pyridine or triethyl amine, and 4-(N,N-dimethylamino)pyridine (DMAP). Alternatively the esters can be prepared by esterification of the alcohols with (meth)acrylic acid in the presence of a dehydrating reagent, for example according to Steglich with dicyclohexylcarbodiimide (DCC), N-(3-dimethylaminopropyl)-N'-ethylcarbodiimide (EDC) or N-(3-dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride and DMAP.

[0603] The invention furthermore relates to an LC medium or LC display as described above, wherein the polymerizable compounds, like those of formula M and its subformulae, are present in polymerized form.

[0604] Optionally one or more polymerization initiators are added to the LC medium. Suitable conditions for the polymerization and suitable types and amounts of initiators are known to the person skilled in the art and are described in the literature. Suitable for free-radical polymerization are, for example, the commercially available photoinitiators Irgacure651®, Irgacure184®, Irgacure907®, Irgacure369® or Darocure1173® (Ciba AG). If a polymerization initiator is employed, its proportion is preferably 0.001 to 5% by weight, particularly preferably 0.001 to 1% by weight.

[0605] The polymerizable compounds according to the invention are also suitable for polymerization without an initiator, which is accompanied by considerable advantages, such, for example, lower material costs and in particular less contamination of the LC medium by possible residual amounts of the initiator or degradation products thereof. The polymerization can thus also be carried out without the addition of an initiator. In a preferred embodiment, the LC medium thus does not contain a polymerization initiator.

[0606] The LC medium may also comprise one or more stabilisers in order to prevent undesired spontaneous polymerization of the RMs, for example during storage or transport. Suitable types and amounts of stabilisers are known to the person skilled in the art and are described in the literature. Particularly suitable are, for example, the commercially available stabilisers from the Irganox® series (Ciba AG), such as, for example, Irganox® 1076. If stabilisers are employed, their proportion, based on the total amount of RMs or the polymerizable component (component A), is preferably 10-50,000 ppm, particularly preferably 50-5,000 ppm.

[0607] In a preferred embodiment the LC media contain one or more chiral dopants, preferably in a concentration from 0.01 to 1% by weight, very preferably from 0.05 to 0.5% by weight. The chiral dopants are preferably selected from the group consisting of compounds from Table C below, very preferably from the group consisting of R- or S-1011, R- or S-2011, R- or S-3011, R- or S-4011, and R- or S-5011.

[0608] In another preferred embodiment the LC media contain a racemate of one or more chiral

dopants, which are preferably selected from the chiral dopants mentioned in the previous paragraph.

[0609] In another preferred embodiment of the present invention the LC media contain one or more further stabilisers.

[0610] Preferred stabilisers are selected from the compounds of formula H

##STR00184## [0611] in which [0612] Ar denotes an aromatic or heteroaromatic hydrocarbon group having 4 to 40 C atoms, preferably 6 to 30 C atoms; [0613] Sp denotes a spacer group; [0614] R^{sup}.S denotes H, alkyl having 1 to 12 C atoms or alkenyl having 2 to 12 C atoms; [0615] Z^{sup}.S denotes —O—, —C(O)O—, —(CH_{sub}.2).sub.z— or —(CH_{sub}.2).sub.zO—, or a single bond; [0616] HA denotes

##STR00185## [0617] R^{sup}.H denotes H, O^{sup}., CH_{sub}.3, OH or OR^{sup}.S, preferably H or O^{sup}.; [0618] R^{sup}.S1, R^{sup}.S2, R^{sup}.S3 and R^{sup}.S4, identically or differently, denote alkyl having 1 to 6 C atoms, preferably having 1 to 3 C atoms, very preferably CH_{sub}.3; [0619] G denotes H or R^{sup}.S or a group Z^{sup}.S-HA; [0620] Z is an integer from 1 to 6; and [0621] q is 3 or 4.

[0622] The compounds of formula H are described in EP3354710 A1 and EP3354709 A1.

[0623] Preferred compounds of formula H are selected from the formulae H-1, H-2 and H-3:

##STR00186## [0624] in which R^{sup}.H has the meanings given above and preferably denotes H or O^{sup}., and n is an integer from 0 to 12, preferably 5, 6, 7, 8 or 9, very preferably 7, and Sp denotes a spacer group, preferably alkylene having 1 to 12 C atoms in which one or more non-adjacent —CH_{sub}.2— groups may be replaced with —O—.

[0625] Preferred compounds of formula H-1 are those of formula H-1-1:

##STR00187## [0626] in which RH has the meanings given above and preferably denotes H or O^{sup}., and n is an integer from 0 to 12, preferably 5, 6, 7, 8 or 9, very preferably 7.

[0627] Very preferred compounds of formula H-1-1 are those of formula H-1-1-1:

##STR00188##

[0628] Preferred compounds of formula H-2 are those of formula H-2-1:

##STR00189## [0629] in which R^{sup}.H has the meanings given above and preferably denotes H or O^{sup}., and n₂, on each occurrence identically or differently, preferably identically, is an integer from 1 to 12, preferably 2, 3, 4, 5, or 6, very preferably 3, and R\$ on each occurrence identically or differently, preferably identically, denotes alkyl having 1 to 6 C atoms, preferably n-butyl.

[0630] Very preferred compounds of formula H-2-1 are those of formula H-2-1-1:

##STR00190##

[0631] Preferred compounds of formula H-3 are selected from the formula H-3-1:

##STR00191## [0632] in which Sp and R^{sup}.H have the meanings given above and R^{sup}.H preferably denotes H or O^{sup}., and n is an integer from 0 to 12, preferably 5, 6, 7, 8 or 9, very preferably 7.

[0633] Further preferred stabilisers are selected from the group consisting of the formulae ST-1 to ST-18:

##STR00192## ##STR00193## ##STR00194## [0634] in which [0635] R^{sup}.ST denotes H, an alkyl or alkoxy radical having 1 to 15 C atoms, wherein, in addition, one or more CH_{sub}.2 groups may each be replaced, independently of one another, by —C≡C—, —CF_{sub}.2O—, —OCF_{sub}.2—, —CH=CH—,

##STR00195## [0636] —O—, —CO—O—, —O—CO— in such a way that O atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by halogen,

##STR00196##

on each occurrence, identically or differently, denotes

##STR00197## [0637] Z^{sup}.ST each, independently of one another, denote —CO—O—, —O—

CO—, —CF.sub.2O—, —OCF.sub.2—, —CH.sub.2O—, —OCH.sub.2—, —CH.sub.2—, —CH.sub.2CH.sub.2—, —(CH.sub.2).sub.4—, —CH=CH—, —CH.sub.2O—, —C.sub.2F.sub.4—, —CH.sub.2CF.sub.2—, —CF.sub.2CH.sub.2—, —CF=CF—, —CH=CF—, —CF=CH—, —CH=CH—, —C≡C— or a single bond, [0638] L.sup.1 and L.sup.2 each, independently of one another, denote F, Cl, CH.sub.3, CF.sub.3 or CHF.sub.2, [0639] p denotes 0, 1 or 2, [0640] q denotes 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10.

[0641] Preferred compounds of formula ST are those selected from the formulae ST-3 and in particular:

##STR00198## [0642] in which n=1, 2, 3, 4, 5, 6 or 7, preferably n=3

##STR00199## [0643] in which n=1, 2, 3, 4, 5, 6 or 7, preferably n=3

##STR00200## [0644] in which n=1, 2, 3, 4, 5, 6 or 7, preferably n=1 or 7

##STR00201##

[0645] In the compounds of the formulae ST-3a and ST-3b, n preferably denotes 3. In the compounds of the formula ST-2a, n preferably denotes 7.

[0646] Very preferred stabilisers are selected from the group of the compounds of the formulae ST-2a-1, ST-3a-1, ST-3b-1, ST-8-1, ST-9-1 and ST-12:

##STR00202## ##STR00203##

[0647] In another preferred embodiment the LC medium comprises one or more stabilisers selected from Table D below.

[0648] Preferably the proportion of stabilisers in the LC medium is from 10 to 500 ppm, very preferably from 20 to 100 ppm.

[0649] In another preferred embodiment the LC medium according to the present invention contains a self alignment (SA) additive, preferably in a concentration of 0.1 to 2.5%.

[0650] In a preferred embodiment the SA-VA display according to the present invention does not contain a polyimide alignment layer. In another preferred embodiment the SA-VA display according to preferred embodiment contains a polyimide alignment layer.

[0651] Preferred SA additives for use in this preferred embodiment are selected from compounds comprising a mesogenic group and a straight-chain or branched alkyl side chain that is terminated with one or more polar anchor groups selected from hydroxy, carboxy, amino or thiol groups.

[0652] Further preferred SA additives contain one or more polymerizable groups which are attached, optionally via spacer groups, to the mesogenic group. These polymerizable SA additives can be polymerized in the LC medium under similar conditions as applied for the RMs in the PSA process.

[0653] Suitable SA additives to induce homeotropic alignment, especially for use in SA-VA mode displays, are disclosed for example in US 2013/0182202 A1, US 2014/0838581 A1, US 2015/0166890 A1 and US 2015/0252265 A1.

[0654] In another preferred embodiment an LC medium or a polymer stabilised SA-VA display according to the present invention contains one or more self alignment additives selected from Table F below.

[0655] In another preferred embodiment the LC medium according to the present invention contains one or more SA additives, preferably selected from formula II or its subformulae or selected from Table F below, in a concentration from 0.1 to 5%, very preferably from 0.2 to 3%, most preferably from 0.2 to 1.5%.

[0656] The invention furthermore relates to an LC display comprising an LC medium according to the invention as described above and below, which is a preferably a PSA or SA display, very preferably a PS-VA, PS-IPS, PS-FFS or SA-VA display.

[0657] The invention furthermore relates to an LC display comprising an LC medium as described above and below wherein the polymerizable compounds are present in polymerized form, which is preferably a PSA or SA display, very preferably a PS-VA, PS-IPS, PS-FFS or SA-VA display.

[0658] For the production of PSA or polymer stabilised SA displays, the polymerizable compounds

contained in the LC medium are polymerized by in-situ polymerization in the LC medium between the substrates of the LC display, preferably while a voltage is applied to the electrodes.

[0659] The structure of the displays according to the invention corresponds to the usual geometry for PSA displays, as described in the prior art cited at the outset. Geometries without protrusions are preferred, in particular those in which, in addition, the electrode on the colour filter side is unstructured and only the electrode on the TFT side has slots.

[0660] Particularly suitable and preferred electrode structures for PS-VA displays are described, for example, in US 2006/0066793 A1.

[0661] A preferred PSA type LC display of the present invention comprises: [0662] a first substrate including a pixel electrode defining pixel areas, the pixel electrode being connected to a switching element disposed in each pixel area and optionally including a micro-slit pattern, and optionally a first alignment layer disposed on the pixel electrode, [0663] a second substrate including a common electrode layer, which may be disposed on the entire portion of the second substrate facing the first substrate, and optionally a second alignment layer, [0664] an LC layer disposed between the first and second substrates and including an LC medium as described above and below, wherein the polymerizable compounds may also be present in polymerized form.

[0665] The first and/or second alignment layer controls the alignment direction of the LC molecules of the LC layer. For example, in PS-VA displays the alignment layer is selected such that it imparts to the LC molecules homeotropic (or vertical) alignment (i.e. perpendicular to the surface) or tilted alignment. Such an alignment layer may for example comprise a polyimide, which may also be rubbed, or may be prepared by a photoalignment method.

[0666] The LC layer with the LC medium can be deposited between the substrates of the display by methods that are conventionally used by display manufacturers, for example the so-called one-drop-filling (ODF) method. The polymerizable component of the LC medium is then polymerized for example by UV photopolymerization. The polymerization can be carried out in one step or in two or more steps.

[0667] The PSA display may comprise further elements, like a colour filter, a black matrix, a passivation layer, optical retardation layers, transistor elements for addressing the individual pixels, etc., all of which are well known to the person skilled in the art and can be employed without inventive skill.

[0668] The electrode structure can be designed by the skilled person depending on the individual display type. For example for PS-VA displays a multi-domain orientation of the LC molecules can be induced by providing electrodes having slits and/or bumps or protrusions in order to create two, four or more different tilt alignment directions.

[0669] Upon polymerization the polymerizable compounds form a copolymer, which causes a certain tilt angle of the LC molecules in the LC medium. Without wishing to be bound to a specific theory, it is believed that at least a part of the crosslinked polymer, which is formed by the polymerizable compounds, will phase-separate or precipitate from the LC medium and form a polymer layer on the substrates or electrodes, or the alignment layer provided thereon. Microscopic measurement data (like SEM and AFM) have confirmed that at least a part of the formed polymer accumulates at the LC/substrate interface.

[0670] The polymerization can be carried out in one step. It is also possible firstly to carry out the polymerization, optionally while applying a voltage, in a first step in order to produce a tilt angle, and subsequently, in a second polymerization step without an applied voltage, to polymerize or crosslink the compounds which have not reacted in the first step ("end curing").

[0671] Suitable and preferred polymerization methods are, for example, thermal or photopolymerization, preferably photopolymerization, in particular UV induced photopolymerization, which can be achieved by exposure of the polymerizable compounds to UV radiation.

[0672] The polymerizable compounds of formula M and its subformulae do in particular show

good UV absorption in, and are therefore especially suitable for, a process of preparing a PSA display including one or more of the following features or any combination thereof: [0673] the polymerizable medium is exposed to UV light in the display in a 2-step process, including a first UV exposure step ("UV1 step"), with application of a voltage to the electrodes of the display, to generate the tilt angle, and a second UV exposure step ("UV2 step"), without application of a voltage to the electrodes of the display, to complete polymerization of the polymerizable compounds, [0674] the polymerizable medium is exposed to UV light in the display generated by an UV-LED lamp, preferably at least in the UV2 step, more preferably both in the UV1 and UV2 step, [0675] the polymerizable medium is exposed to UV light in the display generated by an energy-saving UV lamp (also known as "green UV lamps"). These lamps are characterized by a relative low intensity (1/100-1/10 of a conventional UV1 lamp) in their absorption spectra from 300-380 nm, and are preferably used in the UV2 step, but are optionally also used in the UV1 step when avoiding high intensity is necessary for the process, [0676] the polymerizable medium is exposed to UV light in the display generated by a UV lamp with a radiation spectrum that is shifted to longer wavelengths, preferably ≥ 340 nm, more preferably from 350 to < 370 nm, very preferably from 355 to 368 nm, to avoid short UV light exposure in the PS-VA process.

[0677] Both using lower intensity and a UV shift to longer wavelengths protect the organic layer against damage that may be caused by the UV light.

[0678] A preferred embodiment of the present invention relates to a process for preparing a PSA display as described above and below, comprising one or more of the following features or any combination thereof: [0679] the polymerizable LC medium is irradiated by UV light in a 2-step process, including a first UV exposure step ("UV1 step"), with application of a voltage, to generate the tilt angle, and a second UV exposure step ("UV2 step"), without application of a voltage, to complete polymerization of the polymerizable compounds, [0680] the polymerizable LC medium is irradiated by UV light generated by a UV lamp having an intensity of from 0.5 mW/cm² to 10 mW/cm² in the wavelength range from 300-380 nm, preferably in the UV2 step, and optionally also in the UV1 step, [0681] the polymerizable LC medium is irradiated by UV light having a wavelength of ≥ 340 nm, and preferably ≤ 420 nm, very preferably in the range from 340 to 380 nm, more preferably in the range from 350 to < 370 nm, most preferably in the range from 355 to 368 nm, [0682] the polymerizable LC medium is irradiated by UV light while a voltage is applied to the electrodes of the display, [0683] irradiation by UV light is carried out using a UV-LED lamp.

[0684] This preferred process can be carried out for example by using the desired UV lamps or by using a band pass filter and/or a cut-off filter, which are substantially transmissive for UV light with the respective desired wavelength(s) and are substantially blocking light with the respective undesired wavelengths. For example, when irradiation with UV light of wavelengths λ of 300-400 nm is desired, UV exposure can be carried out using a wide band pass filter being substantially transmissive for wavelengths $300 \text{ nm} < \lambda < 400 \text{ nm}$. When irradiation with UV light of wavelength λ of more than 340 nm is desired, UV irradiation can be carried out using a cut-off filter being substantially transmissive for wavelengths $\lambda > 340 \text{ nm}$.

[0685] In a preferred embodiment of the present invention, UV irradiation is carried out using a UV-LED lamp.

[0686] The use of UV-LED lamps, which have with only one narrow emission peak, in the PSA process provides several advantages, like for example a more effective optical energy transfer to the polymerizable compounds in the LC medium, depending on the choice of the suitable polymerizable compounds that shows absorption at the emission wavelength of the LED lamp. This allows to reduce the UV intensity and/or the UV irradiation time, thus enabling a reduced tact time and savings in energy and production costs. Another advantage is that the narrow emission spectrum of the lamp allows an easier selection of the appropriate wavelength for photopolymerization.

[0687] Very preferably the UV light source is an UV-LED lamp emitting a wavelength in the range from 340 to 400 nm, more preferably in the range from 340 to 380 nm. UV-LED lamps emitting UV light with a wavelength of 365 nm are especially preferred.

[0688] Preferably the UV-LED lamp emits light having an emission peak with a full width half maximum (FWHM) of 30 nm or less.

[0689] UV-LED lamps are commercially available, for example from Dr. Hoenle AG, Germany or Primelite GmbH, Germany, or IST Metz GmbH, Germany, with emission wavelengths e.g. of 365, 385, 395 and 405 nm.

[0690] This preferred process enables the manufacture of displays by using longer UV wavelengths, thereby reducing or even avoiding the hazardous and damaging effects of short UV light components.

[0691] UV radiation energy is in general from 6 to 100 J, depending on the production process conditions.

[0692] The LC medium according to the present invention may additionally comprise one or more further components or additives, preferably selected from the list including but not limited to comonomers, chiral dopants, polymerization initiators, inhibitors, stabilisers, surfactants, wetting agents, lubricating agents, dispersing agents, hydrophobing agents, adhesive agents, flow improvers, defoaming agents, deaerators, diluents, reactive diluents, auxiliaries, colourants, dyes, pigments and nanoparticles.

[0693] Furthermore, it is possible to add to the LC media, for example, 0 to 15% by weight of pleochroic dyes, furthermore nanoparticles, conductive salts, preferably ethyldimethyldodecylammonium 4-hexoxybenzoate, tetrabutylammonium tetraphenylborate or complex salts of crown ethers (cf., for example, Haller et al., Mol. Cryst. Liq. Cryst. 24, 249-258 (1973)), for improving the conductivity, or substances for modifying the dielectric anisotropy, the viscosity and/or the alignment of the nematic phases. Substances of this type are described, for example, in DE-A 22 09 127, 22 40 864, 23 21 632, 23 38 281, 24 50 088, 26 37 430 and 28 53 728.

[0694] The individual components of the above-listed preferred embodiments of the LC media according to the invention are either known or methods for the preparation thereof can readily be derived from the prior art by the person skilled in the relevant art, since they are based on standard methods described in the literature. Corresponding compounds of the formula CY are described, for example, in EP-A-0 364 538. Corresponding compounds of the formula ZK are described, for example, in DE-A-26 36 684 and DE-A-33 21 373.

[0695] The LC media which can be used in accordance with the invention are prepared in a manner conventional per se, for example by mixing one or more of the above-mentioned compounds with one or more polymerizable compounds as defined above, and optionally with further liquid-crystalline compounds and/or additives. In general, the desired amount of the components used in lesser amount is dissolved in the components making up the principal constituent, advantageously at elevated temperature. It is also possible to mix solutions of the components in an organic solvent, for example in acetone, chloroform or methanol, and to remove the solvent again, for example by distillation, after thorough mixing. The invention furthermore relates to the process for the preparation of the LC media according to the invention.

[0696] It goes without saying to the person skilled in the art that the LC media according to the invention may also comprise compounds in which, for example, H, N, O, Cl, F have been replaced by the corresponding isotopes like deuterium etc.

[0697] The following examples explain the present invention without restricting it. However, they show the person skilled in the art preferred mixture concepts with compounds preferably to be employed and the respective concentrations thereof and combinations thereof with one another. In addition, the examples illustrate which properties and property combinations are accessible.

[0698] For the present invention and in the following examples, the structures of the liquid-crystal

compounds are indicated by means of acronyms. Unless stated otherwise, the transformation into chemical formulae is done in accordance with Tables A.1 to A.3 below. All radicals $C_{n+1}H_{2n+1}$, C_mH_{2m+1} and C_lH_{2l+1} or C_nH_{2n} , C_mH_{2m} and C_lH_{2l} are straight-chain alkyl radicals or alkylene radicals, in each case having n , m and l C atoms respectively. Preferably n , m and l are independently of each other 1, 2, 3, 4, 5, 6, or 7. Table A.1 shows the codes for the ring elements of the nuclei of the compound, Table A.2 lists the bridging units, and Table A.3 lists the meanings of the symbols for the left- and right-hand end groups of the molecules. The acronyms are composed of the codes for the ring elements with optional linking groups, followed by a first hyphen and the codes for the left-hand end group, and a second hyphen and the codes for the right-hand end group.

TABLE-US-00001 TABLE A.1 Ring elements C [00204]  D [00205]  A [00206]  P [00207]  G [00208]  U [00209]  Y [00210]  P(F,Cl)Y [00211]  np [00212]  DI [00213]  AI [00214]  GI [00215]  UI [00216]  U(1) [00217]  P(Cl,F)Y [00218]  n3f [00219]  th [00220]  th2f [00221]  o2f [00222]  dh [00223]  B [00224]  O [00225]  K [00226]  L [00227]  F [00228]  nN3fl [00229]  thl [00230]  th2fl [00231]  o2fl [00232]  nf [00233]  B(S) [00234]  S [00235]  KI [00236]  LI [00237]  FI [00238]  Bh [00239]  Bf [00240]  Bfi [00241]  B(P) [00242]  Bh(S) [00243]  Bf(S) [00244]  Bfi(S) [00245]  B(A) [00246] 

TABLE-US-00002 TABLE A.2 Bridging units E —CH₂—CH₂— V —CH=CH— T —C≡C— W —CF₂—CF₂— B —CF=CF— Z —CO—O— ZI —O—CO— X —CF=CH— XI —CH=CF— O —CH₂—O— OI —O—CH₂— Q —CF₂—O— QI —O—CF₂—

TABLE-US-00003 TABLE A.3 End groups On the left individually or in combination —n— $C_{n+1}H_{2n+1}$ — —n— C_nH_{2n+1} —nO— C_nH_{2n+1} —O— —On —O— $C_{n+1}H_{2n+1}$ —m— $C_{m+1}H_{2m+1}$ —Om —O— $C_{m+1}H_{2m+1}$ —V— CH₂=CH— —V— CH=CH₂ —nV— $C_{n+1}H_{2n+1}$ —CH=CH— —nV— C_nH_{2n} —CH=CH₂ —Vn— CH₂=CH— C_nH_{2n} — —Vn —CH=CH— $C_{n+1}H_{2n+1}$ —nVm— $C_{n+1}H_{2n+1}$ —CH=CH— C_mH_{2m} — —nVm — C_nH_{2n} —CH=CH— $C_{m+1}H_{2m+1}$ —N— N≡C— —N— C≡N —S— S=C=N— —S— N=C=S —F— F— —F— F— —CL— Cl— —CL— Cl— —M— CFH₂— —M— CFH₂— —D— CF₂H— —D— CF₂H— —T— CF₃— —T— CF₃— —MO— CFH₂O— —OM— OCFH₂— —DO— CF₂HO— —OD— OCF₂H— —TO— CF₃O— —OT— OCF₃— —A— H—C≡C— —A— C≡C—H —nA— $C_{n+1}H_{2n+1}$ —C≡C— —An —C≡C— C_nH_{2n+1} —NA— N≡C—C≡C— —AN —C≡C—C≡N (cn)— [00247]  —(cn) [00248]  (cn)O— [00249]  —O(cn) [00250]  (cn)m— [00251]  —m(cn) [00252]  (cn)mO— [00253]  —Om(cn) [00254]  (c5en)— [00255]  —(c5en) [00256]  (c5en)O— [00257]  —O(c5en) [00258]  (c5en)m— [00259]  —m(c5en) [00260]  (c5en)mO— [00261]  —Om(c5en) [00262]  On the left only in combination On the right only in combination —. . . n . . . — — C_nH_{2n} — —. . . n . . . — — C_nH_{2n} — —. . . m . . . — — C_mH_{2m} —

—...M...—CFH—...M...—CFH—...D...—CF.sub.2—...D...—
 CF.sub.2—...V...—CH=CH—...V...—CH=CH—...Z...—CO—O—
 —...Z...—CO—O—...ZI...—CO—O—...ZI...—CO—O—...K...
 —CO—...K...—CO—...W...—CF=CF—...W...—CF=CF—...
 O...—O—...O...—O— [0699] in which n and m are each integers, and the three dots
 “...” are placeholders for other abbreviations from this table.

[0700] Table B shows illustrative structures of compounds together with their respective abbreviations.

TABLE-US-00004 TABLE B In Table B, n, m, k and l are, independently of one another, each an integer, preferably 1 to 9 preferably 1 to 7, k and l may also be 0 and are preferably 0 to 4, more preferably 0 or 2 and most preferably 2, n is preferably 1, 2, 3, 4 or 5 or, in the combination “—nO—”, n is preferably 1, 2, 3 or 4, very preferably 2 or 4, m is preferably 1, 2, 3, 4 or 5 or, in the combination “—Om”, m is preferably 1, 2, 3 or 4, more preferably 2 or 4. The combination “—nVm” preferably is “2V1”. (O)C.sub.mH.sub.2m + 1 means C.sub.mH.sub.2m + 1 or OC.sub.mH.sub.2m + 1. [00263]  AIK-n-F [00264]  AIY-n-Om [00265]  AY-n-Om [00266]  B-nO-Om [00267]  B-n-Om [00268]  B-(c3)-Om [00269]  B-(c3)1O-Om [00270]  B-(c4)-Om [00271]  B-(c4)1O-Om [00272]  B-(c5)-Om [00273]  B-(c5)1O-Om [00274]  B(S)-nO-Om [00275]  B(S)-n-Om [00276]  B(S)-(c3)-Om [00277]  B(S)(c3)1O-Om [00278]  B(S)-(c4)-Om [00279]  B(S)-(c4)1O-Om [00280]  B(S)-(c5)-Om [00281]  B(S)-(c5)1O-Om [00282]  CB(S)-n(O)m [00283]  B(P)-nO-Om [00284]  B(A)-nO-Om [00285]  CB-n-m [00286]  CB-n-Om [00287]  COB-n-Om [00288]  APY-n-Om [00289]  COB(S)-n-Om [00290]  PB-n-m [00291]  PB-n-Om [00292]  BCH-nm [00293]  BCH-nmF [00294]  BCN-nm [00295]  C-1V-V1 [00296]  CY-n-Om [00297]  CY(c3)-Om [00298]  CY-(c3)1-Om [00299]  CY(F,Cl)-n-Om [00300]  CY-(c5)-Om [00301]  COY-(c5)-Om [00302]  CY(Cl,F)-n-Om [00303]  CCY-n-Om [00304]  CC1Y-n-Om [00305]  CC1G-V-F [00306]  CCG-V-F [00307]  CAIY-n-Om [00308]  CCY(F,Cl)-n-Om [00309]  CCY(Cl,F)-n-Om [00310]  CCY-n-m [00311]  CCY-V-m [00312]  CCY-Vn-m [00313]  CCY-n-OmV [00314]  CGPC-n-m, CBC-nmF [00315]  CPPC-n-m, CBC-nm [00316]  CCP-V-m [00317]  CCP-VI-m [00318]  CCP-nV-m [00319]  CCP-nVI-m [00320]  CCP-n-m [00321]  CPYP-n(O)m [00322]  CYYC-n-m [00323]  CCYY-n(O)m [00324]  CCY-nO2V [00325]  CCH-nOm, CC-n-Om [00326]  CCC-n-m [00327]  CCC-n-V [00328]  CY-n-m [00329]  CCH-nm, CC-n-m [00330]  CC-n-V [00331]  CC-n-IV [00332]  CC-n-Vm [00333]  CC-V-V [00334]  CC-n-mV [00335]  CC-V-IV [00336]  CC-V-Vm [00337] CC-Vk-IV [00338] CC-n-IVm [00339] CC-nV-Vm [00340] CC-nV-IV [00341] CC-n-VV [00342] CC-n-VVm [00343] CVC-n-m [00344]

 embedded image CVC-n-V [00345]  embedded image CVC-n-Vm [00346]  embedded image CCOC-n-m [00347]  embedded image CCZPC-n-m [00348]  embedded image CCZY-n-Om, CP-nOmFF [00349]  embedded image CCZC-n-m, CH-nm [00350]  embedded image CEY-n-Om [00351]  embedded image CEY-V-n [00352]  embedded image CVY-V-n [00353]  embedded image CY-V-On [00354]  embedded image CY-n-O1V [00355]  embedded image CY-n-OC(CH.sub.3)=CH.sub.2 [00356]  embedded image CCN-nm [00357]  embedded image CY-n-OV [00358]  embedded image CCPC-nm [00359]  embedded image CCY-n-kOm [00360]  embedded image CPY-n-Om [00361]  embedded image CPY-n-m [00362]  embedded image CPY-(c5)-Om [00363]  embedded image CPY-V-Om [00364]  embedded image CPY-n-O1(c3) [00365]  embedded image CQY-n-(O)m [00366]  embedded image CQIY-n-(O)m [00367]  embedded image CCQY-n-(O)m [00368]  embedded image CCQIY-n-(O)m [00369]  embedded image CPQY-n-(O)m [00370]  embedded image CPQIY-n-(O)m [00371]  embedded image CPYG-n-(O)m [00372]  embedded image CCY-V-Om [00373]  embedded image CCY-V2-(O)m [00374]  embedded image CCY-1V2-(O)m [00375]  embedded image CCY-3V-(O)m [00376]  embedded image CCVC-n-V [00377]  embedded image CCVC-n-m [00378]  embedded image CCVC-n-IV [00379]  embedded image CLP-n-m [00380]  embedded image CLP-V-n [00381]  embedded image CPP-n-m [00382]  embedded image CPG-n-m [00383]  embedded image CGP-n-m [00384]  embedded image CCVC-V-V [00385]  embedded image CPGP-n-m [00386]  embedded image CY-nV-(O)m [00387]  embedded image CENaph-n-Om [00388]  embedded image COChrom-n-Om [00389]  embedded image COChrom-n-m [00390]  embedded image CCOChrom-n-Om [00391]  embedded image CCONaph-n-Om [00394]  embedded image CCNaph-n-Om [00395]  embedded image CNaph-n-Om [00396]  embedded image CETNaph-n-Om [00397]  embedded image CTNaph-n-Om [00398]  embedded image CK-n-F [00399]  embedded image CLY-n-Om [00400]  embedded image CLY-(c5)-Om [00401]  embedded image CLY-n-m [00402]  embedded image CLOY-(c5)-Om [00403]  embedded image LOY-n-Om [00404]  embedded image CYLI-n-m [00405]  embedded image COYOICC-n-m [00406]  embedded image COYOIC-n-V [00407]  embedded image CCOY-V-O2V [00408]  embedded image CCOY-V-O3V [00409]  embedded image COY-n-Om [00410]  embedded image CCOY-n-Om [00411]  embedded image CCOY-n-O(c5) [00412]  embedded image CCOY-n-O1(c5) [00413]  embedded image CCOY-n-O(c5en) [00414]  embedded image CCOY-n-O1(c5en) [00415]  embedded image CCEY-n-Om [00416]  embedded image CZYY-n-Om [00417]  embedded image CCQU-n-F [00418]  embedded image D-nOmFF [00419]  embedded image DFDBC-n(O)-(O)m [00420]  embedded image LB(S)-n-T [00421]  embedded image LB(S)-n-OT [00422]  embedded image LY-n-(O)m [00423]  embedded image LY-(c5)-(O)m [00424]  embedded image LP-n-m [00425]  embedded image PGP-n-(Naph) [00426]  embedded image LYLI-n-m [00427]  embedded image PCH-nm, CP-n-m [00428]  embedded image PCH-nOm, CP-n-Om [00429]  embedded image PPGU-n-F [00430]  embedded image PGIGI-n-F [00431]  embedded image PGP-n-m [00432]  embedded image PP-n-(O)m [00433]  embedded image PP-n-2V1 [00434]  embedded image PGP-n-2V1 [00435]  embedded image PPP-n-2V1 [00436]  embedded image PGP-n-2V [00437]  embedded image PYP-n-mV [00438]  embedded image PYP-n-m [00439]  embedded image PYP-n-Om [00440]  embedded image PGIY-n-Om [00441]  embedded image PPYY-n-m [00442]  embedded image YPY-n-m [00443]  embedded image YPY-n-mV [00444]  embedded image PY-n-Om [00445]  embedded image PY-n-m [00446]  embedded image PY(c3)-Om [00447]  embedded image PY-(c3)1-Om [00448]  embedded image PY-V2-Om [00449]  embedded image PY(F,Cl)-n-Om [00450]  embedded image Y-nO-Om [00451]  embedded image Y-nO-OmV [00452]  embedded image Y-nO-OkVm [00453]  embedded image YG-n-Om [00454]  embedded image YG-nO-Om [00455]  embedded image

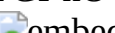
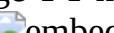
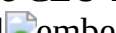
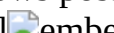
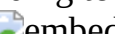
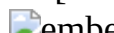
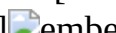
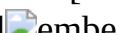
YGI-n-Om [00456]  embedded image YGI-n-O-Om [00457]  embedded image YY-n-Om [00458] 
 embedded image YY-n-O-Om [00459]  embedded image CLU-n-F [00460]  embedded image
CLU(1)-n-F [00461]  embedded image CLG-n-F [00462]  embedded image CLG-n-T
[0701] In a preferred embodiment of the present invention, the LC media according to the
invention comprise one or more compounds selected from the group consisting of compounds from
Table B.

TABLE-US-00005 TABLE C Table C shows possible chiral dopants which can be added to the LC
media according to the invention. [00463]  embedded image C 15 [00464]  embedded image CB
15 [00465]  embedded image CM 21 [00466]  embedded image R/S-811 [00467]
 embedded image CM 44 [00468]  embedded image CM 45 [00469]  embedded image CM 47
[00470]  embedded image CN [00471]  embedded image R/S-1011 [00472]  embedded image
R/S-2011 [00473]  embedded image R/S-3011 [00474]  embedded image R/S-4011 [00475]
 embedded image R/S-5011

[0702] The LC media preferably comprise 0 to 10% by weight, in particular 0.01 to 5% by weight,
particularly preferably 0.1 to 3% by weight, of dopants. The LC media preferably comprise one or
more dopants selected from the group consisting of compounds from Table C.

TABLE-US-00006 TABLE D Table D shows possible stabilisers which can be added to the LC
media according to the invention. Therein n denotes an integer from 1 to 12, preferably 1, 2, 3, 4, 5,
6, 7 or 8, and terminal methyl groups are not shown. [00476]  embedded image [00477]

 embedded image [00478]  embedded image [00479]  embedded image [00480]
 embedded image [00481]  embedded image [00482]  embedded image [00483]
 embedded image [00484]  embedded image [00485]  embedded image [00486]
 embedded image [00487]  embedded image [00488]  embedded image [00489]
 embedded image [00490]  embedded image [00491]  embedded image [00492]
 embedded image [00493]  embedded image [00494]  embedded image [00495]
 embedded image [00496]  embedded image [00497]  embedded image [00498]
 embedded image [00499]  embedded image [00500]  embedded image [00501]
 embedded image [00502]  embedded image [00503]  embedded image [00504]
 embedded image [00505]  embedded image [00506]  embedded image [00507]
 embedded image [00508]  embedded image [00509]  embedded image [00510]
 embedded image [00511]  embedded image [00512]  embedded image [00513]
 embedded image [00514]  embedded image [00515]  embedded image [00516]
 embedded image [00517]  embedded image [00518]  embedded image [00519]
 embedded image [00520]  embedded image [00521]  embedded image [00522]
 embedded image [00523]  embedded image [00524]  embedded image [00525]
 embedded image [00526]  embedded image

[0703] The LC media preferably comprise 0 to 10% by weight, in particular 1 ppm to 5% by
weight, particularly preferably 1 ppm to 1% by weight, of stabilisers. The LC media preferably
comprise one or more stabilisers selected from the group consisting of compounds from Table D.

TABLE-US-00007 TABLE E Table E shows illustrative reactive mesogenic compounds which can
be used in the LC media in accordance with the present invention. [00527]  embedded image RM-

1 [00528]  embedded image RM-2 [00529]  embedded image RM-3 [00530]  embedded image
RM-4 [00531]  embedded image RM-5 [00532]  embedded image RM-6 [00533]
 embedded image RM-7 [00534]  embedded image RM-8 [00535]  embedded image RM-9
[00536]  embedded image RM-10 [00537]  embedded image RM-11 [00538]  embedded image
RM-12 [00539]  embedded image RM-13 [00540]  embedded image RM-14 [00541]
 embedded image RM-15 [00542]  embedded image RM-16 [00543]  embedded image RM-17
[00544]  embedded image RM-18 [00545]  embedded image RM-19 [00546]  embedded image
RM-20 [00547]  embedded image RM-21 [00548]  embedded image RM-22 [00549]
 embedded image RM-23 [00550]  embedded image RM-24 [00551]  embedded image RM-25

[00552]  embedded image RM-26 [00553]  embedded image RM-27 [00554]  embedded image RM-28 [00555]  embedded image RM-29 [00556]  embedded image RM-30 [00557]  embedded image RM-31 [00558]  embedded image RM-32 [00559]  embedded image RM-33 [00560]  embedded image RM-34 [00561]  embedded image RM-35 [00562]  embedded image RM-36 [00563]  embedded image RM-37 [00564]  embedded image RM-38 [00565]  embedded image RM-39 [00566]  embedded image RM-40 [00567]  embedded image RM-41 [00568]  embedded image RM-42 [00569]  embedded image RM-43 [00570]  embedded image RM-44 [00571]  embedded image RM-45 [00572]  embedded image RM-46 [00573]  embedded image RM-47 [00574]  embedded image RM-48 [00575]  embedded image RM-49 [00576]  embedded image RM-50 [00577]  embedded image RM-51 [00578]  embedded image RM-52 [00579]  embedded image RM-53 [00580]  embedded image RM-54 [00581]  embedded image RM-55 [00582]  embedded image RM-56 [00583]  embedded image RM-57 [00584]  embedded image RM-58 [00585]  embedded image RM-59 [00586]  embedded image RM-60 [00587]  embedded image RM-61 [00588]  embedded image RM-62 [00589]  embedded image RM-63 [00590]  embedded image RM-64 [00591]  embedded image RM-65 [00592]  embedded image RM-66 [00593]  embedded image RM-67 [00594]  embedded image RM-68 [00595]  embedded image RM-69 [00596]  embedded image RM-70 [00597]  embedded image RM-71 [00598]  embedded image RM-72 [00599]  embedded image RM-73 [00600]  embedded image RM-74 [00601]  embedded image RM-75 [00602]  embedded image RM-76 [00603]  embedded image RM-77 [00604]  embedded image RM-78 [00605]  embedded image RM-79 [00606]  embedded image RM-80 [00607]  embedded image RM-81 [00608]  embedded image RM-82 [00609]  embedded image RM-83 [00610]  embedded image RM-84 [00611]  embedded image RM-85 [00612]  embedded image RM-86 [00613]  embedded image RM-87 [00614]  embedded image RM-88 [00615]  embedded image RM-89 [00616]  embedded image RM-90 [00617]  embedded image RM-91 [00618]  embedded image RM-92 [00619]  embedded image RM-93 [00620]  embedded image RM-94 [00621]  embedded image RM-95 [00622]  embedded image RM-96 [00623]  embedded image RM-97 [00624]  embedded image RM-98 [00625]  embedded image RM-99 [00626]  embedded image RM-100 [00627]  embedded image RM-101 [00628]  embedded image RM-102 [00629]  embedded image RM-103 [00630]  embedded image RM-104 [00631]  embedded image RM-105 [00632]  embedded image RM-106 [00633]  embedded image RM-107 [00634]  embedded image RM-108 [00635]  embedded image RM-109 [00636]  embedded image RM-110 [00637]  embedded image RM-111 [00638]  embedded image RM-112 [00639]  embedded image RM-113 [00640]  embedded image RM-114 [00641]  embedded image RM-115 [00642]  embedded image RM-116 [00643]  embedded image RM-117 [00644]  embedded image RM-118 [00645]  embedded image RM-119 [00646]  embedded image RM-120 [00647]  embedded image RM-121 [00648]  embedded image RM-122 [00649]  embedded image RM-123 [00650]  embedded image RM-124 [00651]  embedded image RM-125 [00652]  embedded image RM-126 [00653]  embedded image RM-127 [00654]  embedded image RM-128 [00655]  embedded image RM-129 [00656]  embedded image RM-130 [00657]  embedded image RM-131 [00658]  embedded image RM-132 [00659]  embedded image RM-133 [00660]  embedded image RM-134 [00661]  embedded image RM-135 [00662]  embedded image RM-136 [00663]  embedded image RM-137 [00664]  embedded image RM-138 [00665]  embedded image RM-139 [00666]  embedded image RM-140 [00667]  embedded image RM-141 [00668]  embedded image RM-142 [00669]  embedded image RM-143 [00670]  embedded image RM-144 [00671]  embedded image RM-145 [00672]  embedded image RM-146 [00673]  embedded image RM-147 [00674]  embedded image RM-148 [00675]  embedded image RM-149 [00676]  embedded image RM-150 [00677]  embedded image RM-151 [00678]  embedded image RM-152 [00679]  embedded image RM-153 [00680]  embedded image RM-154 [00681]  embedded image RM-

155 [00682]  RM-156 [00683]  RM-157 [00684] 
 RM-158 [00685]  RM-159 [00686]  RM-160 [00687]  RM-161 [00688]  RM-162 [00689] 
 RM-163 [00690]  RM-164 [00691]  RM-165 [00692]  RM-166 [00693]  RM-167 [00694] 
 RM-168 [00695]  RM-169 [00696]  RM-170 [00697]  RM-171 [00698]  RM-172 [00699] 
 RM-173 [00700]  RM-174 [00701]  RM-175 [00702]  RM-176 [00703]  RM-177 [00704] 
 RM-178 [00705]  RM-179 [00706]  RM-180 [00707]  RM-181 [00708]  RM-182 [00709] 
 RM-183

[0704] In a preferred embodiment, the mixtures according to the invention comprise one or more polymerizable compounds, preferably selected from the polymerizable compounds of the formulae RM-1 to RM-182. Of these, compounds RM-1, RM-4, RM-8, RM-17, RM-19, RM-35, RM-37, RM-39, RM-40, RM-41, RM-48, RM-52, RM-54, RM-57, RM-58, RM-64, RM-74, RM-76, RM-88, RM-91, RM-102, RM-103, RM-109, RM-116, RM-117, RM-120, RM-121, RM-122, RM-139, RM-140, RM-142, RM-143, RM-145, RM-146, RM-147, RM-149, RM-156 to RM-163, RM-169, RM-170 and RM-171 to RM-183 are particularly preferred.

TABLE-US-00008 TABLE F Table F shows self-alignment additives for vertical alignment which can be used in LC media for SA-VA and SA-FFS displays according to the present invention together with the polymerizable compound: [00710]  SA-1 [00711]

 SA-2 [00712]  SA-3 [00713]  SA-4 [00714]  SA-5 [00715]  SA-6 [00716]  SA-7 [00717]  SA-8 [00718]  SA-9 [00719]  SA-10 [00720]  SA-11 [00721]  SA-12 [00722] 
 SA-13 [00723]  SA-14 [00724]  SA-15 [00725]  SA-16 [00726]  SA-17 [00727]  SA-18 [00728]  SA-19 [00729]  SA-20 [00730] 
 SA-21 [00731]  SA-22 [00732]  SA-23 [00733]  SA-24 [00734]  SA-25 [00735]  SA-26 [00736]  SA-27 [00737]  SA-28 [00738] 
 SA-29 [00739]  SA-30 [00740]  SA-31 [00741]  SA-32 [00742]  SA-33 [00743]  SA-34 [00744]  SA-35 [00745]  SA-36 [00746] 
 SA-37 [00747]  SA-38 [00748]  SA-39 [00749]  SA-40 [00750]  SA-41 [00751]  SA-42 [00752]  SA-43 [00753]  SA-44 [00754] 
 SA-45 [00755]  SA-46 [00756]  SA-47 [00757]  SA-48

[0705] In a preferred embodiment, the LC media, SA-VA and SA-FFS displays according to the present invention comprise one or more SA additives selected from formulae SA-1 to SA-48, preferably from formulae SA-14 to SA-48, very preferably from formulae SA-20 to SA-34 and SA-44, in combination with one or more RMs.

EXAMPLES

[0706] The following examples explain the present invention without restricting it. However, they show the person skilled in the art preferred mixture concepts with compounds preferably to be employed and the respective concentrations thereof and combinations thereof with one another. In addition, the examples illustrate which properties and property combinations are accessible.

[0707] In addition, the following abbreviations and symbols are used: [0708] V.sub.0 threshold

voltage, capacitive [V] at 20° C., [0709] n.sub.e extraordinary refractive index at 20° C. and 589 nm, [0710] n.sub.o ordinary refractive index at 20° C. and 589 nm, [0711] Δn optical anisotropy at 20° C. and 589 nm, [0712] ϵ .sub. $\perp$ dielectric permittivity perpendicular to the director at 20° C. and 1 kHz, [0713] ϵ .sub. $\parallel$ dielectric permittivity parallel to the director at 20° C. and 1 kHz, [0714] $\Delta\epsilon$ dielectric anisotropy at 20° C. and 1 kHz, [0715] cl.p., T(N,I) clearing point [C], [0716] γ .sub.1 rotational viscosity at 20° C. [mPa.Math.s], [0717] K.sub.1 elastic constant, “splay” deformation at 20° C. [pN], [0718] K.sub.2 elastic constant, “twist” deformation at 20° C. [pN], [0719] K.sub.3 elastic constant, “bend” deformation at 20° C. [pN] [0720] K.sub.av average elastic constant at 20° C. [pN] defined here as $K_{\text{sub.av}} = (3/2 K_{\text{sub.1}} + K_{\text{sub.3}}) / 3 \approx (K_{\text{sub.1}} + K_{\text{sub.2}} + K_{\text{sub.3}}) / 3$, [0721] LTS low-temperature stability of the phase, determined in test cells, [0722] VHR voltage holding ratio.

[0723] Unless explicitly noted otherwise, all concentrations in the present application are quoted in percent by weight and relate to the corresponding mixture as a whole, comprising all solid or liquid-crystalline components, without solvents.

[0724] Unless explicitly noted otherwise, all temperature values indicated in the present application, such as, for example, for the melting point T(C,N), the transition from the smectic (S) to the nematic (N) phase T(S,N) and the clearing point T(N,I), are quoted in degrees Celsius (° C.). M.p. denotes melting point, cl.p.=clearing point. Furthermore, C=crystalline state, N=nematic phase, S=smectic phase and I=isotropic phase and RT=room temperature. The data between these symbols represent the transition temperatures.

[0725] All physical properties are and have been determined in accordance with “Merck Liquid Crystals, Physical Properties of Liquid Crystals”, Status November 1997, Merck KGaA, Germany, and apply for a temperature of 20° C., and Δn is determined at 589 nm and $\Delta\epsilon$ at 1 KHz, unless explicitly indicated otherwise in each case.

[0726] The term “threshold voltage” for the present invention relates to the capacitive threshold (V.sub.0), also known as the Freedericks threshold, unless explicitly indicated otherwise. In the examples, the optical threshold may also, as generally usual, be quoted for 10% relative contrast (V.sub.10).

[0727] Unless stated otherwise, the process of polymerizing the polymerizable compounds in the PSA displays as described above and below is carried out at a temperature where the LC medium exhibits a liquid crystal phase, preferably a nematic phase, and most preferably is carried out at room temperature.

[0728] Unless stated otherwise, methods of preparing test cells and measuring their electrooptical and other properties are carried out by the methods as described hereinafter or in analogy thereto.

[0729] A PSVA display or PSVA test cell used for photopolymerization and measurement of the tilt angles etc. consists of two plane-parallel glass outer plates at a separation of 3-4 μm unless stated otherwise, each of which has on the inside an electrode layer and a polyimide alignment layer on top, where the two polyimide layers are rubbed antiparallel to one another and effect a homeotropic edge alignment of the liquid-crystal molecules. The SAVA display or test cell has the same structure but wherein one or both polyimide layers are omitted.

[0730] Unless stated otherwise, for the PSA process the polymerizable LC medium is filled into PSVA test cells and exposed to UV light in a two-step process, the first step (UV1) for generating a tilt angle and the second step (UV2) for polymerizing any residual monomer that was not polymerized in the first step. In UV1 step a voltage is applied (typically 0.1V step and curing at DC 15V). In UV2 step no voltage is applied. As radiation source typically a fluorescent C-type UV lamp with a cut-off filter of 313 nm is used. The UV intensity and curing times are typically 0.1-2 mW/cm.sup.2 and 30-200s for UV1 step and 0.2-5 mW/cm.sup.2 and 30-120 min for UV2 step.

[0731] The tilt angle is determined using the Mueller Matrix Polarimeter “AxoScan” from Axometrics. A low value (i.e. a large deviation from the 90° angle) corresponds to a large tilt here.

[0732] Unless stated otherwise, the term “tilt angle” means the angle between the LC director and

the substrate, and “LC director” means in a layer of LC molecules with uniform orientation the preferred orientation direction of the optical main axis of the LC molecules, which corresponds, in case of calamitic, uniaxially positive birefringent LC molecules, to their molecular long axis.

Example 1

[0733] The nematic LC host mixture N1 is formulated as follows:

TABLE-US-00009 B(S)-2O-O5 3.0% Cl. p. [° C.]: 75.3 BCH-32 2.4% Δn [589 nm, 20° C.]: 0.1359 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6302 CC-3-V1 9.5% n.sub.o [589 nm, 20° C.]: 1.4943 CCP-3-1 6.4% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.0 CLU-3-F 0.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CPY-2-O2 13.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.6 CPY-3-O2 14.9% γ .sub.1 [mPa s, 20° C.]: 95 PP-1-2V1 9.3% K.sub.1 [pN, 20° C.]: 15.7 PY-3-O2 18% K.sub.3 [pN, 20° C.]: 16.7 PY-4-O2 1.5% PYP-2-3 1.0% Σ 100%

[0734] The mixture N1 contains the compound CLU-3-F of formula L1.

[0735] The chiral nematic LC host mixture Ch1 is formulated by adding 0.9% of the chiral dopant S-4011 to 99.1% of the host mixture N1.

[0736] The polymerizable LC mixture P1 is formulated as follows:

TABLE-US-00010 Ch1 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox ® 1076 0.001% ST-3a-1 0.015% Σ 100% [00758]  embedded image RM-1 (formula IBD1) [00759]  embedded image RM-145 (formula IA1) [00760]  embedded image RM-163 (formula IC45) [00761]  embedded image ST-3a-1

Example 2

[0737] The nematic LC host mixture N2 is formulated as follows:

TABLE-US-00011 B(S)-2O-O5 4.5% Cl. p. [° C.]: 74.9 CC-3-V 10.0% Δn [589 nm, 20° C.]: 0.1128 CC-3-V1 7.5% n.sub.e [589 nm, 20° C.]: 1.5992 CCH-23 10.0% n.sub.o [589 nm, 20° C.]: 1.4864 CCP-3-1 7.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.5 CCY-3-O2 10.0% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CLU-3-F 0.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 7.0 CPY-2-O2 2.0% γ .sub.1 [mPa s, 20° C.]: 112 CPY-3-O2 14.0% K.sub.1 [pN, 20° C.]: 14.5 CY-3-O2 14.0% K.sub.3 [pN, 20° C.]: 16.5 CY-3-O4 8.5% PGIY-2-O4 0.5% PP-1-2V1 7.5% PYP-2-3 4.0% Σ 100%

[0738] The mixture N2 contains the compound CLU-3-F of formula L1.

[0739] The polymerizable LC mixture P2 is formulated as follows:

TABLE-US-00012 N2 99.435% RM-64 0.050% RM-171 0.500% ST-3a-1 0.015% Σ 100% [00762]  embedded image RM-64 (formula IBD23) [00763]  embedded image RM-171 (formula IBD6)

Example 3

[0740] The nematic LC host mixture N3 is formulated as follows:

TABLE-US-00013 B(S)-2O-O4 4.0% Cl. p. [° C.]: 75.2 B(S)-2O-O5 5.0% Δn [589 nm, 20° C.]: 0.1341 B(S)-2O-O6 1.5% n.sub.e [589 nm, 20° C.]: 1.6290 BCH-32 8.5% n.sub.o [589 nm, 20° C.]: 1.4949 CC-3-V 28.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.9 CC-3-V1 7.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.5 CPY-3-O4 8.0% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.4 CPY-3-O2 15.0% γ .sub.1 [mPa s, 20° C.]: 89 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.5 PP-1-2V1 10.0% K.sub.3 [pN, 20° C.]: 16.0 PY-2-O2 11.0% PYP-2-3 1.0% Σ 100%

[0741] The polymerizable LC mixture P3 is formulated as follows:

TABLE-US-00014 N3 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 4

[0742] The nematic LC host mixture N4 is formulated as follows:

TABLE-US-00015 CC-3-V 28.0% Cl. p. [° C.]: 75 CC-3-V1 7.0% Δn [589 nm, 20° C.]: 0.1163 CCH-35 4.8% n.sub.e [589 nm, 20° C.]: 1.6045 COB(S)-2-O4 8.5% n.sub.o [589 nm, 20° C.]: 1.4882 CPY-2-O2 13.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.4 CPY-3-O2 14.9% ϵ .sub.∥ [1 kHz, 20° C.]: 3.7 CLU-3-F 0.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 7.1 PP-1-2V1 4.5% γ .sub.1 [mPa s, 20° C.]: 95 PY-2-O2

8.0% K.sub.1 [pN, 20° C.]: 13.9 PYP-2-3 1.5% K.sub.3 [pN, 20° C.]: 14.6 Σ 100%

[0743] The polymerizable LC mixture P4 is formulated as follows:

TABLE-US-00016 N4 99.684% RM-1 0.300% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 5

[0744] The nematic LC host mixture N5 is formulated as follows:

TABLE-US-00017 B(S)-2O-O5 2.0% Cl. p. [° C.]: 73.3 CC-3-V 20.0% Δn [589 nm, 20° C.]: 0.1158 CC-3-V1 10.0% n.sub.e [589 nm, 20° C.]: 1.6050 CCP-3-1 14.0% n.sub.o [589 nm, 20° C.]: 1.4892 COB(S)-2-O4 4.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.7 CPY-2-O2 4.3% ϵ .sub.∥ [1 kHz, 20° C.]: 3.8 CPY-3-O2 14.9% ϵ .sub.⊥ [1 kHz, 20° C.]: 7.5 CY-3-O2 11.0% γ .sub.1 [mPa s, 20° C.]: 101 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 13.9 PY-2-O2 12.0% K.sub.3 [pN, 20° C.]: 15.6 PY-3-O2 6.3% PYP-2-3 1.0% Σ 100%

[0745] The polymerizable LC mixture P5 is formulated as follows:

TABLE-US-00018 N5 99.685% RM-1 0.300% ST-3a-1 0.015% Σ 100%

Example 6

[0746] The nematic LC host mixture N6 is formulated as follows:

TABLE-US-00019 B(S)-2O-O4 2.0% Cl. p. [° C.]: 74.7 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1409 BCH-32 2.9% n.sub.e [589 nm, 20° C.]: 1.6402 CC-3-V 14.2% n.sub.o [589 nm, 20° C.]: 1.4993 CC-3-V1 9.5% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.8 CCP-3-1 14.9% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CLU-3-F 0.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.5 CPY-2-O2 4.6% γ .sub.1 [mPa s, 20° C.]: 100 CPY-3-O2 14.9% K.sub.1 [pN, 20° C.]: 14.4 PP-1-2V1 11.0% K.sub.3 [pN, 20° C.]: 16.5 PY-1-O2 10.0% PY-2-O2 10.0% PYP-2-3 2.5% Σ 100%

[0747] The polymerizable LC mixture P6 is formulated as follows:

TABLE-US-00020 N6 99.684% RM-19 0.300% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100% [00764]  RM-19 (formula IBD3)

Example 7

[0748] The nematic LC host mixture N7 is formulated as follows:

TABLE-US-00021 B(S)-2O-O4 1.1% Cl. p. [° C.]: 76.3 B(S)-2O-O5 5.0% Δn [589 nm, 20° C.]: 0.1554 B(S)-2O-O6 4.0% n.sub.e [589 nm, 20° C.]: 1.6584 BCH-32 9.5% n.sub.o [589 nm, 20° C.]: 1.5030 BCH-52 8.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.1 CC-3-V 8.8% ϵ .sub.∥ [1 kHz, 20° C.]: 3.7 CC-3-V1 10.0% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.8 CPY-3-O4 3.9% γ .sub.1 [mPa s, 20° C.]: 113 CPY-3-O2 14.9% K.sub.1 [pN, 20° C.]: 16.3 CLU-3-F 0.5% K.sub.3 [pN, 20° C.]: 16.5 PP-1-2V1 12.8% PY-1-O2 10.0% PY-2-O2 10.0% PYP-2-3 1.5% Σ 100%

[0749] The polymerizable LC mixture P7 is formulated as follows:

TABLE-US-00022 N7 99.685% RM-19 0.300% ST-3a-1 0.015% Σ 100%

Example 8

[0750] The nematic LC host mixture N8 is formulated as follows:

TABLE-US-00023 B(S)-2O-O5 3.5% Cl. p. [° C.]: 75.3 BCH-32 3.0% Δn [589 nm, 20° C.]: 0.1355 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6300 CC-3-V1 8.0% n.sub.o [589 nm, 20° C.]: 1.4945 CCP-3-1 7.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.1 CLU-3-F 0.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 COY-3-O2 2.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.7 CPY-2-O2 13.5% γ .sub.1 [mPa s, 20° C.]: 96 CPY-3-O2 14.0% K.sub.1 [pN, 20° C.]: 15.7 PP-1-2V1 10.0% K.sub.3 [pN, 20° C.]: 16.7 PY-3-O2 15.0% PY-4-O2 2.0% PYP-2-3 1.0% Σ 100%

[0751] The polymerizable LC mixture P8 is formulated as follows:

TABLE-US-00024 N8 99.484% RM-1 0.300% RM-145 0.200% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 9

[0752] The nematic LC host mixture N9 is formulated as follows:

TABLE-US-00025 B(S)-2O-O5 3.5% Cl. p. [° C.]: 75.2 BCH-32 3.0% Δn [589 nm, 20° C.]: 0.1354 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6299 CC-3-V1 8.0% n.sub.o [589 nm, 20° C.]:

1.4945 CCP-3-1 7.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.0 CLU-3-F 0.5% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.6 CEY-3-O2 2.5% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.6 CPY-2-O2 13.5% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 96 CPY-3-O2 14.0% $K_{\text{sub.1}}$ [pN, 20° C.]: 15.8 PP-1-2V1 10.0% $K_{\text{sub.3}}$ [pN, 20° C.]: 16.7 PY-3-O2 15.0% PY-4-O2 2.0% PYP-2-3 1.0% Σ 100%

[0753] The polymerizable LC mixture P9 is formulated as follows:

TABLE-US-00026 N9 99.485% RM-1 0.300% RM-145 0.200% ST-3b1 0.015% Σ 100%

[00765] ST-3b-1

Example 10

[0754] The nematic LC host mixture N10 is formulated as follows:

TABLE-US-00027 B(S)-2O-O5 3.0% Cl. p. [° C.]: 75.0 BCH-32 3.8% Δn [589 nm, 20° C.]: 0.1355 CC-3-V 20.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6315 CC-3-V1 9.5% $n_{\text{sub.o}}$ [589 nm, 20° C.]: 1.4960 CCOY-3-O2 2.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.0 CCP-3-1 5.0% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.7 CLU-3-F 0.5% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.7 CPY-2-O2 12.5% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 103 CPY-3-O2 13.9% $K_{\text{sub.1}}$ [pN, 20° C.]: 15.7 PP-1-2V1 9.3% $K_{\text{sub.3}}$ [pN, 20° C.]: 16.7 PY-3-O2 18.0% PY-4-O2 1.5% PYP-2-3 1.0% Σ 100%

[0755] The polymerizable LC mixture P10 is formulated as follows:

TABLE-US-00028 N10 99.484% RM-19 0.300% RM-145 0.200% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 11

[0756] The nematic LC host mixture N11 is formulated as follows:

TABLE-US-00029 B(S)-2O-O5 3.5% Cl. p. [° C.]: 76.5 BCH-32 3.0% Δn [589 nm, 20° C.]: 0.1328 CC-3-V 20.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6264 CC-3-V1 8.0% $n_{\text{sub.o}}$ [589 nm, 20° C.]: 1.4936 CCOY-3-O2 4.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.2 CCP-3-1 7.0% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.5 CLU-3-F 0.5% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.7 COY-3-O2 2.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 100 CPY-2-O2 12.0% $K_{\text{sub.1}}$ [pN, 20° C.]: 15.7 CPY-3-O2 12.0% $K_{\text{sub.3}}$ [pN, 20° C.]: 16.9 PP-1-2V1 10.0% PY-3-O2 12.0% PY-4-O2 5.0% PYP-2-3 1.0% Σ 100%

[0757] The polymerizable LC mixture P11 is formulated as follows:

TABLE-US-00030 N11 99.485% RM-19 0.300% RM-145 0.200% ST-8-1 0.015% Σ 100%

[00766] ST-8-1

Example 12

[0758] The nematic LC host mixture N12 is formulated as follows:

TABLE-US-00031 B(S)-2O-O5 3.0% Cl. p. [° C.]: 76.0 BCH-32 3.0% Δn [589 nm, 20° C.]: 0.1354 CC-3-V 20.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6292 CC-3-V1 9.5% $n_{\text{sub.o}}$ [589 nm, 20° C.]: 1.4938 CCP-3-1 4.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.0 CCY-3-O2 2.0% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.7 CLU-3-F 0.5% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.7 CPY-2-O2 13.6% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 104 CPY-3-O2 14.9% $K_{\text{sub.1}}$ [pN, 20° C.]: 15.8 CY-3-O2 2.0% $K_{\text{sub.3}}$ [pN, 20° C.]: 16.9 PP-1-2V1 10.0% PY-3-O2 16.5% PYP-2-3 1.0% Σ 100%

[0759] The polymerizable LC mixture P12 is formulated as follows:

TABLE-US-00032 N12 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 13

[0760] LC host mixture N13 is formulated as follows:

TABLE-US-00033 B(S)-2O-O5 3.0% Cl. p. [° C.]: 75.8 BCH-32 5.0% Δn [589 nm, 20° C.]: 0.1355 CC-3-V 20.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6295 CC-3-V1 9.5% $n_{\text{sub.o}}$ [589 nm, 20° C.]: 1.4940 CCOY-3-O2 2.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.0 CCP-3-1 2.5% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.7 CCY-3-O2 2.0% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.7 CLU-3-F 0.5% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 104 CPY-2-O2 11.5% $K_{\text{sub.1}}$ [pN, 20° C.]: 15.8 CPY-3-O2 14.0% $K_{\text{sub.3}}$ [pN, 20° C.]: 16.9 CY-3-O2 2.0% PP-1-2V1 10.0% PY-3-O2 16.5% PYP-2-3 1.5% Σ 100%

[0761] The polymerizable LC mixture P13 is formulated as follows:

TABLE-US-00034 N13 99.435% RM-1 0.300% RM-145 0.200% RM-163 0.050% ST-9-1 0.015% Σ 100% [00767]

Example 14

[0762] The nematic LC host mixture N14 is formulated as follows:

TABLE-US-00035 B(S)-2O-O5 3.5% Cl. p. [° C.]: 74.7 BCH-32 3.5% Δn [589 nm, 20° C.]: 0.1361 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6312 CC-3-V1 9.5% n.sub.o [589 nm, 20° C.]: 1.4951 CCP-3-1 5.5% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.9 CLU-3-F 0.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CPY-2-O2 13.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.5 CPY-3-O2 15.0% γ .sub.1 [mPa s, 20° C.]: 99 PP-1-2V1 5.0% K.sub.1 [pN, 20° C.]: 15.4 PP-1-3 5.0% K.sub.3 [pN, 20° C.]: 16.2 PY-3-O2 18.0% PYP-2-3 1.0% Σ 100%

[0763] The polymerizable LC mixture P14 is formulated as follows:

TABLE-US-00036 N14 99.434% RM-19 0.300% RM-145 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 15

[0764] The nematic LC host mixture N15 is formulated as follows:

TABLE-US-00037 B(S)-2O-O5 5.0% Cl. p. [° C.]: 76.3 BCH-32 5.5% Δn [589 nm, 20° C.]: 0.1363 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6313 CC-3-V1 8.0% n.sub.o [589 nm, 20° C.]: 1.4950 CCP-3-1 5.5% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.1 CLU-3-F 0.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CPY-2-O2 13.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.7 CPY-3-O2 15.0% γ .sub.1 [mPa s, 20° C.]: 104 PP-1-3 8.0% K.sub.1 [pN, 20° C.]: 15.2 PY-3-O2 12.0% K.sub.3 [pN, 20° C.]: 15.5 PY-4-O2 6.0% PYP-2-3 1.0% Σ 100%

[0765] The polymerizable LC mixture P15 is formulated as follows:

TABLE-US-00038 N15 99.435% RM-19 0.300% RM-145 0.200% RM-163 0.050% ST-3a-1 0.015% Σ 100%

Example 16

[0766] The nematic LC host mixture N16 is formulated as follows:

TABLE-US-00039 B(S)-2O-O5 3.0% Cl. p. [° C.]: 75.8 BCH-32 2.0% Δn [589 nm, 20° C.]: 0.1357 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6300 CC-3-V1 9.5% n.sub.o [589 nm, 20° C.]: 1.4943 CCP-3-1 6.5% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.0 CLU-3-F 0.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CPY-2-O2 13.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.6 CPY-3-O2 15.0% γ .sub.1 [mPa s, 20° C.]: 96 PP-1-2V1 9.0% K.sub.1 [pN, 20° C.]: 15.7 PY-3-O2 18.0% K.sub.3 [pN, 20° C.]: 16.7 PY-4-O2 1.5% PYP-2-3 1.0% CLP-V-1 0.5 Σ 100%

[0767] The polymerizable LC mixture P16 is formulated as follows:

TABLE-US-00040 N16 99.484% RM-1 0.300% RM-171 0.200% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 17

[0768] The nematic LC host mixture N17 is formulated as follows:

TABLE-US-00041 B(S)-2O-O5 3.0% Cl. p. [° C.]: 75.6 BCH-32 2.0% Δn [589 nm, 20° C.]: 0.1355 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6299 CC-3-V1 9.5% n.sub.o [589 nm, 20° C.]: 1.4944 CCP-3-1 6.5% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.9 CLU-3-F 0.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.7 CPY-2-O2 13.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.6 CPY-3-O2 15.0% γ .sub.1 [mPa s, 20° C.]: 96 PP-1-2V1 9.0% K.sub.1 [pN, 20° C.]: 15.7 PY-3-O2 18.0% K.sub.3 [pN, 20° C.]: 16.7 PY-4-O2 1.5% PYP-2-3 1.0% CLP-3-T 0.5% Σ 100%

[0769] The polymerizable LC mixture P17 is formulated as follows:

TABLE-US-00042 N17 99.485% RM-1 0.300% RM-171 0.200% ST-3a-1 0.015% Σ 100%

Example 18

[0770] The nematic LC host mixture N18 is formulated as follows:

TABLE-US-00043 B(S)-2O-O4 4.0% Cl. p. [° C.]: 75.3 B(S)-2O-O5 5.0% Δn [589 nm, 20° C.]: 0.1340 B(S)-2O-O6 1.5% n.sub.e [589 nm, 20° C.]: 1.6290 BCH-32 8.0% n.sub.o [589 nm, 20°

C.]: 1.4950 CC-3-V 28.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -2.9 CC-3-V1 7.5% $\epsilon_{\text{sub.}}$ || [1 kHz, 20° C.]: 3.5
CPY-3-O4 8.0% $\epsilon_{\text{sub.}}$ $\perp$ [1 kHz, 20° C.]: 6.4 CPY-3-O2 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 89 CLU-
3-F 0.5% K.sub.1 [pN, 20° C.]: 15.5 PP-1-2V1 10.0% K.sub.3 [pN, 20° C.]: 16.0 PY-2-O2
11.0% PYP-2-3 1.0% CLP-V-1 0.5% Σ 100%

[0771] The polymerizable LC mixture P18 is formulated as follows:

TABLE-US-00044 N18 99.484% RM-19 0.300% RM-171 0.200% Irganox ®1076 0.001% ST-3a-
1 0.015% Σ 100%

Example 19

[0772] The nematic LC host mixture N19 is formulated as follows:

TABLE-US-00045 B(S)-2O-O4 4.0% Cl. p. [° C.]: 75.3 B(S)-2O-O5 5.0% Δn [589 nm, 20° C.]:
0.1340 B(S)-2O-O6 1.5% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6290 BCH-32 8.0% $n_{\text{sub.o}}$ [589 nm, 20°
C.]: 1.4950 CC-3-V 28.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -2.9 CC-3-V1 7.5% $\epsilon_{\text{sub.}}$ || [1 kHz, 20° C.]: 3.5
CPY-3-O4 8.0% $\epsilon_{\text{sub.}}$ $\perp$ [1 kHz, 20° C.]: 6.4 CPY-3-O2 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 89 CLU-
3-F 0.5% K.sub.1 [pN, 20° C.]: 15.5 PP-1-2V1 10.0% K.sub.3 [pN, 20° C.]: 16.0 PY-2-O2
11.0% PYP-2-3 1.0% CLP-3-T 0.5% Σ 100%

[0773] The polymerizable LC mixture P19 is formulated as follows:

TABLE-US-00046 N19 99.485% RM-19 0.300% RM-171 0.200% ST-3a-1 0.015% Σ 100%

Example 20

[0774] The nematic LC host mixture N20 is formulated as follows:

TABLE-US-00047 B(S)-2O-O4 2.0% Cl. p. [° C.]: 75.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]:
0.1411 BCH-32 3.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6404 CC-3-V 14.0% $n_{\text{sub.o}}$ [589 nm, 20° C.]:
1.4993 CC-3-V1 9.5% $\Delta\epsilon$ [1 kHz, 20° C.]: -2.7 CCP-V-1 15.0% $\epsilon_{\text{sub.}}$ || [1 kHz, 20° C.]: 3.7
CLU-3-F 0.5% $\epsilon_{\text{sub.}}$ $\perp$ [1 kHz, 20° C.]: 6.4 CPY-2-O2 4.5% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 100 CPY-3-
O2 14.5% K.sub.1 [pN, 20° C.]: 14.4 PP-1-2V1 11.0% K.sub.3 [pN, 20° C.]: 16.6 PY-1-O2
10.0% PY-2-O2 10.0% PYP-2-3 2.5% CLP-V-1 0.5 Σ 100%

[0775] The polymerizable LC mixture P20 is formulated as follows:

TABLE-US-00048 N20 99.434% RM-1 0.300% RM-171 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3c-1 0.015% Σ 100% [00768] ST-3c-1

Example 21

[0776] The nematic LC host mixture N21 is formulated as follows:

TABLE-US-00049 B(S)-2O-O4 2.0% Cl. p. [° C.]: 75.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]:
0.1411 BCH-32 3.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6404 CC-3-V 14.0% $n_{\text{sub.o}}$ [589 nm, 20° C.]:
1.4993 CC-3-V1 9.5% $\Delta\epsilon$ [1 kHz, 20° C.]: -2.7 CCP-V-1 15.0% $\epsilon_{\text{sub.}}$ || [1 kHz, 20° C.]: 3.7
CLU-3-F 0.5% $\epsilon_{\text{sub.}}$ $\perp$ [1 kHz, 20° C.]: 6.4 CPY-2-O2 4.5% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 100 CPY-3-
O2 14.5% K.sub.1 [pN, 20° C.]: 14.4 PP-1-2V1 11.0% K.sub.3 [pN, 20° C.]: 16.6 PY-1-O2
10.0% PY-2-O2 10.0% PYP-2-3 2.5% CLP-3-T 0.5 Σ 100%

[0777] The polymerizable LC mixture P21 is formulated as follows:

TABLE-US-00050 N21 99.435% RM-1 0.300% RM-171 0.200% RM-163 0.050% ST-3b-1
0.015% Σ 100%

Example 22

[0778] The nematic LC host mixture N22 is formulated as follows:

TABLE-US-00051 B(S)-2O-O5 4.5% Cl.p. [° C.]: 75.4 CC-3-V 10.0% Δn [589 nm, 20° C.]:
0.1135 CC-3-V1 7.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.5995 CCH-23 10.0% $n_{\text{sub.o}}$ [589 nm, 20° C.]:
1.4860 CCP-3-1 7.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.5 CCY-3-O2 10.0% $\epsilon_{\text{sub.}}$ || [1 kHz, 20° C.]: 3.6
CLU-3-F 0.5% $\epsilon_{\text{sub.}}$ $\perp$ [1 kHz, 20° C.]: 7.0 CPY-2-O2 2.0% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 114 CPY-3-
O2 14.0% K.sub.1 [pN, 20° C.]: 14.5 CY-3-O2 14.0% K.sub.3 [pN, 20° C.]: 16.5 CY-3-O4 8.5%
PGIY-2-O4 0.5% PP-1-2V1 7.5% PYP-2-3 4.0 CLP-V-1 0.5 Σ 100%

[0779] The polymerizable LC mixture P22 is formulated as follows:

TABLE-US-00052 N22 99.434% RM-19 0.300% RM-171 0.200% RM-163 0.050% Irganox

®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 23

[0780] The nematic LC host mixture N23 is formulated as follows:

TABLE-US-00053 B(S)-2O-O5 4.5% Cl.p. [° C.]: 75.4 CC-3-V 10.0% Δn [589 nm, 20° C.]: 0.1135 CC-3-V1 7.0% n.sub.e [589 nm, 20° C.]: 1.5995 CCH-23 10.0% n.sub.o [589 nm, 20° C.]: 1.4860 CCP-3-1 7.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.5 CCY-3-O2 10.0% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CLU-3-F 0.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 7.0 CPY-2-O2 2.0% γ .sub.1 [mPas, 20° C.]: 114 CPY-3-O2 14.0% K.sub.1 [pN, 20° C.]: 14.5 CY-3-O2 14.0% K.sub.3 [pN, 20° C.]: 16.5 CY-3-O4 8.5% PGIY-2-O4 0.5% PP-1-2V1 7.5% PYP-2-3 4.0 CLP-3-T 0.5 Σ 100%

[0781] The polymerizable LC mixture P23 is formulated as follows:

TABLE-US-00054 N23 99.435% RM-19 0.300% RM-171 0.200% RM-163 0.050% ST-3b-1 0.015% Σ 100%

Example 24

[0782] The nematic LC host mixture N24 is formulated as follows:

TABLE-US-00055 B(S)-2O-O5 3.0% Cl.p. [° C.]: 75.3 BCH-32 2.4% Δn [589 nm, 20° C.]: 0.1359 CC-3-V 20.0% n.sub.e [589 nm, 20° C.]: 1.6302 CC-3-V1 9.5% n.sub.o [589 nm, 20° C.]: 1.4943 CCP-3-1 6.4% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.0 CLU-3-1 0.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CPY-2-O2 13.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.6 CPY-3-O2 14.9% γ .sub.1 [mPas, 20° C.]: 95 PP-1-2V1 9.3% K.sub.1 [pN, 20° C.]: 15.7 PY-3-O2 18.0% K.sub.3 [pN, 20° C.]: 16.7 PY-4-O2 1.5% PYP-2-3 1.0% Σ 100%

[0783] The mixture N24 contains the compound CLU-3-1 of formula L1.

[0784] The polymerizable LC mixture P24 is formulated as follows:

TABLE-US-00056 N24 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3c-1 0.015% Σ 100%

Example 25

[0785] The nematic LC host mixture N25 is formulated as follows:

TABLE-US-00057 B(S)-2O-O4 4.0% Cl.p. [° C.]: 75.4 B(S)-2O-O5 5.0% Δn [589 nm, 20° C.]: 0.1238 BCH-32 6.6% $\Delta\epsilon$ [1 kHz, 20° C.]: -1.9 CC-3-V 34.0% ϵ .sub.∥ [1 kHz, 20° C.]: 3.3 CC-3-V1 9.8% ϵ .sub.⊥ [1 kHz, 20° C.]: 5.2 CCP-V-1 1.0% γ .sub.1 [mPa s, 20° C.]: 72 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.7 CPY-3-O2 15.0% V.sub.0 [V, 20° C.]: 3.03 CPY-3-O4 5.0% PP-1-2V1 12.6% PY-2-O2 5.0% PYP-2-3 1.5% Σ 100%

[0786] The polymerizable LC mixture P25 is formulated as follows:

TABLE-US-00058 N25 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3c-1 0.015% Σ 100%

Example 26

[0787] The nematic LC host mixture N26 is formulated as follows:

TABLE-US-00059 B(S)-2O-O4 2.0% Cl.p. [° C.]: 74.9 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1230 CC-3-V 23.0% n.sub.e [589 nm, 20° C.]: 1.6132 CC-3-V1 9.6% n.sub.o [589 nm, 20° C.]: 1.4902 CC-4-V1 7.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -2.2 CCP-V-1 7.5% ϵ .sub.∥ [1 kHz, 20° C.]: 3.4 CLU-3-F 0.5% ϵ .sub.⊥ [1 kHz, 20° C.]: 5.6 CPY-3-O2 15.0% γ .sub.1 [mPas, 20° C.]: 80 CPY-3-O4 8.2% K.sub.1 [pN, 20° C.]: 14.9 PP-1-2V1 10.0% K.sub.3 [pN, 20° C.]: 15.7 PY-2-O2 6.0% K.sub.3/K.sub.1 [20° C.] 1.05 PY-3-O2 7.2% V.sub.0 [V, 20° C.]: 3.03 PYP-2-3 1.0% Σ 100%

[0788] The polymerizable LC mixture P26 is formulated as follows:

TABLE-US-00060 N26 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 27

[0789] The nematic LC host mixture N27 is formulated as follows:

TABLE-US-00061 B(S)-2O-O4 3.0% Cl.p. [° C.]: 74.6 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1160 CC-3-V 22.0% n.sub.e [589 nm, 20° C.]: 1.6046 CC-3-V1 9.0% n.sub.o [589 nm, 20° C.]:

1.4886 CCP-3-1 11.4% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.6 CPY-2-O2 10.0% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.8
CPY-3-O2 15.0% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 7.4 CY-3-O2 13.7% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 97 CLU-3-
F 0.5% K.sub.1 [pN, 20° C.]: 14.0 PP-1-2V1 2.4% K.sub.3 [pN, 20° C.]: 15.6 PY-2-O2 10.0%
K.sub.3/K.sub.1 [20° C.] 1.11 V.sub.0 [V, 20° C.]: 2.20 Σ 100%

[0790] The polymerizable LC mixture P27 is formulated as follows:

TABLE-US-00062 N27 99.454% RM-1 0.300% RM-145 0.200% RM-163 0.030% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 28

[0791] The nematic LC host mixture N28 is formulated as follows:

TABLE-US-00063 B(S)-2O-O4 3.0% Cl.p. [° C.]: 75.2 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]:
0.1161 CC-3-V 22.1% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6050 CC-3-V1 9.5% $n_{\text{sub.o}}$ [589 nm, 20° C.]:
1.4889 CCP-3-1 11.3% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.5 CPY-2-O2 9.8% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.8
CPY-3-O2 15.3% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 7.3 CY-3-O2 12.8% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 97 CLU-
3-F 0.5% K.sub.1 [pN, 20° C.]: 14.2 PP-1-2V1 2.7% K.sub.3 [pN, 20° C.]: 15.8 PY-2-O2 10.0%
K.sub.3/K.sub.1 [20° C.] 1.11 V.sub.0 [V, 20° C.]: 2.23 Σ 100%

[0792] The polymerizable LC mixture P28 is formulated as follows:

TABLE-US-00064 N28 99.454% RM-1 0.300% RM-145 0.200% RM-163 0.030% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 29

[0793] The nematic LC host mixture N29 is formulated as follows:

TABLE-US-00065 B(S)-2O-O4 2.0% Cl.p. [° C.]: 74.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]:
0.1347 BCH-32 1.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6318 CC-3-V 17.2% $n_{\text{sub.o}}$ [589 nm, 20° C.]:
1.4971 CC-3-V1 9.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -2.8 CCP-V-1 15.0% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.6
CLU-3-F 0.5% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.4 CPY-2-O2 6.6% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 91 CPY-3-O2
14.9% K.sub.1 [pN, 20° C.]: 14.8 PP-1-2V1 10.0% K.sub.3 [pN, 20° C.]: 15.9 PY-2-O2 9.4%
K.sub.3/K.sub.1 [20° C.] 1.07 PY-3-O2 9.9% V.sub.0 [V, 20° C.]: 2.53 PYP-2-3 1.5% Σ 100%

[0794] The polymerizable LC mixture P29 is formulated as follows:

TABLE-US-00066 N29 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 30

[0795] The nematic LC host mixture N30 is formulated as follows:

TABLE-US-00067 B(S)-2O-O5 3.0% Cl.p. [° C.]: 75.4 BCH-32 2.4% Δn [589 nm, 20° C.]: 0.1359
CC-3-V 20.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6302 CC-3-V1 9.5% $n_{\text{sub.o}}$ [589 nm, 20° C.]: 1.4943
CCP-3-1 6.4% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.0 CLG-3-F 0.5% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.6 CPY-2-O2
13.5% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.6 CPY-3-O2 14.9% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 95 PP-1-2V1 9.3%
K.sub.1 [pN, 20° C.]: 15.7 PY-3-O2 18.0% K.sub.3 [pN, 20° C.]: 16.7 PY-4-O2 1.5% PYP-2-3
1.0% Σ 100%

[0796] The mixture N30 contains the compound CLG-3-F of formula L1.

[0797] The polymerizable LC mixture P30 is formulated as follows:

TABLE-US-00068 N30 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015%

Example 31

[0798] The nematic LC host mixture N31 is formulated as follows:

TABLE-US-00069 B(S)-2O-O5 3.0% Cl.p. [° C.]: 75.1 BCH-32 2.5% Δn [589 nm, 20° C.]: 0.1363
CC-3-V 20.0% $n_{\text{sub.e}}$ [589 nm, 20° C.]: 1.6305 CC-3-V1 9.5% $n_{\text{sub.o}}$ [589 nm, 20° C.]: 1.4942
CCP-3-1 6.0% $\Delta\epsilon$ [1 kHz, 20° C.]: -3.0 CLG-3-T 0.5% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.6 CPY-2-O2
13.5% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.6 CPY-3-O2 15.0% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 95 PP-1-2V1 9.5%
K.sub.1 [pN, 20° C.]: 15.7 PY-3-O2 18.0% K.sub.3 [pN, 20° C.]: 16.6 PY-4-O2 1.5% PYP-2-3
1.0% Σ 100%

[0799] The mixture N31 contains the compound CLG-3-T of formula L1.

[0800] The polymerizable LC mixture P31 is formulated as follows:

TABLE-US-00070 N31 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015%

Example 32

[0801] The nematic LC host mixture N32 is formulated as follows:

TABLE-US-00071 B(S)-2O-O4 2.0% Cl.p. [° C.]: 75.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]:
0.1354 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.9 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 3.6 CC-3-
V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: 6.5 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 93 CLU-3-F 0.5%
K.sub.1 [pN, 20° C.]: 14.8 APY-3-O2 6.6% K.sub.3 [pN, 20° C.]: 16.3 CPY-3-O2 14.9% PP-1-
2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0802] The polymerizable LC mixture P32 is formulated as follows:

TABLE-US-00072 N32 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 33

[0803] The nematic LC host mixture N33 is formulated as follows:

TABLE-US-00073 B(P)-2O-O3 2.0% Cl.p. [° C.]: 75.5 B(P)-2O-O4 3.0% Δn [589 nm, 20° C.]:
0.1361 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 3.6 CC-3-
V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: 6.2 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 93 CLU-3-F 0.5%
K.sub.1 [pN, 20° C.]: 14.7 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.8 CPY-3-O2 14.9% PP-1-
2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0804] The polymerizable LC mixture P33 is formulated as follows:

TABLE-US-00074 N33 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 34

[0805] The nematic LC host mixture N34 is formulated as follows:

TABLE-US-00075 B(A)-2O-O2 2.0% Cl.p. [° C.]: 74.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]:
0.1336 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.7 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 3.6 CC-3-
V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: 6.3 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 92 CLU-3-F 0.5%
K.sub.1 [pN, 20° C.]: 14.6 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.9 CPY-3-O2 14.9% PP-1-
2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0806] The polymerizable LC mixture P34 is formulated as follows:

TABLE-US-00076 N34 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 35

[0807] The nematic LC host mixture N35 is formulated as follows:

TABLE-US-00077 B(S)-2O-O1(c5) 2.0% Cl.p. [° C.]: 74.0 B(S)-4O-O1(c5) 3.0% Δn [589 nm, 20°
C.]: 0.1339 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.9 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 3.6
CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: 6.5 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 95 CLU-3-F
0.5% K.sub.1 [pN, 20° C.]: 14.7 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.7 CPY-3-O2 14.9%
PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0808] The polymerizable LC mixture P35 is formulated as follows:

TABLE-US-00078 N35 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox
®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 36

[0809] The nematic LC host mixture N36 is formulated as follows:

TABLE-US-00079 B(S)-1V1O-O1(c5) 2.0% Cl.p. [° C.]: 74.5 B(S)-4O-O5 3.0% Δn [589 nm, 20°
C.]: 0.1343 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.8 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 3.6
CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: 6.4 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPas, 20° C.]: 93 CLU-3-F

0.5% K.sub.1 [pN, 20° C.]: 14.5 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.7 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0810] The polymerizable LC mixture P36 is formulated as follows:

TABLE-US-00080 N36 99.435% RM-1 0.300% RM-145 0.200% RM-163 0.050% ST-3a-1 0.015% Σ 100%

Example 37

[0811] The nematic LC host mixture N37 is formulated as follows:

TABLE-US-00081 B(S)-2O-O4 2.0% Cl.p. [° C.]: 75.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1349 CPP-3-2V1 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.8 CC-3-V 17.2% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CC-3-V1 9.0% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.4 CCP-V-1 15.0% γ .sub.1 [mPa s, 20° C.]: 92 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 14.9 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 16.0 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0812] The polymerizable LC mixture P37 is formulated as follows:

TABLE-US-00082 N37 99.484% RM-1 0.300% RM-35 0.200% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 38

[0813] The nematic LC host mixture N38 is formulated as follows:

TABLE-US-00083 B(S)-2O-O4 2.0% Cl.p. [° C.]: 73.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1287 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -3.0 CC-3-V 17.2% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CC-3-V1 9.0% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.6 CCP-V-1 15.0% γ .sub.1 [mPa s, 20° C.]: 97 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 14.6 CAIY-3-O2 6.6% K.sub.3 [pN, 20° C.]: 16.5 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0814] The polymerizable LC mixture P38 is formulated as follows:

TABLE-US-00084 N38 99.485% RM-1 0.300% RM-35 0.200% ST-3a-1 0.015% Σ 100%

Example 39

[0815] The nematic LC host mixture N39 is formulated as follows:

TABLE-US-00085 B(S)-2O-O4 2.0% Cl.p. [° C.]: 70.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1351 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.8 CC-3-V 9.2% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CC-1-2V1 9.0% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.4 CCP-V-1 15.0% γ .sub.1 [mPa s, 20° C.]: 91 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 14.2 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.4 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% CC-V-V1 8.0% Σ 100%

[0816] The polymerizable LC mixture P39 is formulated as follows:

TABLE-US-00086 N39 99.484% RM-19 0.300% RM-35 0.200% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 40

[0817] The nematic LC host mixture N40 is formulated as follows:

TABLE-US-00087 B(S)-2O-O4 2.0% Cl.p. [° C.]: 74.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1303 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.9 CC-3-V 11.2% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CC-3-V1 9.0% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.5 CCP-V-1 15.0% γ .sub.1 [mPa s, 20° C.]: 98 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 16.0 CCEY-3-O2 6.6% K.sub.3 [pN, 20° C.]: 16.9 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% CCH-25 6.0% Σ 100%

[0818] The polymerizable LC mixture P40 is formulated as follows:

TABLE-US-00088 N40 99.485% RM-19 0.300% RM-35 0.200% ST-3a-1 0.015% Σ 100%

Example 41

[0819] The nematic LC host mixture N41 is formulated as follows:

TABLE-US-00089 B(S)-2O-O4 2.0% Cl.p. [° C.]: 74.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1353 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: -2.8 CC-3-V 17.2% ϵ .sub.∥ [1 kHz, 20° C.]: 3.6 CC-3-V1 9.0% ϵ .sub.⊥ [1 kHz, 20° C.]: 6.4 CCP-V-1 8.0% γ .sub.1 [mPa s, 20° C.]: 95 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.3 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 16.3 CPY-3-O2 14.9% PP-1-

2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% CCP-1-2V1 7.0% Σ 100%

[0820] The polymerizable LC mixture P41 is formulated as follows:

TABLE-US-00090 N41 99.434% RM-1 0.300% RM-35 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3c-1 0.015% Σ 100%

Example 42

[0821] The nematic LC host mixture N42 is formulated as follows:

TABLE-US-00091 B(S)-2O-O4 2.0% Cl.p. [$^{\circ}$ C.]: 75.5 B(S)-2O-O5 3.0% Δn [589 nm, 20 $^{\circ}$ C.]: 0.1363 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20 $^{\circ}$ C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20 $^{\circ}$ C.]: 6.7 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20 $^{\circ}$ C.]: -3.1 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20 $^{\circ}$ C.]: 88 CLU-3-F 0.5% K.sub.1 [pN, 20 $^{\circ}$ C.]: 14.9 CPY-2-O2 6.6% K.sub.3 [pN, 20 $^{\circ}$ C.]: 15.1 CPY-V-O2 7.9% CPY-V-O4 7.0% PP-1-2V1 10.0% PY-V2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0822] The polymerizable LC mixture P42 is formulated as follows:

TABLE-US-00092 N42 99.435% RM-1 0.300% RM-35 0.200% RM-163 0.050% ST-3c-1 0.015% Σ 100%

Example 43

[0823] The nematic LC host mixture N43 is formulated as follows:

TABLE-US-00093 B(S)-2O-O4 2.0% Cl.p. [$^{\circ}$ C.]: 78.5 B(S)-2O-O5 3.0% Δn [589 nm, 20 $^{\circ}$ C.]: 0.1322 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20 $^{\circ}$ C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20 $^{\circ}$ C.]: 6.6 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20 $^{\circ}$ C.]: -3.0 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20 $^{\circ}$ C.]: 107 CLU-3-F 0.5% K.sub.1 [pN, 20 $^{\circ}$ C.]: 16.8 CPY-2-O2 6.6% K.sub.3 [pN, 20 $^{\circ}$ C.]: 16.9 CPY-3-O2 14.9% PP-1-2V1 10.0% LY-(c5)-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0824] The polymerizable LC mixture P43 is formulated as follows:

TABLE-US-00094 N43 99.434% RM-19 0.300% RM-35 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 44

[0825] The nematic LC host mixture N44 is formulated as follows:

TABLE-US-00095 B(S)-2O-O4 2.0% Cl.p. [$^{\circ}$ C.]: 79.5 B(S)-2O-O5 3.0% Δn [589 nm, 20 $^{\circ}$ C.]: 0.1359 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20 $^{\circ}$ C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20 $^{\circ}$ C.]: 6.7 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20 $^{\circ}$ C.]: -3.1 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20 $^{\circ}$ C.]: 111 CLU-3-F 0.5% K.sub.1 [pN, 20 $^{\circ}$ C.]: 16.0 CPY-2-O2 6.6% K.sub.3 [pN, 20 $^{\circ}$ C.]: 16.3 CPY-3-O2 14.9% PP-1-2V1 10.0% YG-2O-O4 4.4% YG-2O-O4 5.0% YG-2O-O4 4.0% PY-3-O2 5.9% PYP-2-3 1.5% Σ 100%

[0826] The polymerizable LC mixture P44 is formulated as follows:

TABLE-US-00096 N44 99.435% RM-19 0.300% RM-35 0.200% RM-163 0.050% ST-3a-1 0.015% Σ 100%

Example 45

[0827] The nematic LC host mixture N45 is formulated as follows:

TABLE-US-00097 B(S)-2O-O4 2.0% Cl.p. [$^{\circ}$ C.]: 74.5 B(S)-2O-O5 3.0% Δn [589 nm, 20 $^{\circ}$ C.]: 0.1348 BCH-31 1.0% $\Delta \epsilon$ [1 kHz, 20 $^{\circ}$ C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20 $^{\circ}$ C.]: 6.4 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20 $^{\circ}$ C.]: -2.8 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20 $^{\circ}$ C.]: 91 CLU-3-F 0.5% K.sub.1 [pN, 20 $^{\circ}$ C.]: 14.9 CPY-2-O2 6.6% K.sub.3 [pN, 20 $^{\circ}$ C.]: 15.9 CPY-3-O2 14.9% V.sub.0 [V, 20 $^{\circ}$ C.]: 2.53 PP-1-2V1 10.0% LTS bulk [h, -20 $^{\circ}$ C.]: 1000 PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0828] The polymerizable LC mixture P45 is formulated as follows:

TABLE-US-00098 N45 99.684% RM-35 0.300% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 46

[0829] The nematic LC host mixture N46 is formulated as follows:

TABLE-US-00099 B(S)-2O-O4 2.0% Cl.p. [$^{\circ}$ C.]: 78.0 B(S)-2O-O5 3.0% Δn [589 nm, 20 $^{\circ}$ C.]:

0.1334 BCH-32 1.0% $\Delta\epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 6.4 CC-3-V1 9.0% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: -2.8 CCP-V-1 11.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 93 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.4 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 16.0 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% CCC-3-V 4.0% Σ 100%

[0830] The polymerizable LC mixture P46 is formulated as follows:

TABLE-US-00100 N46 99.685% RM-35 0.300% ST-3a-1 0.015% Σ 100%

Example 47

[0831] The nematic LC host mixture N47 is formulated as follows:

TABLE-US-00101 B(S)-2O-O4 2.0% Cl.p. [° C.]: 76.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1178 BCH-32 1.0% $\Delta\epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 7.0 CC-3-V1 9.0% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: -3.4 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 107 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.7 CCOY-2-O(c5) 6.6% K.sub.3 [pN, 20° C.]: 16.7 CCOY-3-O1(c5) 6.9% CPY-3-O2 8.0% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0832] The polymerizable LC mixture P47 is formulated as follows:

TABLE-US-00102 N47 99.634% RM-35 0.300% RM-64 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 48

[0833] The nematic LC host mixture N48 is formulated as follows:

TABLE-US-00103 B(S)-2O-O4 2.0% Cl.p. [° C.]: 77.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1280 BCH-32 1.0% $\Delta\epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 6.4 CC-3-V1 9.0% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: -2.8 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 104 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 16.5 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 16.4 CPY-3-O2 14.9% PP-1-2V1 10.0% CY-(c5)-O4 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0834] The polymerizable LC mixture P48 is formulated as follows:

TABLE-US-00104 N48 99.635% RM-35 0.300% RM-64 0.050% ST-3b-1 0.015% Σ 100%

Example 49

[0835] The nematic LC host mixture N49 is formulated as follows:

TABLE-US-00105 B(S)-2O-O4 2.0% Cl.p. [° C.]: 77.5 B(S)-2O-O4 2.0% Δn [589 nm, 20° C.]: 74.5 B(S)-2O-O5 3.0% $\Delta\epsilon$ [1 kHz, 20° C.]: 0.1346 BCH-32 1.0% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: 6.4 CC-3-V1 9.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: -2.8 CCP-V-1 15.0% 92 CLU-3-F 0.5% CPY-2-O2 6.6% CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-(c5) 1.5% Σ 100%

[0836] The polymerizable LC mixture P49 is formulated as follows:

TABLE-US-00106 N49 99.484% RM-171 0.500% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 50

[0837] The nematic LC host mixture N50 is formulated as follows:

TABLE-US-00107 B(S)-2O-O4 2.0% Cl.p. [° C.]: 74.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1345 BCH-32 1.0% $\Delta\epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 6.4 CC-3-V1 9.0% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: -2.8 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 92 CLU-3-F 0.5% CPY-2-O2 6.6% CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-1(c3) 1.5% Σ 100%

[0838] The polymerizable LC mixture P50 is formulated as follows:

TABLE-US-00108 N50 99.484% RM-171 0.500% ST-3a-1 0.015% Σ 100%

Example 51

[0839] The nematic LC host mixture N51 is formulated as follows:

TABLE-US-00109 B(S)-2O-O4 2.0% Cl.p. [° C.]: 73.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1340 BCH-32 1.0% $\Delta\epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}}\parallel$ [1 kHz, 20° C.]: 6.5 CC-3-V1 9.0% $\epsilon_{\text{sub.}}\perp$ [1 kHz, 20° C.]: -2.9 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 91 CLU-3-F

0.5% K.sub.1 [pN, 20° C.]: 14.9 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.2 CPY-3-O2 14.9%
PP-1-2V 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0840] The polymerizable LC mixture P51 is formulated as follows:

TABLE-US-00110 N51 99.434% RM-171 0.500% RM-64 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 52

[0841] The nematic LC host mixture N52 is formulated as follows:

TABLE-US-00111 B(S)-2O-O4 2.0% Cl. p. [° C.]: 69.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1296 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{sub. \parallel}$ [1 kHz, 20° C.]: 6.5 CC-3-V1 9.0% $\epsilon_{sub. \perp}$ [1 kHz, 20° C.]: -2.9 CCP-V-1 15.0% $\gamma_{sub.1}$ [mPa s, 20° C.]: 83 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 14.3 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 14.9 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 5.4% PY-3-O2 9.9% PYP-2-3 1.5% Y-4O-O4 4.0% Σ 100%

[0842] The polymerizable LC mixture P52 is formulated as follows:

TABLE-US-00112 N52 99.435% RM-171 0.500% RM-64 0.050% ST-3a-1 0.015% Σ 100%

Example 53

[0843] The nematic LC host mixture N53 is formulated as follows:

TABLE-US-00113 B(S)-2O-O4 2.0% Cl. p. [° C.]: 77.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1327 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{sub. \parallel}$ [1 kHz, 20° C.]: 6.5 CC-3-V1 9.0% $\epsilon_{sub. \perp}$ [1 kHz, 20° C.]: -2.9 CCP-V-1 15.0% $\gamma_{sub.1}$ [mPa s, 20° C.]: 93 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.6 CLY-3-O2 6.6% K.sub.3 [pN, 20° C.]: 17.0 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0844] The polymerizable LC mixture P53 is formulated as follows:

TABLE-US-00114 N53 99.684% RM-1 0.300% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 54

[0845] The nematic LC host mixture N54 is formulated as follows:

TABLE-US-00115 B(S)-2O-O4 2.0% Cl. p. [° C.]: 77.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1330 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{sub. \parallel}$ [1 kHz, 20° C.]: 6.4 CC-3-V1 9.0% $\epsilon_{sub. \perp}$ [1 kHz, 20° C.]: -2.8 CCP-V-1 15.0% $\gamma_{sub.1}$ [mPa s, 20° C.]: 97 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 16.0 CLY-(c5)-O2 6.6% K.sub.3 [pN, 20° C.]: 17.1 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0846] The polymerizable LC mixture P54 is formulated as follows:

TABLE-US-00116 N54 99.685% RM-1 0.300% ST-3a-1 0.015% Σ 100%

Example 55

[0847] The nematic LC host mixture N55 is formulated as follows:

TABLE-US-00117 B(S)-2O-O4 2.0% Cl. p. [° C.]: 72.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1297 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{sub. \parallel}$ [1 kHz, 20° C.]: 6.7 CC-3-V1 9.0% $\epsilon_{sub. \perp}$ [1 kHz, 20° C.]: -3.1 CCP-V-1 15.0% $\gamma_{sub.1}$ [mPa s, 20° C.]: 97 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.1 CLOY-3-O2 6.6% K.sub.3 [pN, 20° C.]: 16.7 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0848] The polymerizable LC mixture P55 is formulated as follows:

TABLE-US-00118 N55 99.684% RM-35 0.300% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 56

[0849] The nematic LC host mixture N56 is formulated as follows:

TABLE-US-00119 B(S)-2O-O4 2.0% Cl. p. [° C.]: 72.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1299 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{sub. \parallel}$ [1 kHz, 20° C.]: 6.7 CC-3-V1 9.0% $\epsilon_{sub. \perp}$ [1 kHz, 20° C.]: -3.1 CCP-V-1 15.0% $\gamma_{sub.1}$ [mPa s, 20° C.]: 102 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.1 CLOY-(c5)-O2 6.6% K.sub.3 [pN, 20° C.]: 16.8 CPY-3-O2

14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PYP-2-3 1.5% Σ 100%

[0850] The polymerizable LC mixture P56 is formulated as follows:

TABLE-US-00120 N56 99.685% RM-1 0.300% ST-3a-1 0.015% Σ 100%

Example 57

[0851] The nematic LC host mixture N57 is formulated as follows:

TABLE-US-00121 B(S)-2O-O4 2.0% Cl. p. [° C.]: 74.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1321 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 6.4 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: -2.8 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 91 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 14.9 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 16.1 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 4.9% PYP-2-3 1.5% LY-3-O2 5.0% Σ 100%

[0852] The polymerizable LC mixture P57 is formulated as follows:

TABLE-US-00122 N57 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 58

[0853] The nematic LC host mixture N58 is formulated as follows:

TABLE-US-00123 B(S)-2O-O4 2.0% Cl. p. [° C.]: 71.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1304 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 6.6 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: -3.0 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 91 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 14.4 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.6 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 4.9% PYP-2-3 1.5% LOY-3-O2 5.0% Σ 100%

[0854] The polymerizable LC mixture P58 is formulated as follows:

TABLE-US-00124 N58 99.435% RM-1 0.300% RM-145 0.200% RM-163 0.050% ST-3a-1 0.015% Σ 100%

Example 59

[0855] The nematic LC host mixture N59 is formulated as follows:

TABLE-US-00125 B(S)-2O-O4 2.0% Cl. p. [° C.]: 75.0 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1345 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 6.5 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: -2.9 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 92 CLU-3-F 0.5% K.sub.1 [pN, 20° C.]: 15.1 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.8 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PPY-4-O3 1.5% Σ 100%

[0856] The polymerizable LC mixture P59 is formulated as follows:

TABLE-US-00126 N59 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 60

[0857] The nematic LC host mixture N60 is formulated as follows:

TABLE-US-00127 B(S)-2O-O4 2.0% Cl. p. [° C.]: 74.5 B(S)-2O-O5 3.0% Δn [589 nm, 20° C.]: 0.1347 BCH-32 1.0% $\Delta \epsilon$ [1 kHz, 20° C.]: 3.6 CC-3-V 17.2% $\epsilon_{\text{sub.}} \parallel$ [1 kHz, 20° C.]: 6.4 CC-3-V1 9.0% $\epsilon_{\text{sub.}} \perp$ [1 kHz, 20° C.]: -2.8 CCP-V-1 15.0% $\gamma_{\text{sub.1}}$ [mPa s, 20° C.]: 91 CLU(1)-3-F 0.5% K.sub.1 [pN, 20° C.]: 14.8 CPY-2-O2 6.6% K.sub.3 [pN, 20° C.]: 15.9 CPY-3-O2 14.9% PP-1-2V1 10.0% PY-2-O2 9.4% PY-3-O2 9.9% PPY-2-3 1.5% Σ 100%

[0858] The polymerizable LC mixture P60 is formulated as follows:

TABLE-US-00128 N60 99.435% RM-1 0.300% RM-145 0.200% RM-163 0.050% ST-3a-1 0.015% Σ 100%

Example 61

[0859] The polymerizable LC mixture P61 is formulated as follows:

TABLE-US-00129 N1 99.084% RM-1 0.300% SA-23 0.600% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100% [00769]SA-23

Example 62

[0860] The polymerizable LC mixture P62 is formulated as follows:

TABLE-US-00130 N25 99.085% RM-1 0.300% SA-32 0.600% ST-3a-1 0.015% Σ 100%
[00770] SA-32

Example 63

[0861] The polymerizable LC mixture P63 is formulated as follows:

TABLE-US-00131 N1 99.084% RM-19 0.300% SA-23 0.600% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 64

[0862] The polymerizable LC mixture P64 is formulated as follows:

TABLE-US-00132 N25 99.085% RM-19 0.300% SA-32 0.600% ST-3a-1 0.015% Σ 100%

Example 65

[0863] The polymerizable LC mixture P65 is formulated as follows:

TABLE-US-00133 N26 98.884% RM-1 0.300% RM-145 0.200% SA-23 0.600% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 66

[0864] The polymerizable LC mixture P66 is formulated as follows:

TABLE-US-00134 N27 98.885% RM-19 0.300% RM-145 0.200% SA-32 0.600% ST-3a-1 0.015% Σ 100%

Example 67

[0865] The polymerizable LC mixture P67 is formulated as follows:

TABLE-US-00135 N28 98.884% RM-19 0.300% RM-145 0.200% SA-23 0.600% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 68

[0866] The polymerizable LC mixture P68 is formulated as follows:

TABLE-US-00136 N29 98.885% RM-19 0.300% RM-145 0.200% SA-32 0.600% ST-3a-1 0.015% Σ 100%

Example 69

[0867] The polymerizable LC mixture P69 is formulated as follows:

TABLE-US-00137 N1 98.584% RM-1 0.300% RM-145 0.200% SA-23 0.900% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 70

[0868] The polymerizable LC mixture P70 is formulated as follows:

TABLE-US-00138 N1 98.585% RM-1 0.300% RM-145 0.200% SA-32 0.900% ST-3a-1 0.015% Σ 100%

Example 71

[0869] The polymerizable LC mixture P71 is formulated as follows:

TABLE-US-00139 N30 98.584% RM-19 0.300% RM-145 0.200% SA-23 0.900% Irganox ®1076 0.001% ST-3b-1 0.015% Σ 100%

Example 72

[0870] The polymerizable LC mixture P72 is formulated as follows:

TABLE-US-00140 N31 98.585% RM-19 0.300% RM-145 0.200% SA-23 0.900% ST-3b-1 0.015% Σ 100%

Example 73

[0871] The polymerizable LC mixture P73 is formulated as follows:

TABLE-US-00141 N1 98.534% RM-1 0.300% RM-145 0.200% RM-163 0.050% SA-32 0.900% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 74

[0872] The polymerizable LC mixture P74 is formulated as follows:

TABLE-US-00142 N1 98.535% RM-1 0.300% RM-145 0.200% RM-163 0.050% SA-32 0.900% ST-3a-1 0.015% Σ 100%

Example 75

[0873] The polymerizable LC mixture P75 is formulated as follows:

TABLE-US-00143 N1 98.534% RM-19 0.300% RM-145 0.200% RM-163 0.050% SA-32 0.900% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 76

[0874] The polymerizable LC mixture P76 is formulated as follows:

TABLE-US-00144 N1 98.535% RM-19 0.300% RM-145 0.200% RM-163 0.050% SA-32 0.900% ST-3a-1 0.015% Σ 100%

Example 77

[0875] The polymerizable LC mixture P77 is formulated as follows:

TABLE-US-00145 N1 97.934% RM-1 0.300% RM-145 0.200% RM-163 0.050% SA-32 1.500% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 78

[0876] The polymerizable LC mixture P78 is formulated as follows:

TABLE-US-00146 N1 97.935% RM-1 0.300% RM-145 0.200% RM-163 0.050% SA-32 1.500% ST-3a-1 0.015% Σ 100%

Example 79

[0877] The polymerizable LC mixture P79 is formulated as follows:

TABLE-US-00147 N1 98.584% RM-1 0.300% RM-171 0.200% SA-23 0.900% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 80

[0878] The polymerizable LC mixture P80 is formulated as follows:

TABLE-US-00148 N1 98.585% RM-19 0.300% RM-171 0.200% SA-23 0.900% ST-3a-1 0.015% Σ 100%

Example 81

[0879] The polymerizable LC mixture P81 is formulated as follows:

TABLE-US-00149 N27 98.534% RM-1 0.300% RM-171 0.200% RM-163 0.050% SA-32 0.900% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 82

[0880] The polymerizable LC mixture P82 is formulated as follows:

TABLE-US-00150 N27 98.535% RM-1 0.300% RM-171 0.200% RM-163 0.050% SA-32 0.900% ST-3a-1 0.015% Σ 100%

Example 83

[0881] The polymerizable LC mixture P83 is formulated as follows:

TABLE-US-00151 N28 97.934% RM-1 0.300% RM-171 0.200% RM-163 0.050% SA-23 1.500% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 84

[0882] The polymerizable LC mixture P84 is formulated as follows:

TABLE-US-00152 N28 97.935% RM-1 0.300% RM-171 0.200% RM-163 0.050% SA-23 1.500% ST-3a-1 0.015% Σ 100%

Example 85

[0883] The polymerizable LC mixture P85 is formulated as follows:

TABLE-US-00153 N29 99.084% RM-35 0.300% SA-32 0.600% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 86

[0884] The polymerizable LC mixture P86 is formulated as follows:

TABLE-US-00154 N29 99.085% RM-35 0.300% SA-32 0.600% ST-3a-1 0.015% Σ 100%

Example 87

[0885] The polymerizable LC mixture P87 is formulated as follows:

TABLE-US-00155 N28 99.084% RM-171 0.300% SA-32 0.600% Irganox ®1076 0.001% ST-3b-1 0.015% Σ 100%

Example 88

[0886] The polymerizable LC mixture P88 is formulated as follows:

TABLE-US-00156 N28 99.085% RM-171 0.300% SA-32 0.600% ST-3b-1 0.015% Σ 100%

Example 89

[0887] The polymerizable LC mixture P89 is formulated as follows:

TABLE-US-00157 N25 99.084% RM-171 0.300% SA-32 0.600% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 90

[0888] The polymerizable LC mixture P90 is formulated as follows:

TABLE-US-00158 N25 99.085% RM-171 0.300% SA-32 0.600% ST-3a-1 0.015% Σ 100%

Example 91

[0889] The polymerizable LC mixture P91 is formulated as follows:

TABLE-US-00159 N26 98.884% RM-1 0.300% RM-35 0.200% SA-23 0.600% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 92

[0890] The polymerizable LC mixture P92 is formulated as follows:

TABLE-US-00160 N26 98.885% RM-1 0.300% RM-35 0.200% SA-32 0.600% ST-3a-1 0.015% Σ 100%

Example 93

[0891] The polymerizable LC mixture P93 is formulated as follows:

TABLE-US-00161 N2 98.534% RM-1 0.300% RM-35 0.200% RM-163 0.050% SA-32 0.900% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 94

[0892] The polymerizable LC mixture P94 is formulated as follows:

TABLE-US-00162 N2 98.535% RM-1 0.300% RM-171 0.200% RM-163 0.050% SA-32 0.900% ST-3a-1 0.015% Σ 100%

Example 95

[0893] The polymerizable LC mixture P95 is formulated as follows:

TABLE-US-00163 N25 97.934% RM-1 0.300% RM-35 0.200% RM-163 0.050% SA-32 1.500% Irganox ®1076 0.001% ST-3a-1 0.015% Σ 100%

Example 96

[0894] The polymerizable LC mixture P96 is formulated as follows:

TABLE-US-00164 N25 97.935% RM-1 0.300% RM-35 0.200% RM-163 0.050% SA-32 1.500% ST-3a-1 0.015% Σ 100%

Example 97

[0895] The polymerizable LC mixture P97 is formulated as follows:

TABLE-US-00165 N25 99.434% RM-1 0.300% RM-145 0.200% RM-163 0.050% Irganox ®1076 0.001% H-1-1-1 0.015% Σ 100% [00771]H-1-1-1

Example 98

[0896] The polymerizable LC mixture P98 is formulated as follows:

TABLE-US-00166 N25 99.435% RM-1 0.300% RM-145 0.200% RM-163 0.050% H-2-1-1 0.015% Σ 100% [00772]H-2-1-1

Example 99

[0897] The polymerizable LC mixture P99 is formulated as follows:

TABLE-US-00167 N1 99.435% RM-1 0.300% RM-120 0.200% RM-163 0.050% ST-3a-1 0.015% Σ 100% [00773]RM-120 (formula IBT1)

Example 100

[0898] The polymerizable LC mixture P100 is formulated as follows:

TABLE-US-00168 N27 99.434% RM-1 0.300% RM-142 0.200% RM-163 0.050% Irganox ®1076 0.001% ST-3c-1 0.015% Σ 100% [00774]RM-142 (formula IBT35)

Example 101

[0899] The polymerizable LC mixture P101 is formulated as follows:

TABLE-US-00169 N28 99.435% RM-171 0.300% RM-143 0.200% RM-163 0.050% ST-3b-1 0.015% Σ 100% [00775]  RM-143 (formula IBT22)

Example 102

[0900] The polymerizable LC mixture P102 is formulated as follows:

TABLE-US-00170 N26 99.584% RM-35 0.300% RM-58 0.100% Irganox® 1076 0.001% ST-3b-1 0.015% Σ 100% [00776]  RM-58 (formula IBD19)

Example 103

[0901] The polymerizable LC mixture P103 is formulated as follows:

TABLE-US-00171 N27 99.435% RM-1 0.300% RM-169 0.200% RM-163 0.050% ST-3b-1 0.015% Σ 100% [00777]  RM-169

Example 104

[0902] The polymerizable LC mixture P104 is formulated as follows:

TABLE-US-00172 N1 99.7% RM-19 0.3% Σ 100%

Example 105

[0903] The polymerizable LC mixture P105 is formulated as follows:

TABLE-US-00173 N1 99.7% RM-35 0.3% Σ 100%

Example 106

[0904] The polymerizable LC mixture P106 is formulated as follows:

TABLE-US-00174 N31 99.7% RM-171 0.3% Σ 100%

Example 107

[0905] The polymerizable LC mixture P107 is formulated as follows:

TABLE-US-00175 N25 99.5% RM-171 0.5% Σ 100%

Example 108

[0906] The polymerizable LC mixture P65 is formulated as follows:

TABLE-US-00176 N30 99.6% RM-171 0.4% Σ 100%

Example 109

[0907] The polymerizable LC mixture P109 is formulated as follows:

TABLE-US-00177 N2 99.5% RM-1 0.3% RM-171 0.2% Σ 100%

Example 110

[0908] The polymerizable LC mixture P110 is formulated as follows:

TABLE-US-00178 N3 99.5% RM-1 0.3% RM-35 0.2% Σ 100%

Example 111

[0909] The polymerizable LC mixture P111 is formulated as follows:

TABLE-US-00179 N25 99.5% RM-1 0.3% RM-120 0.2% Σ 100%

Claims

1. An LC medium having negative dielectric anisotropy and comprising one or more compounds of formula L1: ##STR00778## wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings R^{sup}.L1 straight chain alkyl having 1 to 15 C atoms or branched or cyclic alkyl having 3 to 15 C atoms, wherein one or more non-adjacent CH₂-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, CR^{sup}.0=CR^{sup}.00—, —C≡C—, ##STR00779## in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F or Cl, X^{sup}.L CH₃, F, Cl, CF₃ or CHF₂, Y^{sup}.L H, F, Cl or CH₃, and Y^{sup}.0 H, F, Cl, CF₃, CHF₂ or CH₃.
2. The LC medium according to claim 1, wherein the one or more compounds of formula L1 are selected from the following subformulae: ##STR00780## wherein R^{sup}.L1 denotes straight chain

alkyl having 1 to 15 C atoms or branched or cyclic alkyl having 3 to 15 C atoms, wherein one or more non-adjacent CH₂-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, CR^{sup.0}=CR^{sup.00}—, —C≡C—, ##STR00781## in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F or Cl.

3. The LC medium according to claim 1, wherein it further comprises one or more compounds selected from the group consisting of compounds of the formulae IIA, IIB, IIC, IID, IIE and IIF: ##STR00782## in which the individual radicals, on each occurrence identically or differently, and each, independently of one another, have the following meaning: R^{sup.21}, R^{sup.22} H, an alkyl or alkoxy radical having 1 to 15 C atoms or an alkenyl radical having 2 to 15 C atoms which is unsubstituted or monosubstituted by F, Cl, CN or CF₃ and where, in addition, one or more CH₂ groups in these radicals may be replaced by —O—, —S—, —C≡C—, —CF₂O—, —OCF₂—, —OC—O—, —O—CO— ##STR00783## in such a way that O- and/or S-atoms are not linked directly to one another, wherein in formula IID R^{sup.21} is different from cyclic alkyl or alkoxy, L^{sup.1} to L^{sup.5} F, Cl, CF₃ or CHF₂, Y H, F, Cl, CF₃, CHF₂ or CH₃, Z^{sup.1}, Z^{sup.2} a single bond, —CH₂CH₂—, —CH=CH—, —CF₂O—, —OCF₂—, —CH₂O—, —OCH₂—, —COO—, —OCO—, —C₂F₄—, —CF=CF—, —CH=CHCH₂O—, p 0, 1 or 2, and q 0 or 1.

4. The LC medium according to any claim 1, wherein it additionally comprises one or more compounds of formula III ##STR00784## in which R^{sup.31} and R^{sup.32} each, independently of one another, denote H, an alkyl or alkoxy radical having 1 to 15 C atoms or an alkenyl radical having 2 to 15 C atoms which is unsubstituted, monosubstituted by F, Cl, CN or CF₃ or at least monosubstituted by halogen, where, in addition, one or more CH₂ groups in these radicals may be replaced by —O—, —S—, ##STR00785## —C≡C—, —CF₂O—, —OCF₂—, —OC—O— or —O—CO— in such a way that O atoms are not linked directly to one another, Y^{sup.1}, Y^{sup.2} H, F, Cl, CF₃, CHF₂, CH₃ or OCH₃, A^{sup.3} on each occurrence, independently of one another, denotes a) 1,4-cyclohexenylene or 1,4-cyclohexylene radical, in which one or two non-adjacent CH₂ groups may be replaced by —O— or —S—, b) a 1,4-phenylene radical, in which one or two CH groups may be replaced by N, or c) a radical selected from the group consisting of spiro[3.3]heptane-2,6-diyl, 1,4-bicyclo[2.2.2]octylene, naphthalene-2,6-diyl, decahydronaphthalene-2,6-diyl, 1,2,3,4-tetrahydro-naphthalene-2,6-diyl, phenanthrene-2,7-diyl and fluorene-2,7-diyl, wherein the radicals a), b) and c) may be mono- or polysubstituted by halogen atoms, n denotes 0, 1 or 2, Z^{sup.1} on each occurrence independently of one another denotes —CO—O—, —O—CO—, —CF₂O—, —OCF₂—, —CH₂O—, —OCH₂—, —CH₂—, —CH₂CH₂—, —(CH₂)₄—, —CH=CH—CH₂O—, —C₂F₄—, —CH₂CF₂—, —CF₂CH₂—, —CF=CF—, —CH=CF—, —CF=CH—, —CH=CH—, —C≡C— or a single bond, and L^{sup.11} and L^{sup.12} each, independently of one another, denote F, Cl, CF₃ or CHF₂, and W denotes O or S.

5. The LC medium according to claim 1, wherein it additionally comprises one or more compounds of formula IV: ##STR00786## in which R^{sup.41} denotes an unsubstituted alkyl radical having 1 to 7 C atoms where, in addition, one or more CH₂ groups may be replaced by ##STR00787## or an unsubstituted alkenyl radical having 2 to 7 C atoms, and R^{sup.42} denotes an unsubstituted alkyl radical having 1 to 7 C atoms or an unsubstituted alkoxy radical having 1 to 6 C atoms or an unsubstituted alkenyl radical having 2 to 7 C atoms.

6. The LC medium according to claim 1, wherein it additionally comprises one, two or more polymerizable compounds selected of formula M:

R^{sup.a}—B^{sup.1}—(Z^{sup.m}—B^{sup.2})_{sub.m}—R^{sup.b} M wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings: P a polymerizable group, Sp a spacer group or a single bond, which is optionally substituted by L^{sup.a}, R^{sup.a}, R^{sup.b} P, P-Sp-, H, F, Cl, Br, I, —CN, —NO₂, —NCO, —

NCS, —OCN, —SCN, SF.sub.5 or straight-chain alkyl having 1 to 25 C atoms or branched alkyl having 3 to 25 C atoms, in which, in addition, one or more non-adjacent CH.sub.2 groups may each be replaced, independently of one another, by —C(R.sup.0)=C(R.sup.00)—, —C≡C—, —N(R.sup.00)—, —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, Br, I, CN, P or P-Sp-, where, if B.sup.1 and/or B.sup.2 contain a saturated C atom, R.sup.a and/or R.sup.b may also denote a radical which is spiro-linked to this saturated C atom, wherein at least one of the radicals R.sup.a and R.sup.b denotes or contains a group P or P-Sp-, B.sup.1, B.sup.2 an aromatic, heteroaromatic, alicyclic or heterocyclic group, which may also contain fused rings, and which is unsubstituted, or mono- or polysubstituted by L, Z.sup.m —O—, —S—, —CO—, —CO—O—, —OCO—, —O—CO—O—, —OCH.sub.2—, —CH.sub.2O—, —SCH.sub.2—, —CH.sub.2S—, —CF.sub.2O—, —OCF.sub.2—, —CF.sub.2S—, —SCF.sub.2—, —(CH.sub.2).sub.n1—, —CF.sub.2CH.sub.2—, —CH.sub.2CF.sub.2—, —(CF.sub.2).sub.n1—, —CH=CH—, —CF=CF—, —CH=CF—, —CF=CH—, —C≡C—, —CH=CH—COO—, —OCO—CH=CH—, —CH2CH2—CO—O—, O—CO—CH.sub.2—CH.sub.2—, —CR.sup.0R.sup.00— or a single bond, R.sup.0, R.sup.00 H or alkyl having 1 to 12 C atoms, m 0, 1, 2, 3 or 4, n1 1, 2, 3 or 4, L P, P-Sp-, OH, CH.sub.2OH, F, Cl, Br, I, —CN, —NO.sub.2, —NCO, —NCS, —OCN, —SCN, —C(=O)N(R.sup.x).sub.2, —C(=O)Y.sup.1, —C(=O)R.sup.x, —N(R.sup.x).sub.2, optionally substituted silyl, optionally substituted aryl having 6 to 20 C atoms, or straight-chain alkyl or alkoxy having 1 to 25 C atoms, straight-chain alkylcarbonyl, alkoxycarbonyl, alkylcarbonyloxy or alkoxycarbonyloxy having 2 to 25 C atoms, or branched alkyl, alkoxy, alkylcarbonyl, alkoxycarbonyl, alkylcarbonyloxy or alkoxycarbonyloxy having 3 to 25 C atoms, in which, in addition, one or more H atoms may be replaced by F, Cl, P or P-Sp-, Y.sup.1 halogen, R.sup.x P, P-Sp-, H, halogen, straight-chain alkyl having 1 to 25 C atoms or branched or cyclic alkyl having 3 to 25 C atoms, in which, in addition, one or more non-adjacent CH.sub.2 groups may be replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, P or P-Sp-, an optionally substituted aryl or aryloxy group having 6 to 40 C atoms, or an optionally substituted heteroaryl or heteroaryloxy group having 2 to 40 C atoms.

7. The LC medium according to claim 6, wherein the one, two or more polymerizable compounds M comprises one, two or more polymerizable compounds selected from formulae M1 to M32: ##STR00788## ##STR00789## ##STR00790## ##STR00791## ##STR00792## in which the individual radicals, on each occurrence identically or differently, and each, independently of one another, have the following meaning: P.sup.1, P.sup.2, P.sup.3 a polymerizable group, Sp.sup.1, Sp.sup.2, Sp.sup.3 a single bond or a spacer group where, in addition, one or more of the radicals P.sup.1-Sp.sup.1-, P.sup.2-Sp.sup.2- and P.sup.3-Sp.sup.3- may denote R.sup.M, with the proviso that at least one of the radicals P.sup.1-Sp.sup.1-, P.sup.2-Sp.sup.2- and P.sup.3-Sp.sup.3-present is different from R.sup.M, R.sup.M H, F, Cl, CN or straight-chain or branched alkyl having 1 to 25 C atoms or branched alkyl having 3 to 25 C atoms, in which, in addition, one or more non-adjacent CH.sub.2 groups may each be replaced, independently of one another, by —C(R.sup.0)=C(R.sup.00)—, —C≡C—, —N(R.sup.0)—, —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, CN or P.sup.1-Sp.sup.1-, and wherein R.sup.M does not denote or contain a group P.sup.1, P.sup.2 or P.sup.3, R.sup.0, R.sup.00 H or alkyl having 1 to 12 C atoms, R.sup.y, R.sup.z H, F, CH.sub.3 or CF.sub.3, X.sup.1, X.sup.2, X.sup.3 —CO—O—, —O—CO— or a single bond, Z.sup.M1 —O—, —CO—, —C(R.sup.yR.sup.z)— or —CF.sub.2CF.sub.2—, Z.sup.M2, Z.sup.M3 —CO—O—, —O—CO—, —CH.sub.2O—, —OCH.sub.2—, —CF.sub.2O—, —OCF.sub.2— or —(CH.sub.2).sub.n—, where n is 2, 3 or 4, L F, Cl, CN or straight-chain, optionally mono- or polyfluorinated alkyl,

alkoxy, or thioalkyl having 1 to 12 C atoms, straight-chain, optionally mono- or polyfluorinated alkenyl, alkynyl, alkylcarbonyl, alkoxy carbonyl, alkylcarbonyloxy or alkoxy carbonyloxy having 2 to 12 C atoms, or branched, optionally mono- or polyfluorinated alkyl, alkoxy, thioalkyl, alkenyl, alkynyl, alkylcarbonyl, alkoxy carbonyl, alkylcarbonyloxy or alkoxy carbonyloxy having 3 to 12 C atoms, L', L'' H, F or Cl, k 0 or 1, r 0, 1, 2, 3 or 4, s 0, 1, 2 or 3, t 0, 1 or 2, and x 0 or 1.

8. The LC medium according to claim 6, wherein the one, two or more polymerizable compounds of formula M are selected from formulae IA, IB and IC:

P-Sp-M.sup.1-Sp-P IA

P-Sp-M.sup.2-Sp-P IB

P-Sp-M.sup.3-Sp-P IC in which the individual radicals, on each occurrence identically or differently, and each, independently of one another, have the following meanings: M.sup.1, M.sup.2, M.sup.3 a group each individually selected from the following formulae ##STR00793## wherein the benzene rings are optionally substituted by one or more groups L or P-Sp-, L L.sup.a, L.sup.b, F, Cl, —CN, P-Sp-, or straight chain alkyl having 1 to 25 C atoms or branched or cyclic alkyl having 3 to 25 C atoms, wherein one or more non-adjacent CH.sub.2-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by P, F or Cl, L.sup.a —C(R.sup.aa)(R.sup.bb)OH, R.sup.aa, R.sup.bb straight-chain alkyl with 1 to 6 C atoms, L.sup.b straight-chain or branched alkenyl with 2 to 7 C atoms, wherein in the compounds of formula IA the group M.sup.1 and/or at least one spacer group Sp is at least monosubstituted with L.sup.a, and wherein in the compounds of formula IC the group M.sup.3 is at least monosubstituted with L.sup.b.

9. The LC medium according to claim 1, wherein it additionally comprises one or more additives selected from the group consisting of stabilisers, chiral dopants, polymerization initiators and self-alignment additives.

10. A process of preparing an LC medium, comprising the steps of mixing one or more one or more compounds of formula L1: ##STR00794## wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings R.sup.L1 straight chain alkyl having 1 to 15 C atoms or branched or cyclic alkyl having 3 to 15 C atoms, wherein one or more non-adjacent CH.sub.2-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, CR.sup.0=CR.sup.00—, —C≡C—, ##STR00795## in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F or Cl, X.sup.L CH.sub.3, F, Cl, CF.sub.3 or CHF.sub.2, Y.sup.L H, F, Cl or CH.sub.3, and Y.sup.0 H, F, Cl, CF.sub.3, CHF.sub.2 or CH.sub.3, and optionally one, two or more polymerizable compounds of formula M: R.sup.a—B.sup.1—(Z.sup.m—B.sup.2).sub.m—R.sup.b M wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings: P a polymerizable group, Sp a spacer group or a single bond, which is optionally substituted by L.sup.a, R.sup.a, R.sup.b P, P-Sp-, H, F, Cl, Br, I, —CN, —NO.sub.2, —NCO, —NCS, —OCN, —SCN, SF.sub.5 or straight-chain alkyl having 1 to 25 C atoms or a branched alkyls having 3 to 25 C atoms, in which, in addition, one or more non-adjacent CH.sub.2 groups may each be replaced, independently of one another, by —C(R.sup.0)=C(R.sup.00)—, —C≡C—, —N(R.sup.00)—, —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, Br, I, CN, P or P-Sp-, where, if B.sup.1 and/or B.sup.2 contain a saturated C atom, R.sup.a and/or R.sup.b may also denote a radical which is spiro-linked to this saturated C atom, wherein at least one of the radicals R.sup.a and R.sup.b denotes or contains a group P or P-Sp-, B.sup.1, B.sup.2 an aromatic, heteroaromatic, alicyclic or heterocyclic group, which may also contain fused rings, and which is unsubstituted, or mono- or polysubstituted by L, Z.sup.m —O—, —S—, —CO—, —CO—O—, —OCO—, —O—CO—O—, —OCH.sub.2

—CH.sub.2O— , —SCH.sub.2— , —CH.sub.2S— , —CF.sub.2O— , —OCF.sub.2— , —CF.sub.2S— , —SCF.sub.2— , $\text{—(CH.sub.2).sub.n1—}$, $\text{—CF.sub.2CH.sub.2—}$, $\text{—CH.sub.2CF.sub.2—}$, $\text{—(CF.sub.2).sub.n1—}$, —CH=CH— , —CF=CF— , —CH=CF— , —CF=CH— , $\text{—C}\equiv\text{C—}$, —CH=CH—COO— , —OCO—CH=CH— , $\text{—CH}_2\text{CH}_2\text{—CO—O—}$, $\text{O—CO—CH.sub.2—CH.sub.2—}$, $\text{—CR.sup.0R.sup.00—}$ or a single bond, R⁰, R⁰⁰ H or alkyl having 1 to 12 C atoms, m 0, 1, 2, 3 or 4, n₁ 1, 2, 3 or 4, L P, P-Sp-, OH, CH₂OH, F, Cl, Br, I, —CN— , —NO.sub.2— , —NCO— , —NCS— , —OCN— , —SCN— , $\text{—C(=O)N(R.sup.x).sub.2—}$, —C(=O)Y.sup.1— , —C(=O)R.sup.x— , $\text{—N(R.sup.x).sub.2—}$, optionally substituted silyl, optionally substituted aryl having 6 to 20 C atoms, or straight-chain alkyl or alkoxy having 1 to 25 C atoms, straight-chain alkylcarbonyl, alkoxy carbonyl, alkylcarbonyloxy or alkoxy carbonyloxy having 2 to 25 C atoms, or branched alkyl, alkoxy, alkylcarbonyl, alkoxy carbonyl, alkylcarbonyloxy or alkoxy carbonyloxy having 3 to 25 C atoms, in which, in addition, one or more H atoms may be replaced by F, Cl, P or P-Sp-, Y¹ halogen, R^x P, P-Sp-, H, halogen, straight-chain alkyl having 1 to 25 C atoms or branched or cyclic alkyl having 3 to 25 C atoms, in which, in addition, one or more non-adjacent CH₂ groups may be replaced by —O— , —S— , —CO— , —CO—O— , —O—CO— , —O—CO—O— in such a way that O and/or S atoms are not linked directly to one another, and in which, in addition, one or more H atoms may be replaced by F, Cl, P or P-Sp-, an optionally substituted aryl or aryloxy group having 6 to 40 C atoms, or an optionally substituted heteroaryl or heteroaryloxy group having 2 to 40 C atoms, with one or more compounds of formula II, III, IV and/or V: ##STR00796## wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings R²¹ and R²²H, straight chain, branched or cyclic alkyl or alkoxy having 1 to 20 C atoms, wherein one or more non-adjacent CH₂-groups are optionally replaced by —O— , —S— , —CO— , —CO—O— , —O—CO— , —O—CO—O— , CR⁰—CR⁰⁰—, $\text{—C}\equiv\text{C—}$, ##STR00797## in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F, Cl, CN or CF₃, preferably alkyl or alkoxy having 1 to 6 C atoms, R⁰, R⁰⁰ H or alkyl having 1 to 12 C atoms, A¹ and A² a group selected from the following formulae ##STR00798## ##STR00799## Z¹ and Z² $\text{—CH.sub.2CH.sub.2—}$, —CH=CH— , —CF.sub.2O— , —OCF.sub.2— , —CH.sub.2O— , —OCH.sub.2— , —CO—O— , —O—CO— , —C.sub.2F.sub.4— , —CF=CF— , —CH=CH—CH.sub.2O— or a single bond, L¹, L², L³ and L⁴ F, Cl, OCF₃, CF₃, CH₃, CH₂F or CHF₂, Y H, F, Cl, CF₃, CHF₂ or CH₃, L^C CH₃ or OCH₃, a₁ 0, 1 or 2, a₂ 0 or 1, ##STR00800## in which R³¹ and R³² each, independently of one another, denote H, an alkyl or alkoxy radical having 1 to 15 C atoms or an alkenyl radical having 2 to 15 C atoms which is unsubstituted, monosubstituted by F, Cl, CN or CF₃ or at least monosubstituted by halogen, where, in addition, one or more CH₂ groups in these radicals may be replaced by —O— , —S— , ##STR00801## $\text{—C}\equiv\text{C—}$, —CF.sub.2O— , —OCF.sub.2— , —OC—O— or —O—CO— in such a way that O atoms are not linked directly to one another, Y¹, Y² H, F, Cl, CF₃, CHF₂, CH₃ or OCH₃, A³ on each occurrence, independently of one another, denotes a) 1,4-cyclohexenylene or 1,4-cyclohexylene radical, in which one or two non-adjacent CH₂ groups may be replaced by —O— or —S— , b) a 1,4-phenylene radical, in which one or two CH groups may be replaced by N, or c) a radical selected from the group consisting of spiro[3.3]heptane-2,6-diyl, 1,4-bicyclo[2.2.2]octylene, naphthalene-2,6-diyl, decahydronaphthalene-2,6-diyl, 1,2,3,4-tetrahydronaphthalene-2,6-diyl, phenanthrene-2,7-diyl and fluorene-2,7-diyl, wherein the radicals a), b) and c) may be mono- or polysubstituted by halogen atoms, n denotes 0, 1 or 2, Z¹ on each occurrence independently of one another denotes —CO—O— , —O—CO— , —CF.sub.2O— , —OCF.sub.2— , —CH.sub.2O— , —OCH.sub.2— , —CH.sub.2— , $\text{—CH.sub.2CH.sub.2—}$, $\text{—(CH.sub.2).sub.4—}$, —CH=CH—CH.sub.2O— , —C.sub.2F.sub.4— , $\text{—CH.sub.2CF.sub.2—}$, $\text{—CF.sub.2CH.sub.2—}$, —CF=CF— , —CH=CF— , —CF=CH— , —CH=CH— , $\text{—C}\equiv\text{C—}$ or a

single bond, and L.sup.11 and L.sup.12 each, independently of one another, denote F, Cl, CF.sub.3 or CHF.sub.2, and W denotes O or S, ##STR00802## in which R.sup.41 denotes an unsubstituted alkyl radical having 1 to 7 C atoms where, in addition, one or more CH.sub.2 groups may be replaced by ##STR00803## or an unsubstituted alkenyl radical having 2 to 7 C atoms, and R.sup.42 denotes an unsubstituted alkyl radical having 1 to 7 C atoms or an unsubstituted alkoxy radical having 1 to 6 C atoms or an unsubstituted alkenyl radical having 2 to 7 C atoms, ##STR00804## in which R.sup.51 and R.sup.52 independently of one another, denote H, an alkyl, alkoxy or alkenyl radical having up to 15 C atoms which is unsubstituted, monosubstituted by F, Cl, CN or CF.sub.3 or at least monosubstituted by halogen, where, in addition, one or more CH.sub.2 groups in these radicals may be replaced by —O—, —S—, —C≡C—, —CF.sub.2O—, —OCF.sub.2—, —OC—O—, —O—CO— ##STR00805## in such a way that O atoms are not linked directly to one another, ##STR00806## identically or differently, denote ##STR00807## Z.sup.51, Z.sup.52 each, independently of one another, denote —CH.sub.2—CH.sub.2—, —CH.sub.2—O—, —CH=CH—, —C≡C—, —COO— or a single bond, and n is 1 or 2, and optionally with further liquid-crystalline compounds and/or additives, and optionally polymerizing the polymerizable compounds.

11. An LC display comprising an LC medium as defined in claim 1.

12. The LC display according to claim 11, which is a VA, IPS, FFS, UB-FFS, PS-VA, PS-IPS, PS-FFS, PS-UB-FFS or SA-VA display.

13. The LC display according to claim 11, wherein it comprises two substrates, at least one of which is transparent to light, an electrode provided on each substrate or two electrodes provided on only one of the substrates, and located between the substrates a layer of an LC medium having negative dielectric anisotropy and comprising one or more compounds of formula L1:

##STR00808## wherein the individual radicals, independently of each other and on each occurrence identically or differently, have the following meanings R.sup.L1 straight chain alkyl having 1 to 15 C atoms or branched or cyclic alkyl having 3 to 15 C atoms, wherein one or more non-adjacent CH.sub.2-groups are optionally replaced by —O—, —S—, —CO—, —CO—O—, —O—CO—, —O—CO—O—, CR.sup.0=CR.sup.00—, —C≡C—, ##STR00809## in such a manner that O- and/or S-atoms are not directly connected with each other, and wherein one or more H atoms are each optionally replaced by F or Cl, X.sup.L CH.sub.3, F, Cl, CF.sub.3 or CHF.sub.2, Y.sup.L H, F, Cl or CH.sub.3, and Y.sup.0 H, F, Cl, CF.sub.3, CHF.sub.2 or CH.sub.3, wherein the polymerizable compounds are polymerized between the substrates of the display by UV photopolymerization.

14. A process of manufacturing an LC display according to claim 13, comprising the steps of providing the LC medium having negative dielectric anisotropy and comprising one or more compounds of formula L1 between the substrates of the display, and polymerizing the polymerizable compounds by irradiation with UV light.

15. The LC medium according to claim 2, wherein R.sup.L1 denotes alkyl having 1 to 6 C atoms in which a CH.sub.2 group is optionally replaced by cyclopropane-1,2-diyl or cyclopentane-1,2-diyl.

16. The LC medium according to claim 15, wherein R.sup.L1 denotes propyl, pentyl, cyclopropyl, cyclopentyl, cyclopropylmethyl or cyclopentylmethyl.

17. The LC display according to claim 11, comprising an energy-saving LC display.

18. The process according to claim 10, comprising an energy-saving LC display production process.
